

ELECTRICAL SAFETY

Fault Protection in LV system

Fire & Electrocution

IS 732 – 2016 (draft): Code practice of Electrical Wiring Installations

IS 3043: Code practice of Earthing

IEC 60364-5-53, 54 & 4-44 (Low Voltage Electrical Installation: Earthing, Bonding and Protective Devices

CEA: Regulations. Measures relating to safety and electric supply.

IS/IEC 62305: Lightning Protection (Internal and External Protection)

ISO/IEC 30129: Bonding Networks & Earthing for Telecom and DATA Centers

IEC 62561 - 2, 7: Components for Earthing and LPS

Most experienced company in India in the subject of LPS, Structural Protection, SPD, Earthing
We are continuously training the market with the modern ways since 1997



Dr. Chandima Gomes, Dr. Muralidas & Mr. Gopakumar with participants during 2 days seminar at cochin in 2006



Dr. Shriram Sharma, Dr. Chandima Gomes, Mr. Gopakumar & participants during 2 days Lightning Protection workshop in Nepal, June 2016 along with former Prime Minister Hon. Shri Jhala Nath Khanal



Participants during 2 days Seminar at CPRI in 2016



Participants during 5 days EMI/EMC training program at Malaysia



2 days EMI/EMC training program for railway installations at RDSO Lucknow



Mr. Ganesan and Mr. Gopakumar with Railway S&T officials at IRISSET, Secunderabad during 2 days EMI/EMC program on Railway System

We are an UL listed LPS installer.

Trained by UL to handle maximum safety for your installation

We are an UL listed Electrode manufacturer.

Have more than 100 components for LPS and earthing tested to IEC 62561



LIGHTNING PROTECTION
MASTER LABEL® CERTIFICATE SERVICES

Prevent loss from lightning damage with UL's Master Label® Certificate



Accident - Overview

Table 1.9 (Contd...)

Sl. No.	State/UT	No. of Cases	Accidental Fire (Electrical Short Circuit)							
			No. of Persons Injured		No. of Persons Died					
			Male	Female	Male	Female	Male	Female	Male	Female
(1)										
11	JHARKHAND	18	0	0	0	0	15	4	0	19
12	KARNATAKA	118	2	0	0	2	110	8	0	118
13	KERALA	31	2	0	0	2	29	2	0	31
14	MADHYA PRADESH	85	0	0	0	0	61	24	0	85
15	MAHARASHTRA	364	173	63	0	236	118	42	0	160
16	MANIPUR	0	0	0	0	0	0	0	0	0
17	MEGHALAYA	3	0	0	0	0	2	1	0	3
18	MIZORAM	2	0	0	0	0	1	1	0	2
19	NAGALAND	0	0	0	0	0	0	0	0	0
20	ODISHA	82	0	0	0	0	63	19	0	82
21	PUNJAB	44	4	1	0	5	38	8	0	46
22	RAJASTHAN	118	0	0	0	0	96	22	0	118
23	SIKKIM	0	0	0	0	0	0	0	0	0
24	TAMIL NADU	262	0	0	0	0	212	50	0	262
25	TELANGANA	211	1	0	0	0	209	2	0	211
26	TRIPURA	2	0	0	0	0	2	0	0	2
27	UTTAR PRADESH	222	14	0	0	0	208	14	0	222
28	UTTARAKHAND	9	0	0	0	0	9	0	0	9
29	WEST BENGAL	116	12	0	0	0	104	12	0	116
TOTAL (STATES)		2413	230	0	0	0	1833	422	0	2255
UNION TERRITORIES										
30	A & N ISLANDS	2	0	0	0	0	2	0	0	2
31	CHANDIGARH	4	0	0	0	0	4	0	0	4
32	D & N HAVELI	0	0	0	0	0	0	0	0	0
33	DAMAN & DIU	0	0	0	0	0	0	0	0	0
34	DELHI (UT)	61	14	0	0	0	47	14	0	61
35	LAKSHADWEEP	0	0	0	0	0	0	0	0	0
36	PUDUCHERRY	5	0	0	0	0	5	0	0	5
TOTAL (UTs)		72	14	0	0	14	64	8	0	72
TOTAL (ALL INDIA)		2485	244	0	0	324	1897	430	0	2255

Table 1.9 (Contd...)

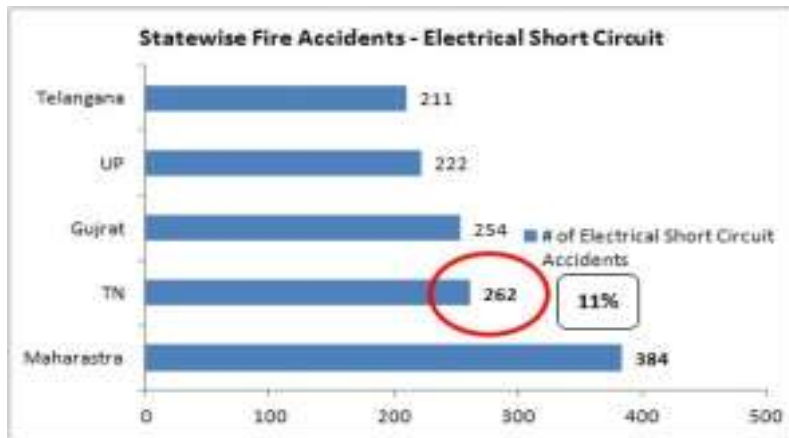
Sl. No.	State/UT	No. of Cases	Electrocution							
			No. of Persons Injured		No. of Persons Died					
			Male	Female	Trans-gender	Total	Male	Female	Trans-gender	Total
11	JHARKHAND	81	2	0	0	2	87	10	0	82
12	KARNATAKA	434	1	0	0	1	366	68	0	434
13	KERALA	217	3	9	0	12	183	34	0	217
14	MADHYA PRADESH	1543	17	0	0	17	1284	261	0	1545
15	MAHARASHTRA	1557	164	46	0	210	1143	218	0	1361
16	MANIPUR	13	0	0	0	0	13	1	0	13
17	MEGHALAYA	8	0	0	0	0	4	3	0	7
18	MIZORAM	7	0	0	0	0	7	1	0	8
19	NAGALAND	2	0	0	0	0	2	0	0	2
20	ODISHA	273	0	0	0	0	225	52	0	277
21	PUNJAB	320	0	0	0	0	304	17	0	321
22	RAJASTHAN	1057	11	3	0	14	909	157	0	1066
23	SIKKIM	11	0	0	0	0	11	1	0	12
24	TAMIL NADU	637	1	0	0	1	548	102	0	651
25	TELANGANA	450	0	0	0	0	400	50	0	450
26	TRIPURA	41	0	0	0	0	34	7	0	41
27	UTTAR PRADESH	756	0	0	0	0	608	97	0	705
28	UTTARAKHAND	43	0	0	0	0	38	4	0	43
29	WEST BENGAL	156	11	137	18	0	156	18	0	156
TOTAL (STATES)		9886	364	8374	1515	0	9886	0	0	9886
UNION TERRITORIES										
30	A & N ISLANDS	1	0	1	0	0	1	0	0	1
31	CHANDIGARH	2	0	2	0	0	2	0	0	2
32	D & N HAVELI	3	0	3	0	0	3	0	0	3
33	DAMAN & DIU	0	0	0	0	0	0	0	0	0
34	DELHI (UT)	81	4	70	11	0	81	11	0	81
35	LAKSHADWEEP	0	0	0	0	0	0	0	0	0
36	PUDUCHERRY	10	0	10	0	0	10	0	0	10
TOTAL (UTs)		97	4	88	11	0	97	11	0	97
TOTAL (ALL INDIA)		10200	278	95	9	364	8461	1526	0	9986

+

1000's of un-reported cases
100's of millions of Rupees Loss
Majority in L.V System

Recent incidents

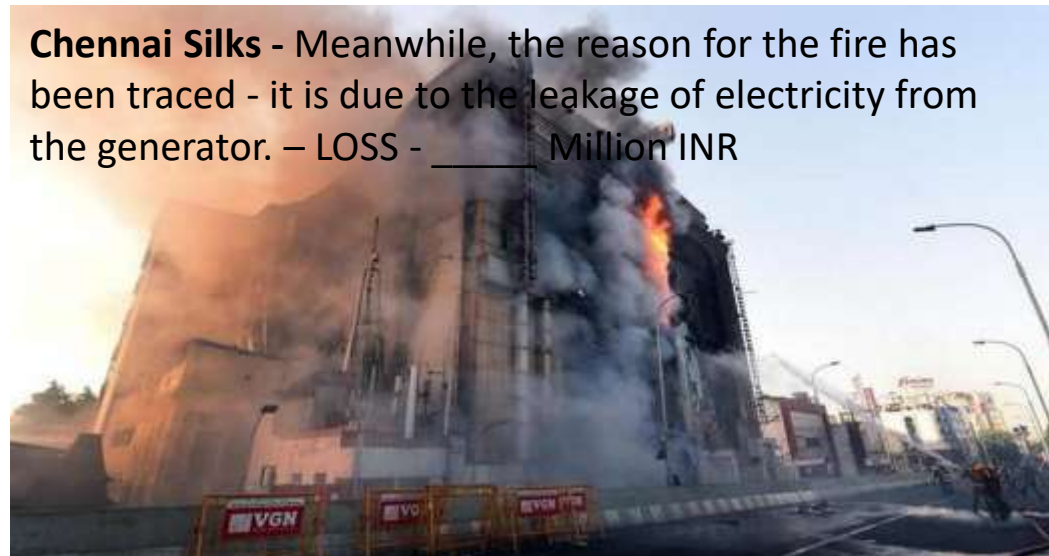
Short circuit has caused 80% of Mumbai fires over past three years, says fire brigade. The city has seen around 15,000 fires in the past three years. P Rahangdale, chief fire officer of the Mumbai fire brigade – Hindustan time – Nov 2017



Mumbai Kamala Mills tragedy: Probe reveals short circuit as cause of fire (____ died, ____ injured)



Chennai Silks - Meanwhile, the reason for the fire has been traced - it is due to the leakage of electricity from the generator. – LOSS - ____ Million INR



Reason

- Faulty and fried wiring
- Too many equipment in Extension cords
- Faulty Electrical appliances
- Outdated circuit breakers
- Light fixtures and decorations
- Electrical switches and outlets
- Unattended charging points (Laptop / TV's in powered condition)



Result - Insulation failure, Over current, Heat
..... fire & electrocution

Protective Measures =

Basic Protection & Fault Protection

also called as

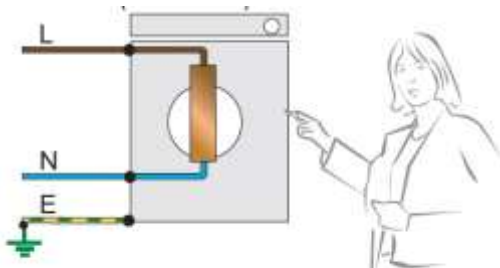
Direct Contact & Indirect Contact

Basic Protection

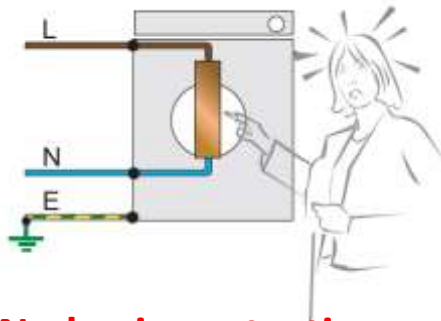
**Insulation of Live Parts
Barriers or Enclosure**

Fault Protection

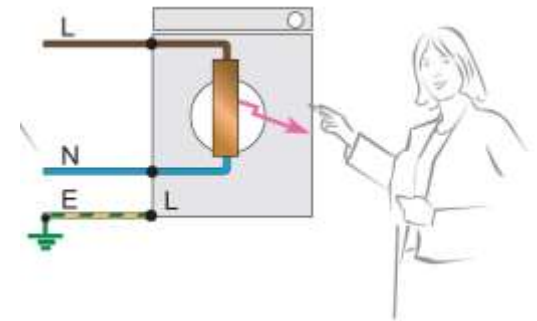
**Automatic Disconnection of supply
Double insulation, Separation, ELV**



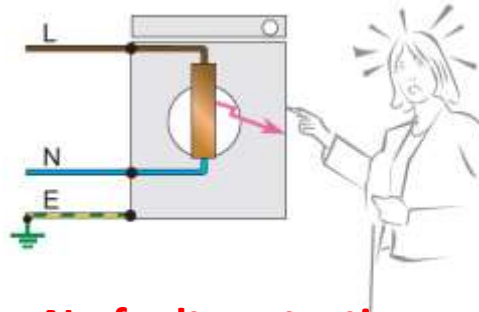
**With basic protection
(enclosure)**



No basic protection



**With fault protection
disconnection of supply
or double insulation**



No fault protection

Preventive Measure

1. Avoid current flow through body
2. Limit amount of current
3. **Automatic Disconnection of Supply**
Disconnection of supply is common possible measure

Protective Measure (only for fire)

1. Smoke / Fire detection
2. Fire Suppression
3. Smoke / Fire sealing
4. Evacuation
5. Rules and regulations.....

- More than 95% of accidents can be avoided if the supply is automatically disconnected within the prescribed time.
- Several Conditions shall be fulfilled for Automatic Disconnection of Supply

STANDARD – is a set of technical definitions and guidelines that function as instruction for designers, manufacturers, operators or users of equipment.

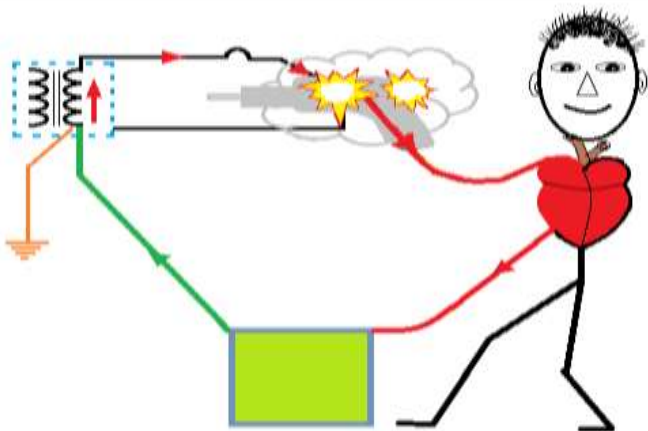
Code – A standard becomes a code when it has been adopted by one of more government bodies and is enforced by law or when it has been incorporated into a business contract.

**Standards for
Switches, cables,
motors, etc**

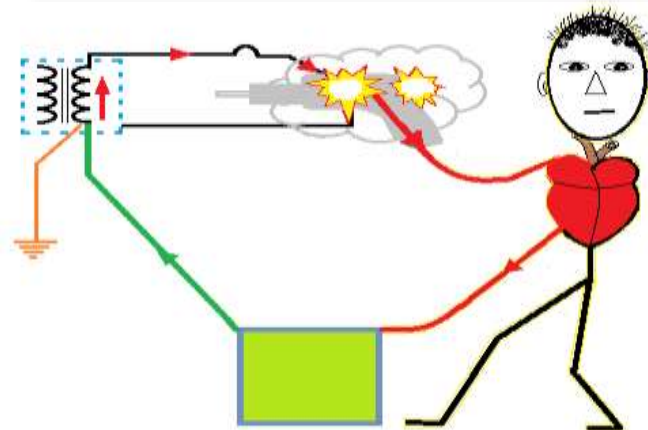
**Code of practice
IS3043 (earthing
IS 732 (wiring)
NEC - 2011**

Electric Shock Values in Humans

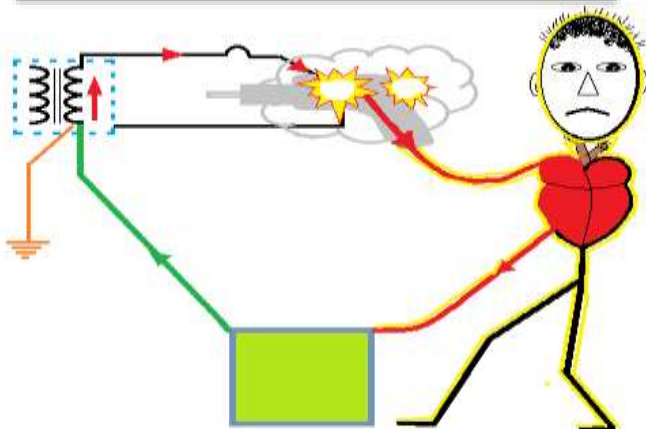
Electrical Sensation – 0.3 mA to 0.4 mA



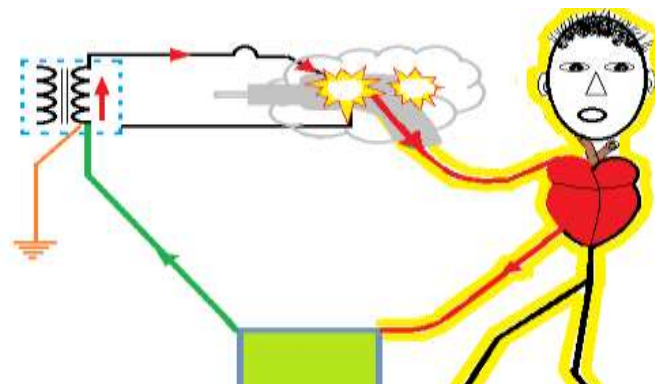
Perception Let- Go - 0.7 mA to 1.1 mA



Maximum Let - Go Level
10 mA (Female); 16 mA (Male)

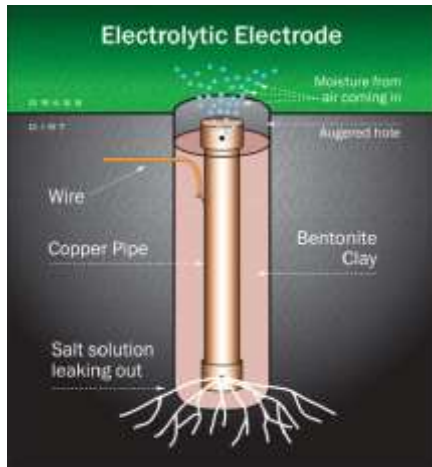


Fibrillation Level
50 mA for 0.2 Secs (female)
75 mA for 0.5 Secs (Male)



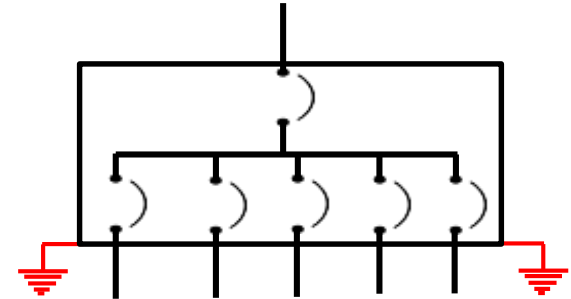
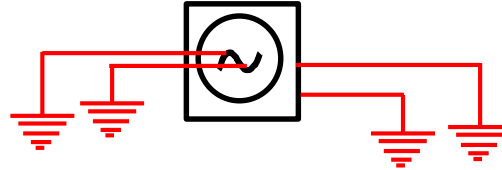
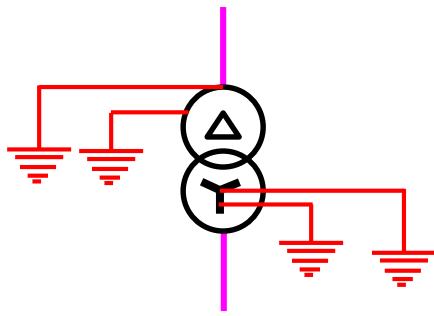
Solution: Automatic Disconnection of Power Supply before the prescribed time

Earthing – General idea



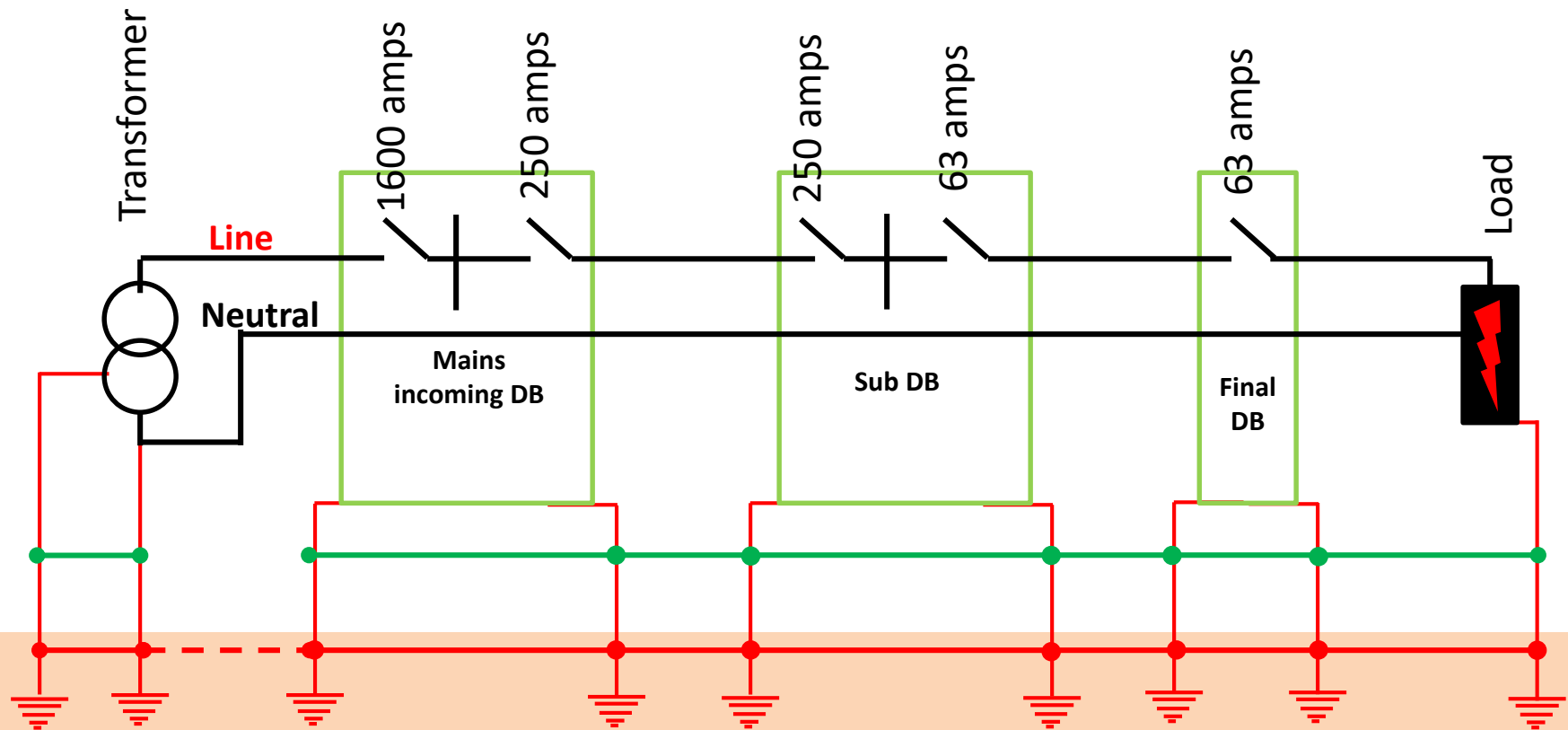
- PLATE, PIPE, Current dissipation, surface area, soil resistivity
- Chemical Earthing, Pipe-in-pipe, Dual pipe technology
- Chemical compound
- & several other names





- **Copper Busbar and Copper plates for Transformer / DG Neutral**
- **GI Busbar and GI pipes for Transformer / DG Body**
- **GI Busbar and GI pipes for Panels**
- **Pipe-in-pipe, chemical earth electrodes..... etc are also used**

Typical Earthing installation in LV Industrial / commercial system



Industrial installation - equipment are connected to one / two vertical earth electrode

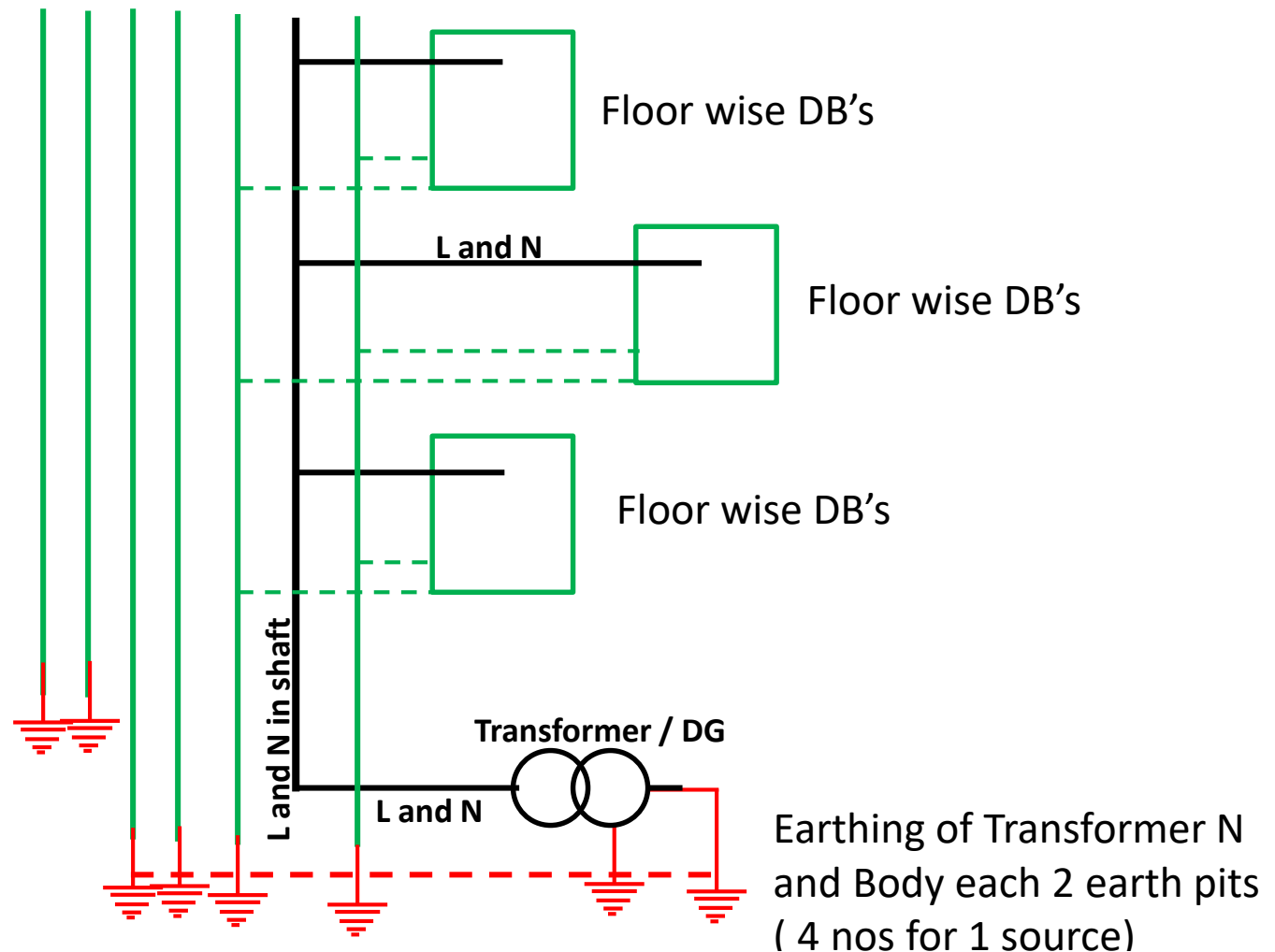
Some places earth electrodes in soil are not interconnected

There is an earth Busbar which runs on the load side (Some time two Bus bars)

Some time these electrodes are interconnected under soil. (based on convenience ???)

Two separate bus bars connected to earth electrodes in soil for lifts. WHY????

Earthing in typical high raise buildings



There are buildings having more than 300 to 400 earth pits. Each tested once in an year and replaced once in 2 to 3 years, Tried hard to achieve 1 ohm resistance.....
Finally a tough job for the electrical engineer
Every failure is attributed to poor quality earthing

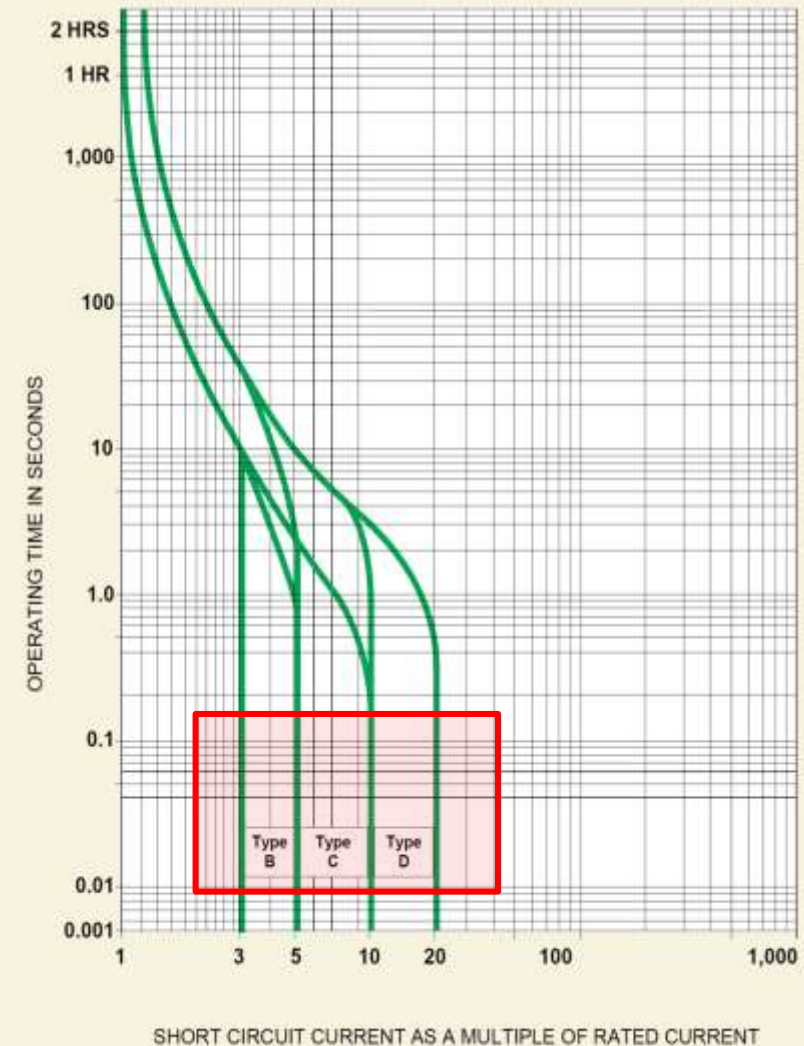
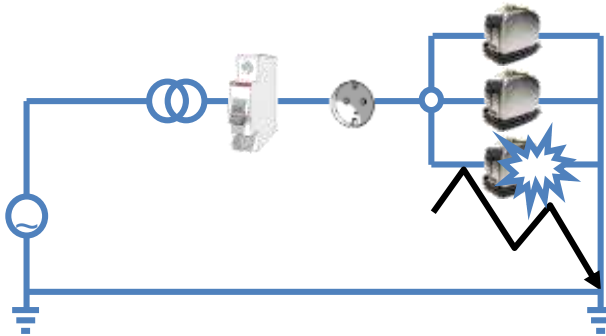
IS 732 & IS 3043: Primary Protective Device is an OCPD

(RCD is additional protective device)

OCPD – Fuse, MCB, MCCB, ACB etc

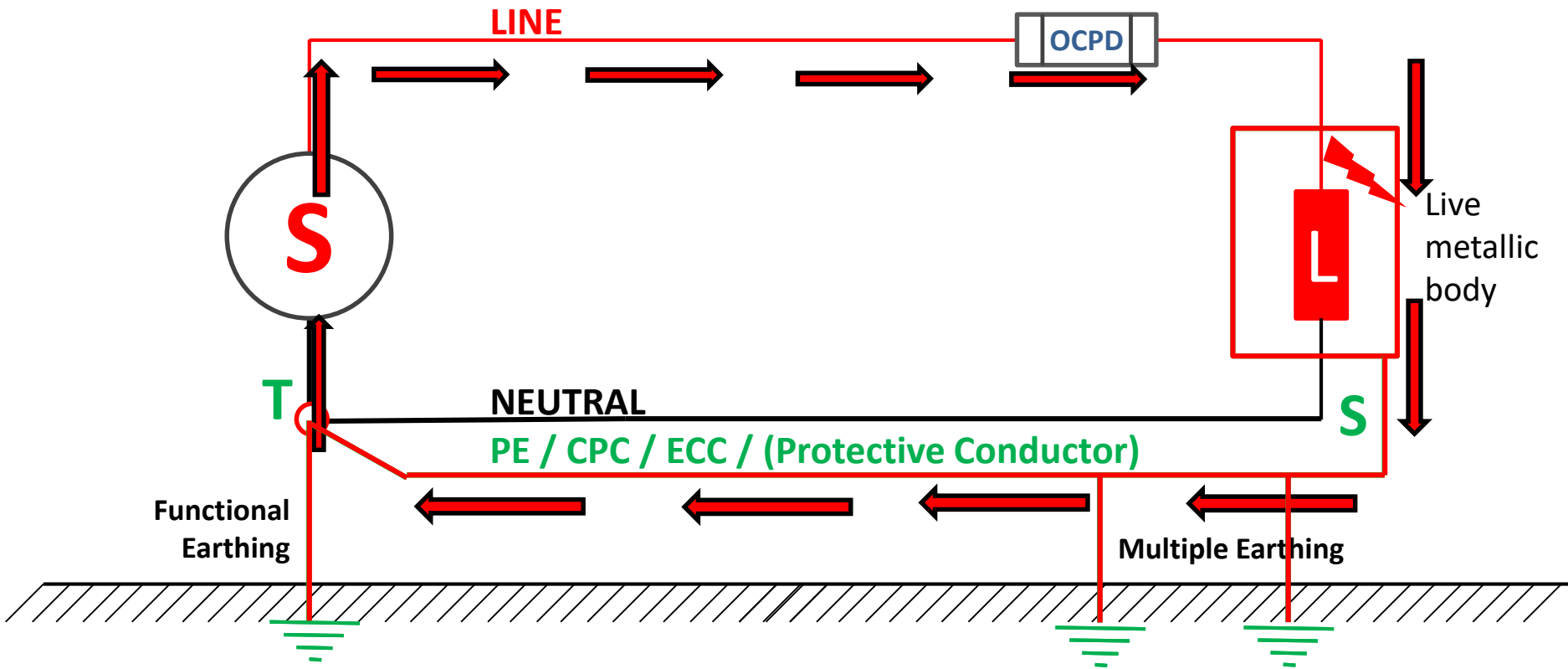
Occurs when any lines come in contact with each other or the earth

Short-circuit



For instantaneous tripping - current through the device shall be 10 times the rated current

Case – 1 TN-S System – Loop impedance (eg earth fault)



Operating Voltage = 230

OCPD = 250 amps MCCB.

$Z_s = 1.0 \Omega$

$I_{sc} = 230$ amps.

Device will never trip.

Danger

$Z_s = 0.5 \Omega$

$I_{sc} = 460$ amps.

Device will trip after long time

Danger

$Z_s = 0.1 \Omega$

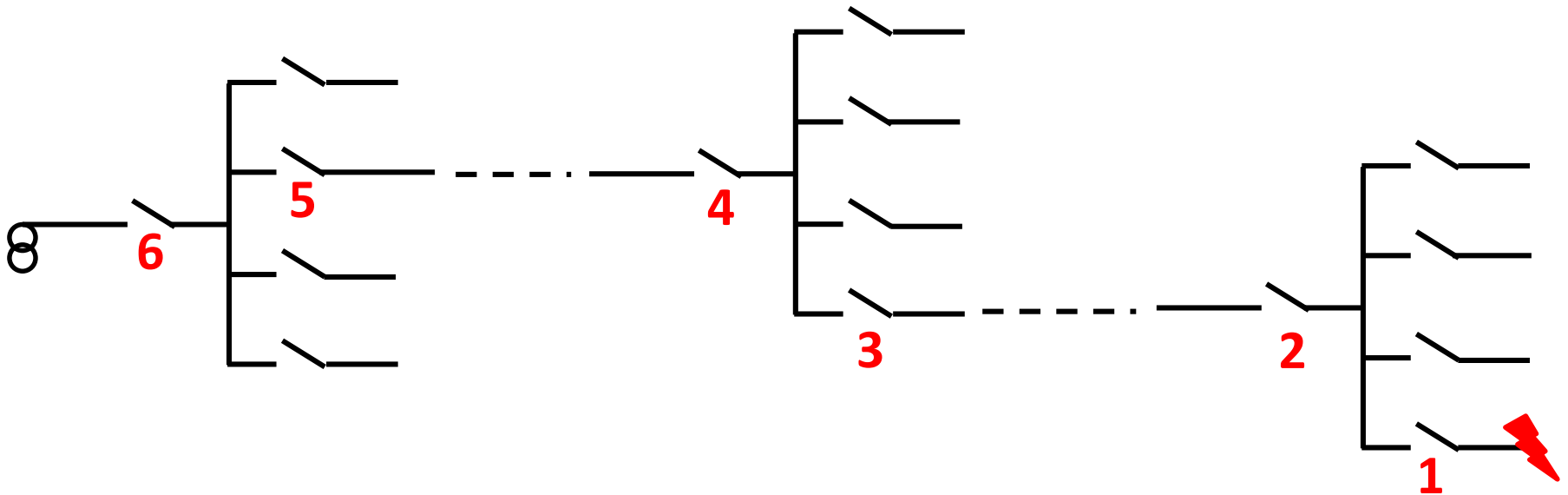
$I_{sc} = 2300$ amps.

Device will trip quickly.

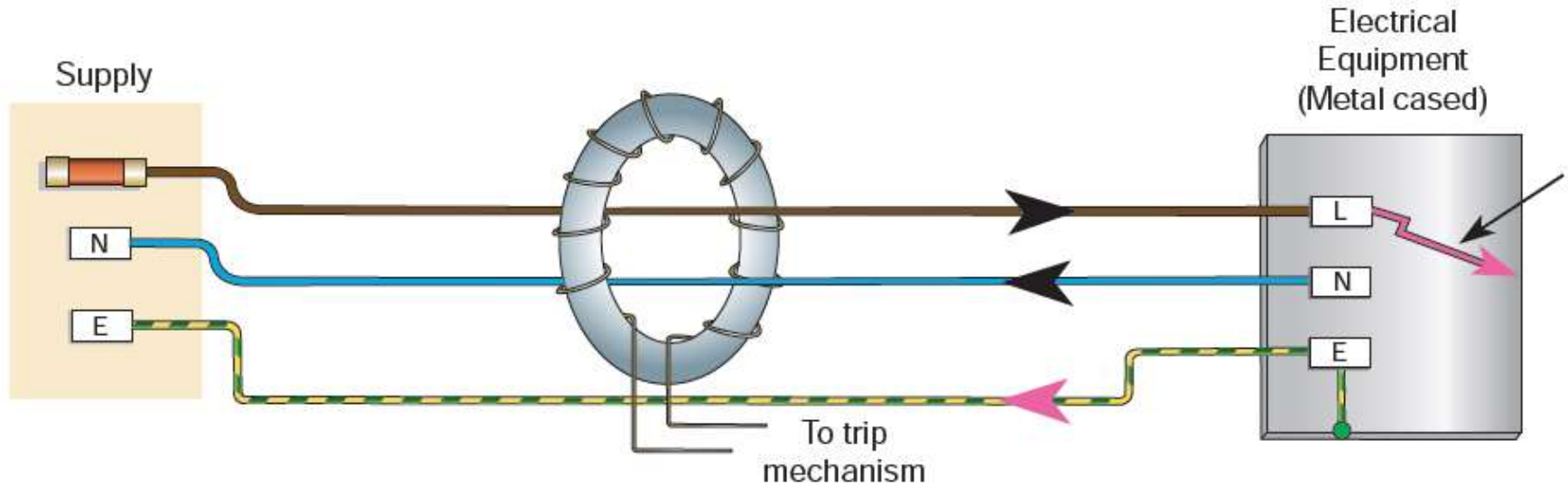
Safe

Low Loop impedance = More Fault current = Fast operation of Protective Device
Resistance of PE conductor plays major role in TN network in an earth fault

Purpose of Coordination (also called as discrimination / Cascading) is to disconnect only the faulty load or feeder from the network and no others, while keeping as much as possible of the installation powered up. Safety can thus be combined with continuity of service, and the fault easily located. Discrimination is said to be total if it is assured irrespective of the value of the fault current, up to the maximum value available in the installation. If this is not the case, it is said to be partial.



RCD: Residual Current Device



Opens the circuit when

Residual current is

- > 30 mA (Human protection) (from 15 mA to 30 mA)
- > 100 mA
- > 300 mA (Fire protection)

3 type of RCD's exist

Type AC is for AC leakage current

Type A is for AC and pulsating DC leakage current

Type B is for AC and DC leakage currents

Type of RCD's

Supply	Form of Residual Current	Recommended type of symbol		
		AC	A	B
Sinusoidal A.C.	 Suddenly applied	✓	✓	✓
	 Slowly rising			
Pulsating D.C.	 Suddenly applied		✓	✓
	 Slowly rising			
Smooth D.C.	 — — — — —			✓

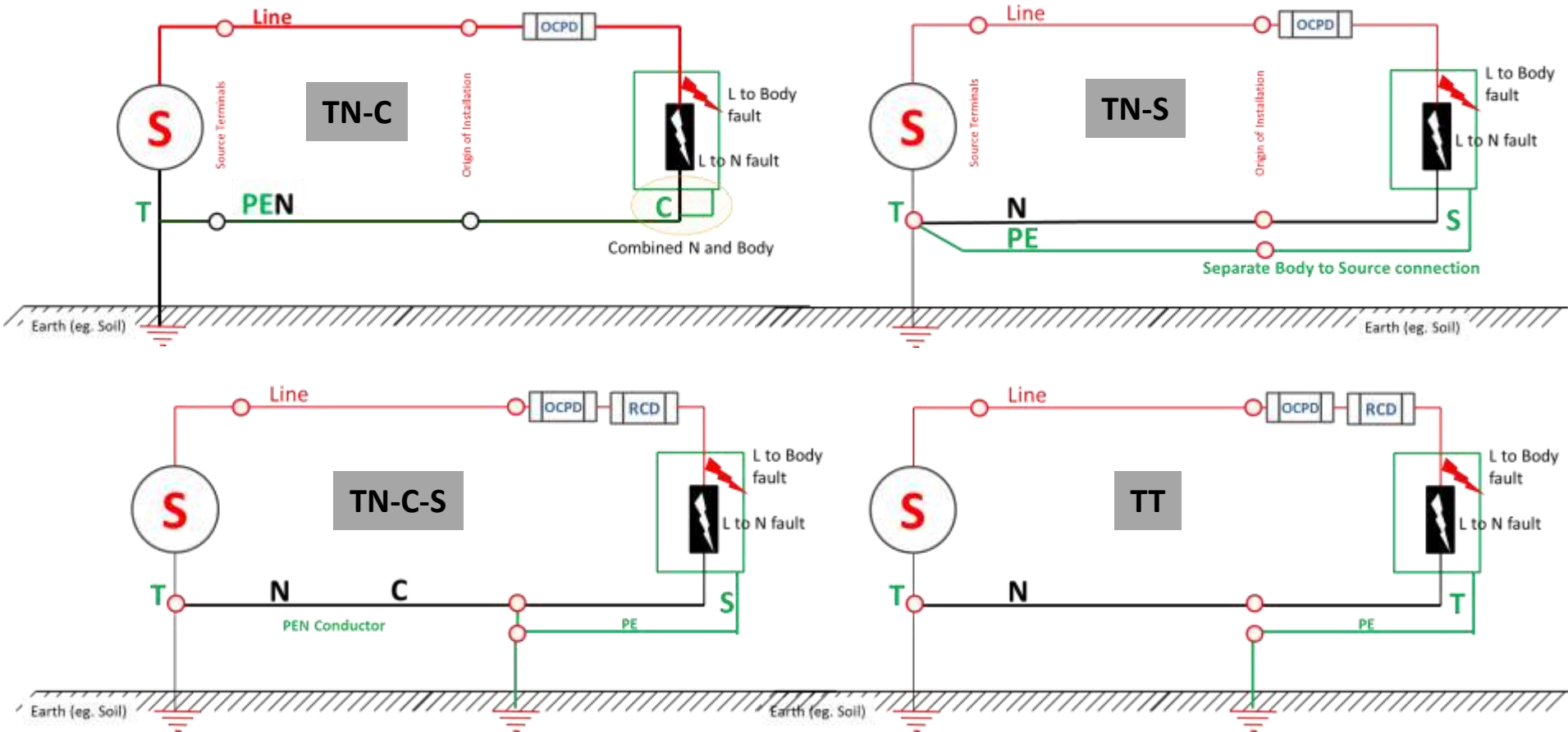
Several Countries Type A or Type B RCD is mandatory by rule (LED lights, SOLAR, Drives, etc)

1. Earthing for safety (Automatic Disconnection of Power Supply)
2. Voltage Reference of System (Neutral)
3. Dissipation of Lightning current
4. Base for Equipotentialisation
5. Shielding against Electro Magnetics

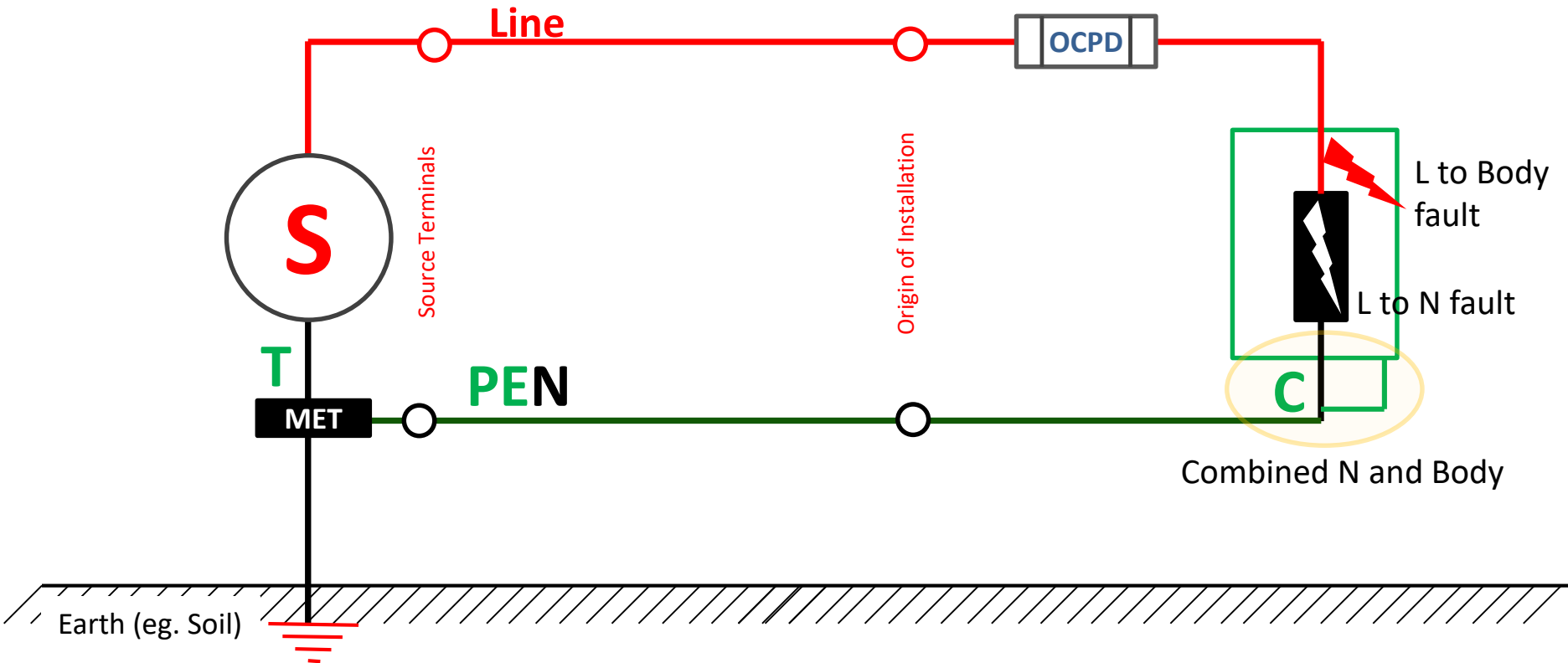
Wrong - Separate Connection to a low resistance earth electrode in soil

Right - Separate and Low resistance connection to an earthing system

LV System Earthing: TN-C, TN-S, TN-C-S, TT, IT

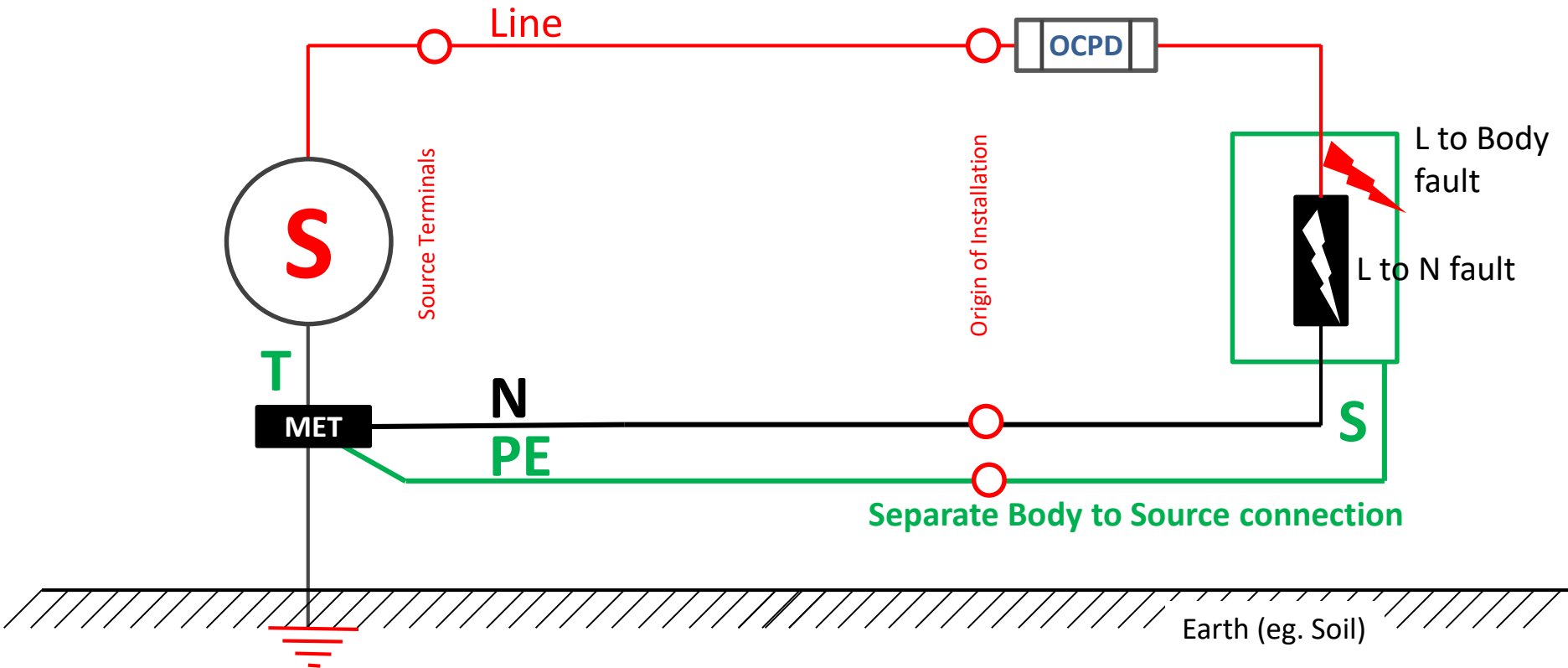


- Safety – Automatic disconnection of Supply during a fault
- How – Loop impedance shall be adequately lesser so that enough fault current flows within a short time so that protective device operate faster



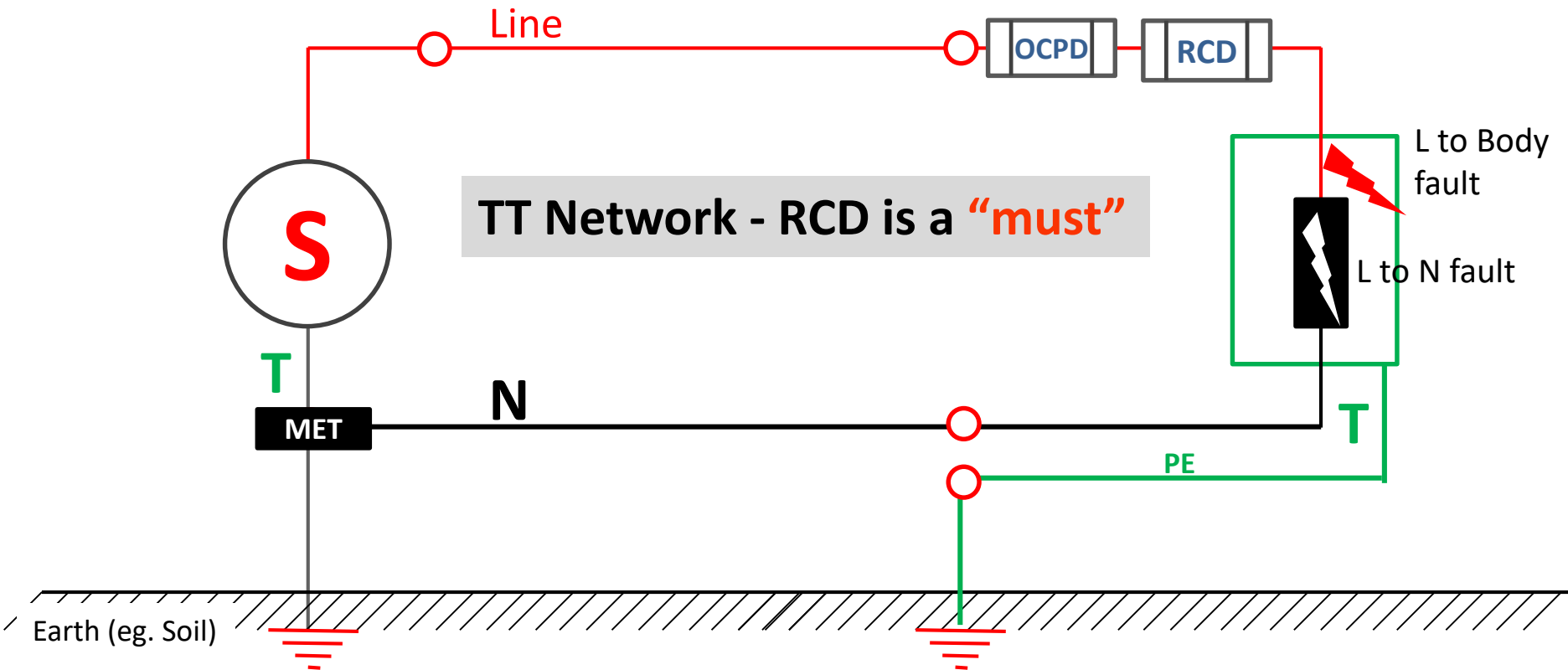
TN-C Network – OCPD is the primary protector

**Earth Fault was solved by connecting the body to Neutral wire.
Circuit Breaker protects against for L to N and Earth fault**



TN-S Network – OCPD is the primary protector

**Separate Body to Earthed Neutral of source solved the problem of “Neutral breakdown”
Fuse / OCPD continued to protect the circuit against L to N and Earth Faults**



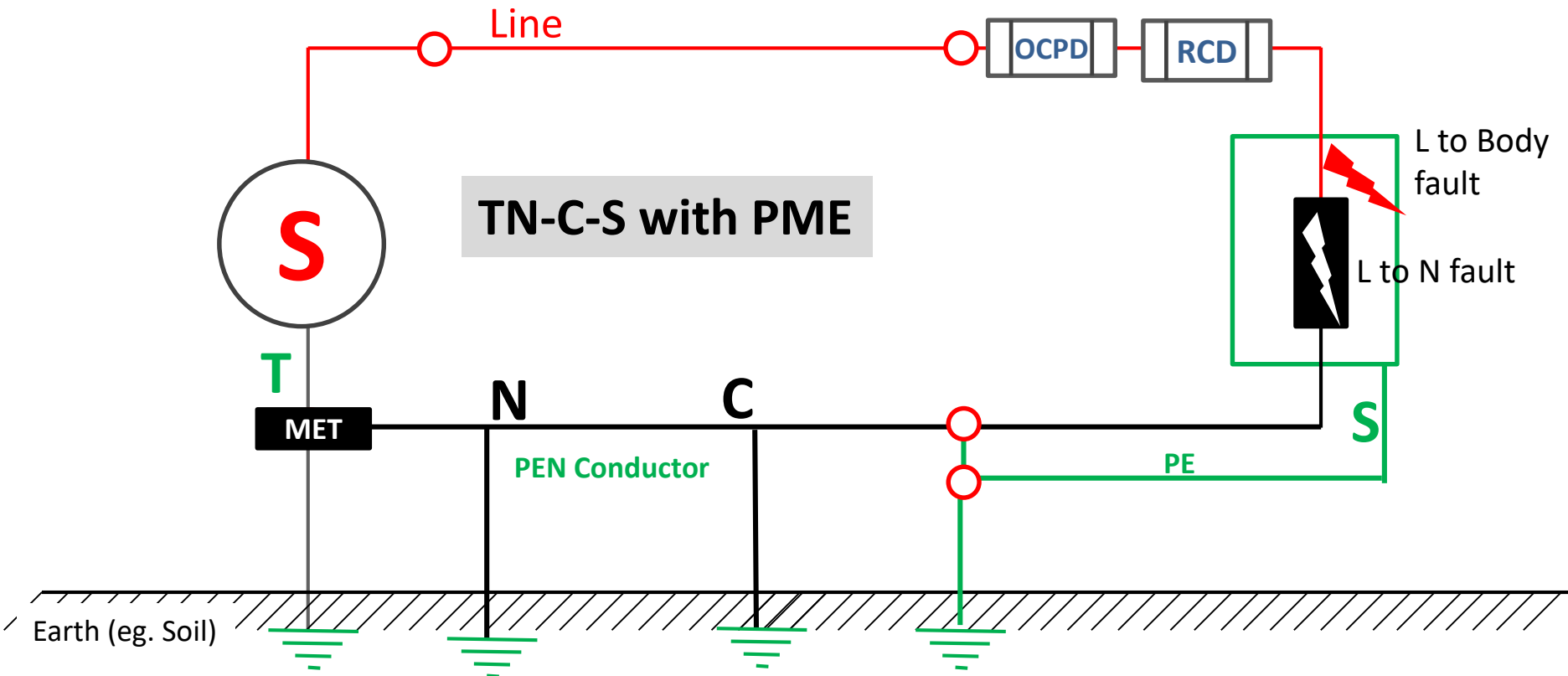
During 1970's RCD (earlier ELCB) has been invented.

RCD (RCCB/ELCB) measures the current in Line and Neutral.

If there is an unbalance (due to leakage) the device trips and opens the circuit

Instead of running 3rd wire (PE) from load to source, PE is connected to the local earth.

During earth fault, **Leakage current flows to local earth**. RCD trips.

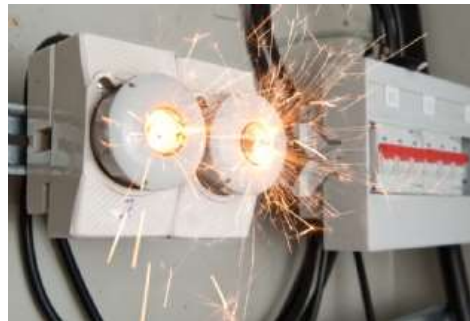
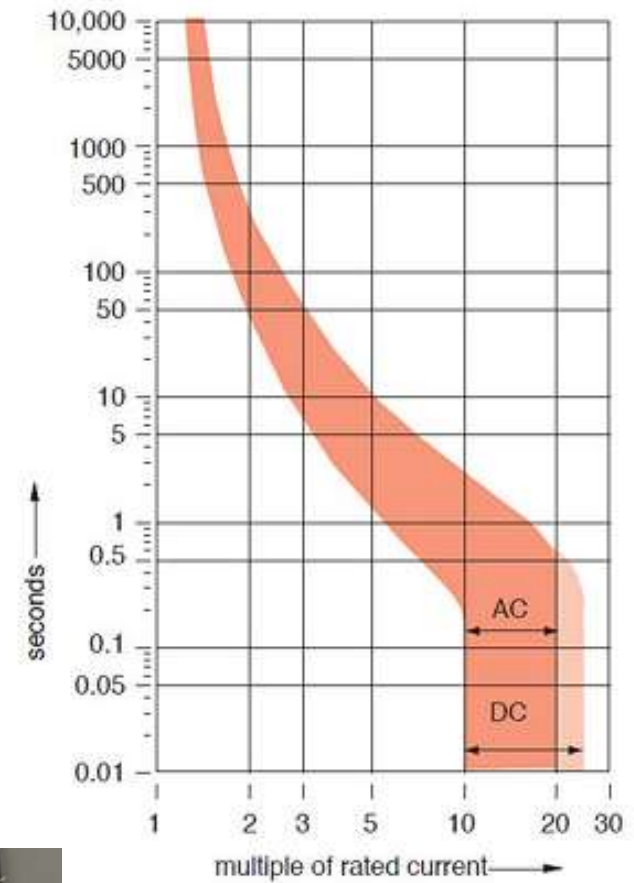
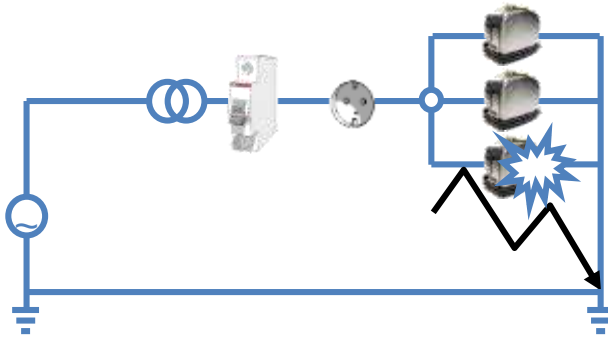


OCPD is the primary protector. RCD is an additional Protector. Earthing at the origin of installation is in the scope of power supplier. RCD required for load more than 2 KW (1 KW in future)

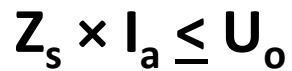
Functionality of a Short-Circuit Protective Device

Short-circuit

Occurs when any lines come in contact with each other or the earth

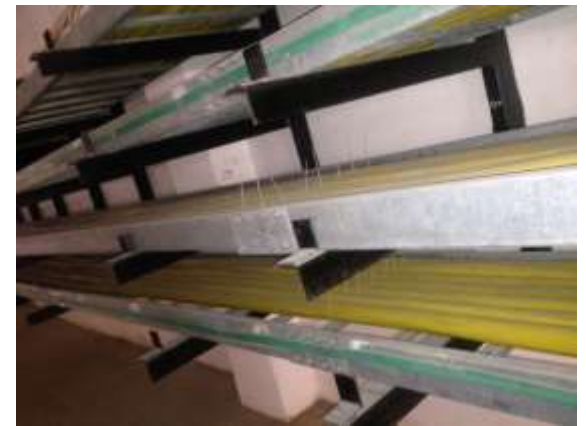


CAPE
ELECTRIC

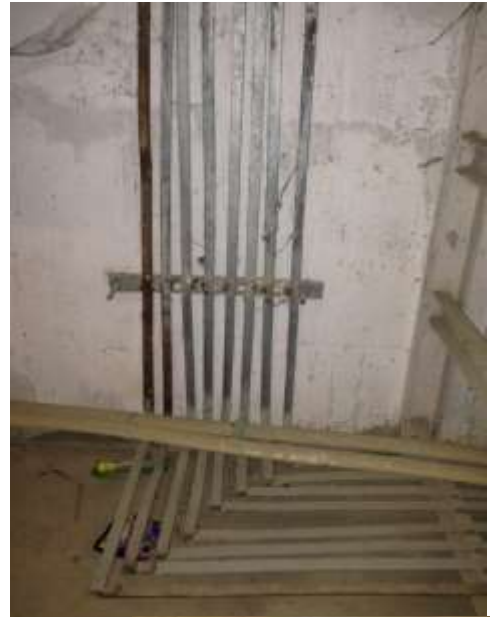

$$Z_s \leq U_o / I_a$$

U_0 = conventional voltage limits.

PE conductor immediately after installation



PE conductor after 6 months of installation

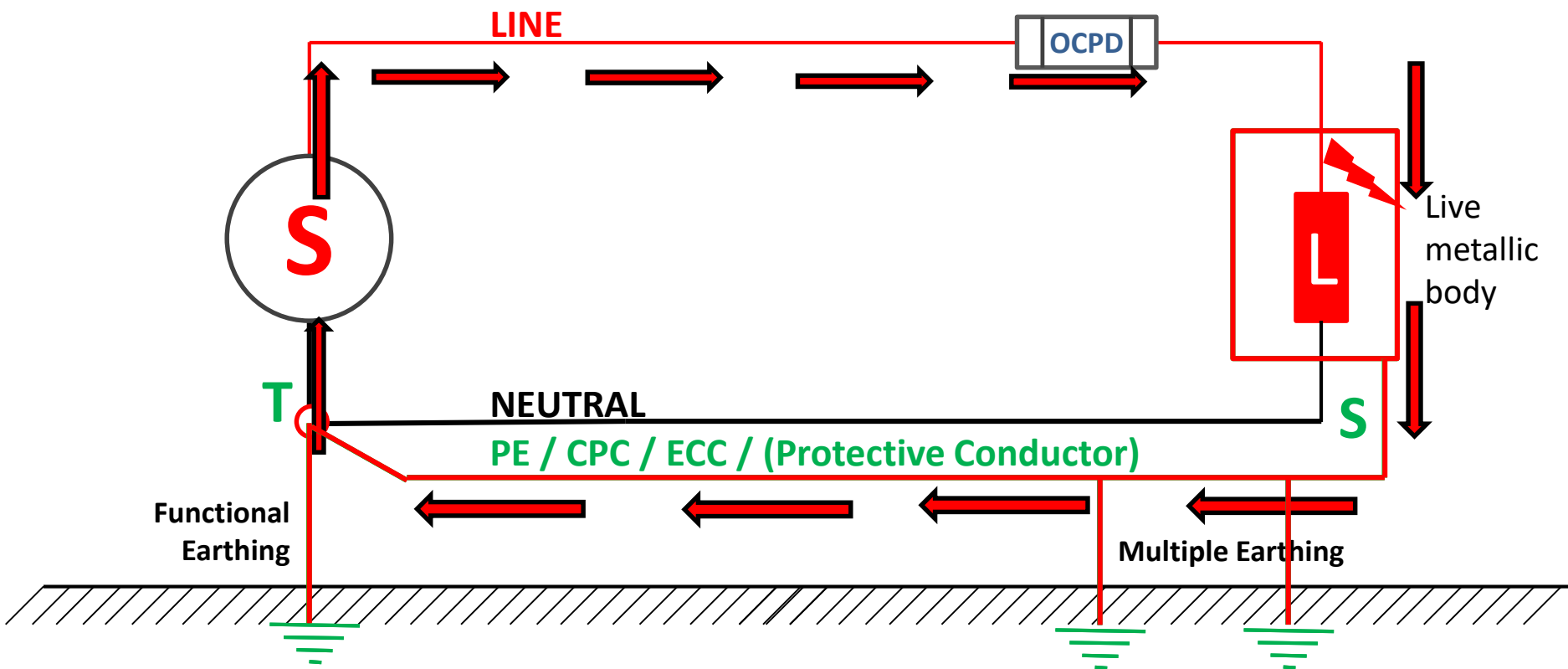


PE conductor connected with NUT & BOLT



Contact Resistance – About $0.050 \Omega = 50 \text{ m} \Omega$

Case – 1 TN-S System – Loop impedance (eg earth fault)



Operating Voltage = 230

OCPD = 250 amps MCCB.

$Z_s = 1.0 \Omega$

$I_{sc} = 230$ amps.

Device will never trip.

Danger

$Z_s = 0.5 \Omega$

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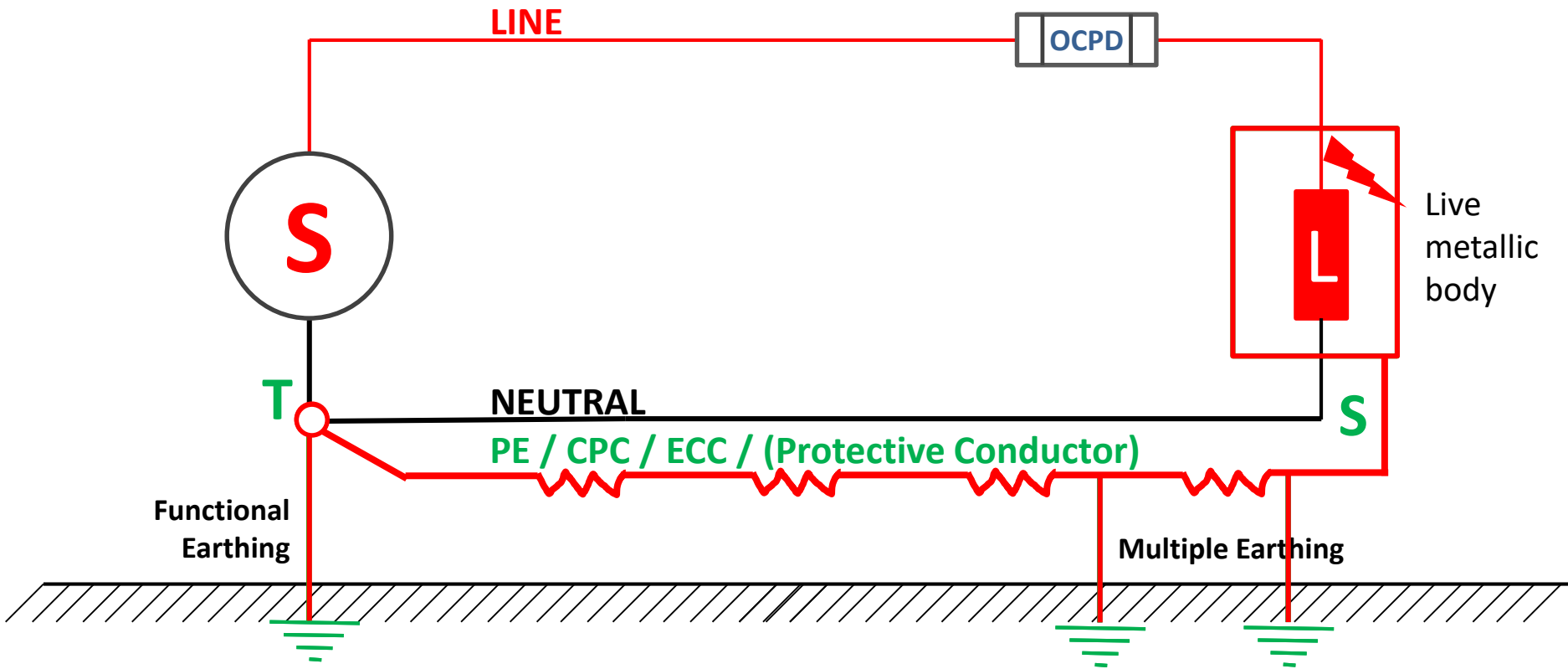
Device will trip quickly.

Safe

Low Loop impedance = More Fault current = Fast operation of Protective Device

Resistance of PE conductor plays major role in TN network in an earth fault

Case – 1 TN-S System – Loop impedance (eg earth fault)



- Bolted joints increase fault loop impedance, resulting in non operation of protective device causing accident
- Welded joints are not strong enough to handle fault current resulting in breakage and accident

Regulation in UK: Loop Impedance

Table 41.3
411.3.2.2
411.3.2.3

▼ **Table 7.3**

Data from Table 41.3 of BS 7671 for maximum earth fault loop impedance (Z_s) for circuit-breakers with U_0 of 230 V, for instantaneous operation giving compliance with the 0.4 s disconnection time of Regulation 411.3.2.2 and the 5 s disconnection time of Regulation 411.3.2.31, together with 80 per cent values² for comparison with measured values taken at an ambient temperature of 20 °C

temperature of 20 °C

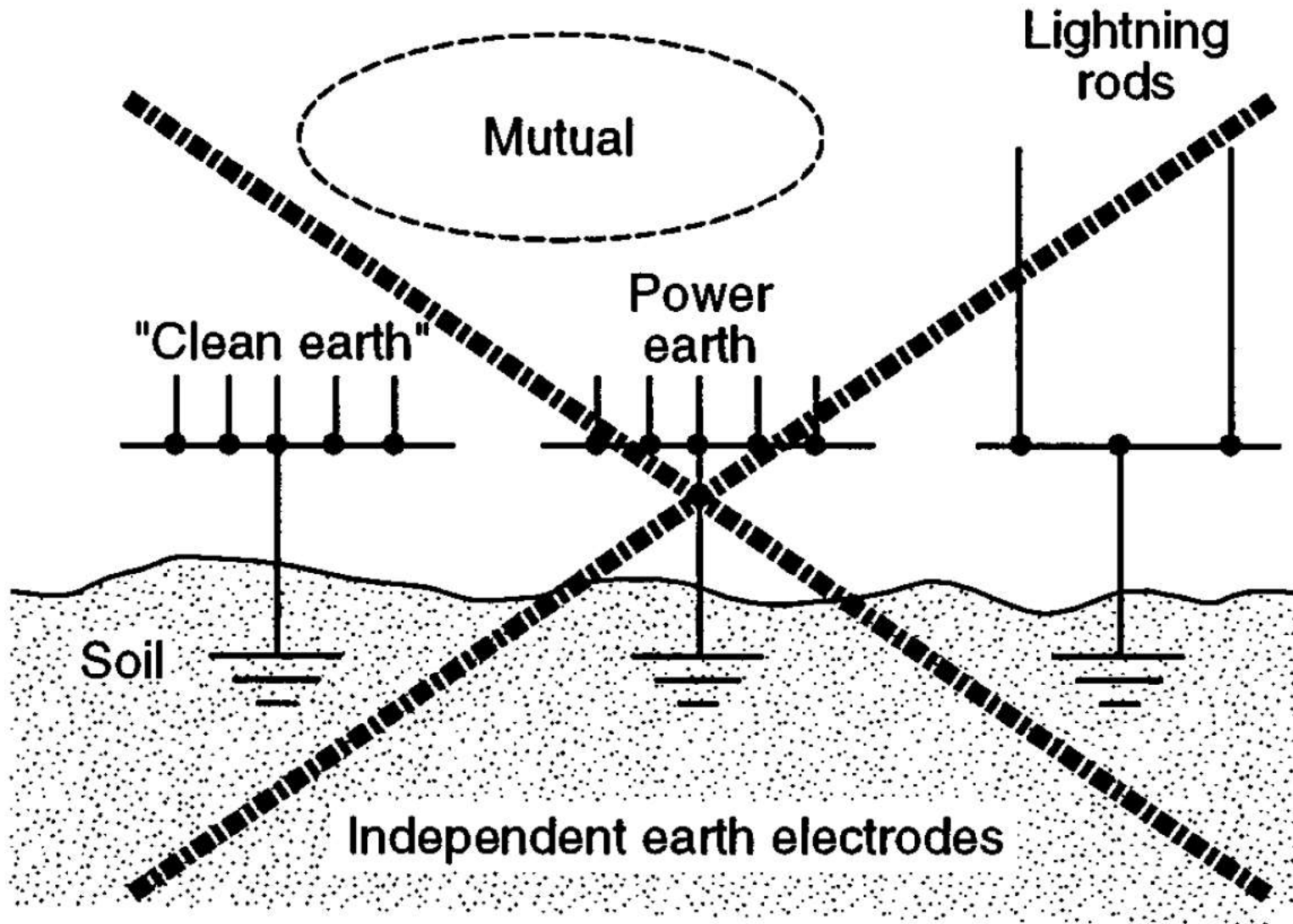
(a) Type B circuit-breakers to BS EN 60898 and the overcurrent characteristics of RCBOs to BS EN 61009														
Rating (A)	3	6	10	16	20	25	32	40	50	63	80	100	125	I_n
$Z_s (\Omega)$	14.57	7.28	4.37	2.73	2.19	1.75	1.37	1.09	0.87	0.69	0.55	0.44	0.35	$230 \times 0.95/(5 I_n)$
$Z_s (\Omega)$ measured ²	11.66	5.82	3.50	2.18	1.75	1.40	1.10	0.87	0.70	0.55	0.44	0.35	0.28	
(b) Type C circuit-breakers to BS EN 60898 and the overcurrent characteristics of RCBOs to BS EN 61009														
Rating (A)	6	10	16	20	25	32	40	50	63	80	100	125	I_n	
$Z_s (\Omega)$	3.64	2.19	1.37	1.09	0.87	0.68	0.55	0.44	0.35	0.27	0.22	0.17	$230 \times 0.95/(10 I_n)$	
$Z_s (\Omega)$ measured ²	2.91	1.75	1.10	0.87	0.70	0.54	0.44	0.35	0.28	0.22	0.18	0.14		
(c) Type D circuit-breakers to BS EN 60898 and the overcurrent characteristics of RCBOs to BS EN 61009														
Rating (A)	6	10	16	20	25	32	40	50	63	80	100	125	I_n	
$Z_s (\Omega), 0.4 \text{ s}$	1.82	1.09	0.68	0.55	0.44	0.34	0.27	0.22	0.17	0.14	0.11	0.09	$230 \times 0.95/(20 I_n)$	
$Z_s (\Omega), 0.4 \text{ s}$ measured ²	1.46	0.87	0.54	0.44	0.35	0.27	0.22	0.18	0.14	0.11	0.09	0.07		
$Z_s (\Omega), 5 \text{ s}$	3.64	2.19	1.37	1.09	0.87	0.68	0.55	0.44	0.35	0.27	0.22	0.17	$230 \times 0.95/(10 I_n)$	
$Z_s (\Omega), 5 \text{ s}$ measured ²	2.91	1.75	1.10	0.87	0.70	0.54	0.44	0.35	0.28	0.22	0.18	0.14		

TT	Over head power distribution for residential / commercial from a common Transformer. Every installation shall have an RCD and an earth electrode at Origin of installation
TN-S	Industrial / commercial / IT Buildings with electronic systems and Transformer with in facility (transformer operated by the owner)
TN-C	Over Head Power Distribution up to Origin of an Installation.
TN-C-S	Over head power distribution for residential / commercial from a common Transformer. RCD and earth electrode at origin of installation are optional.
IT	Hospitals / IT installation for a building or part of a building. Not suitable for different buildings with same supply

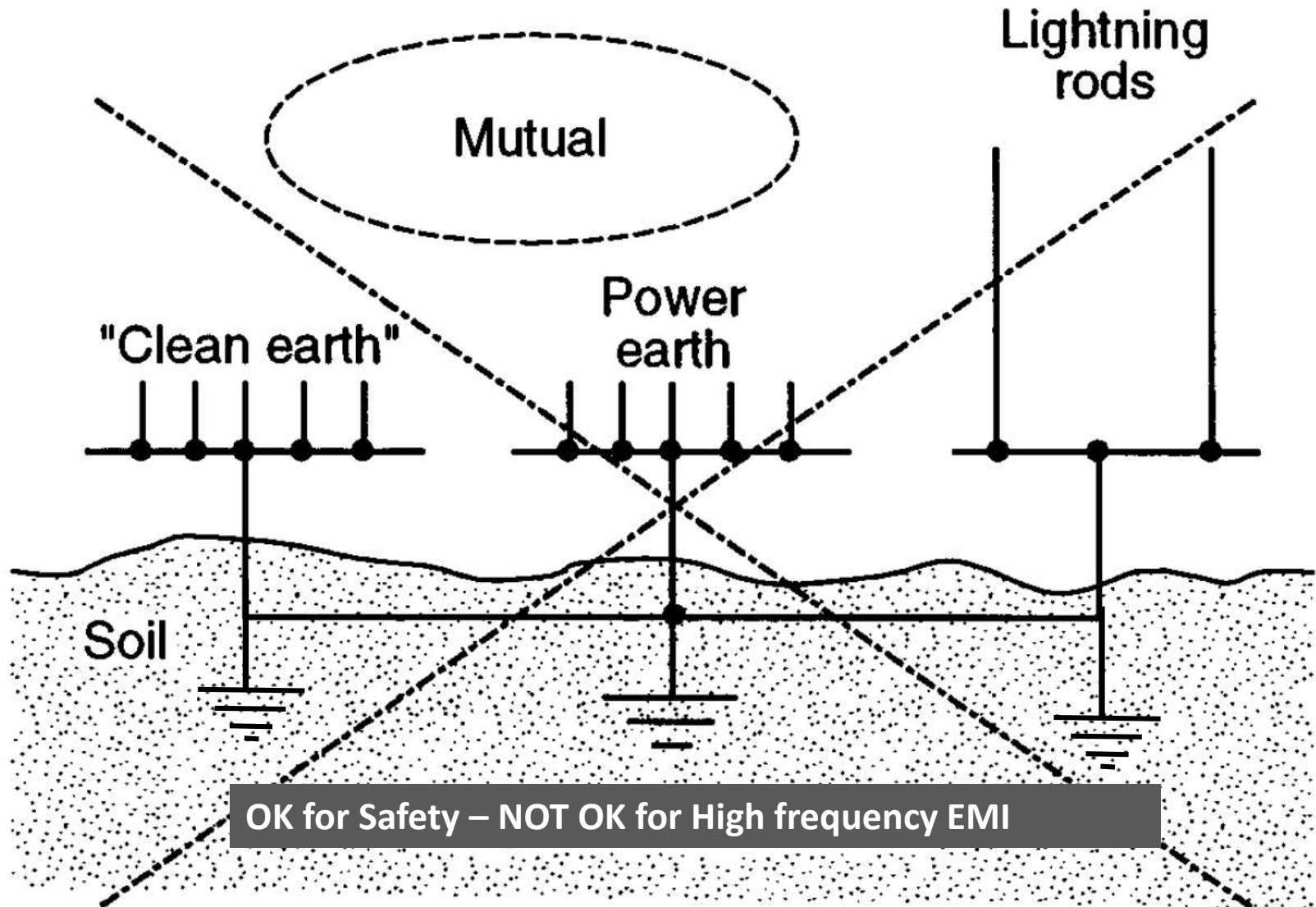
One system One earthing

Earthed Equipotentialisation

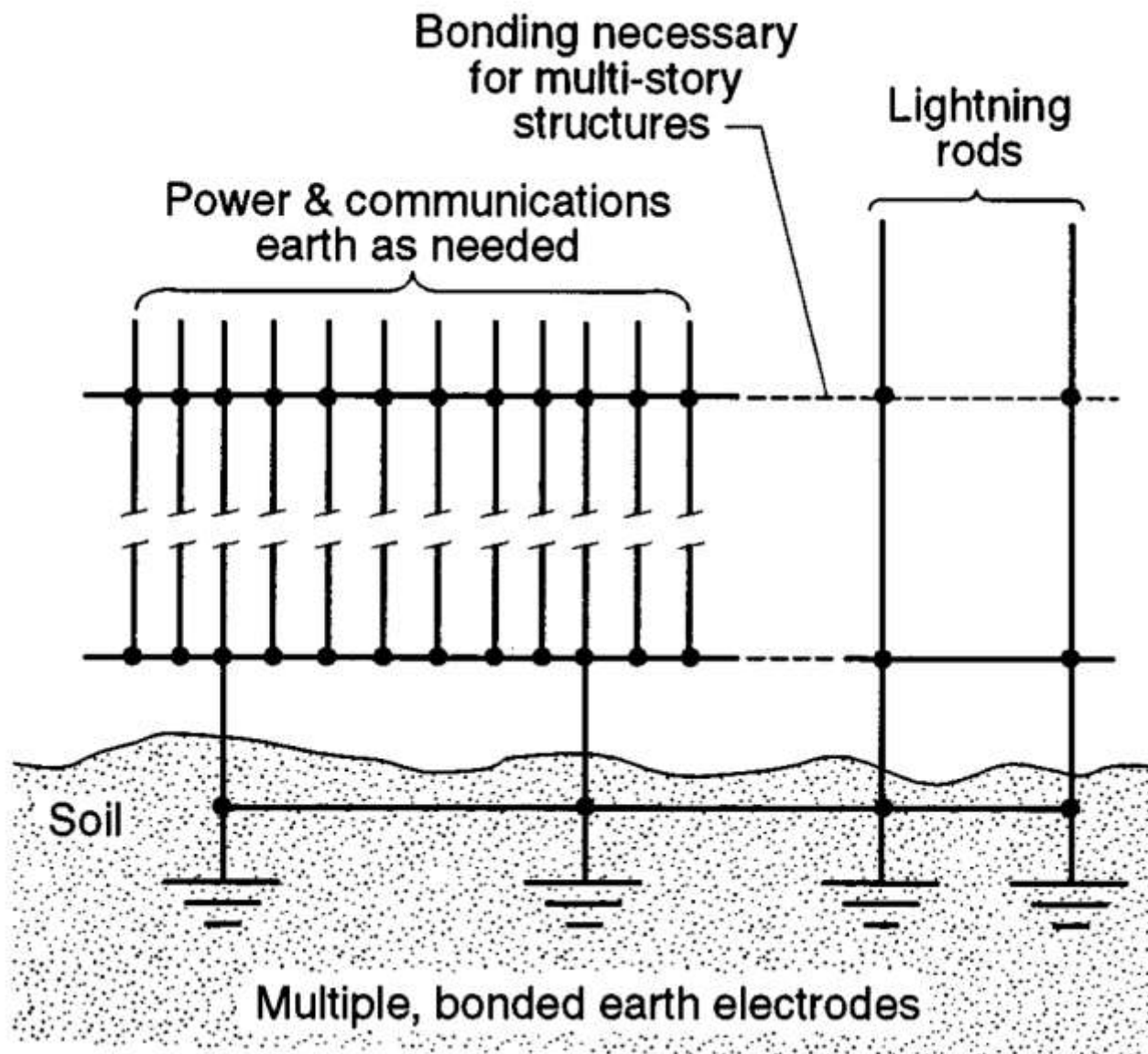
Mis-concept of “dedicated”, “independent”, or “isolated” earth electrodes



The concept of a single earth electrode



Recommended configuration earth electrodes and earthing network for Electronic Installations

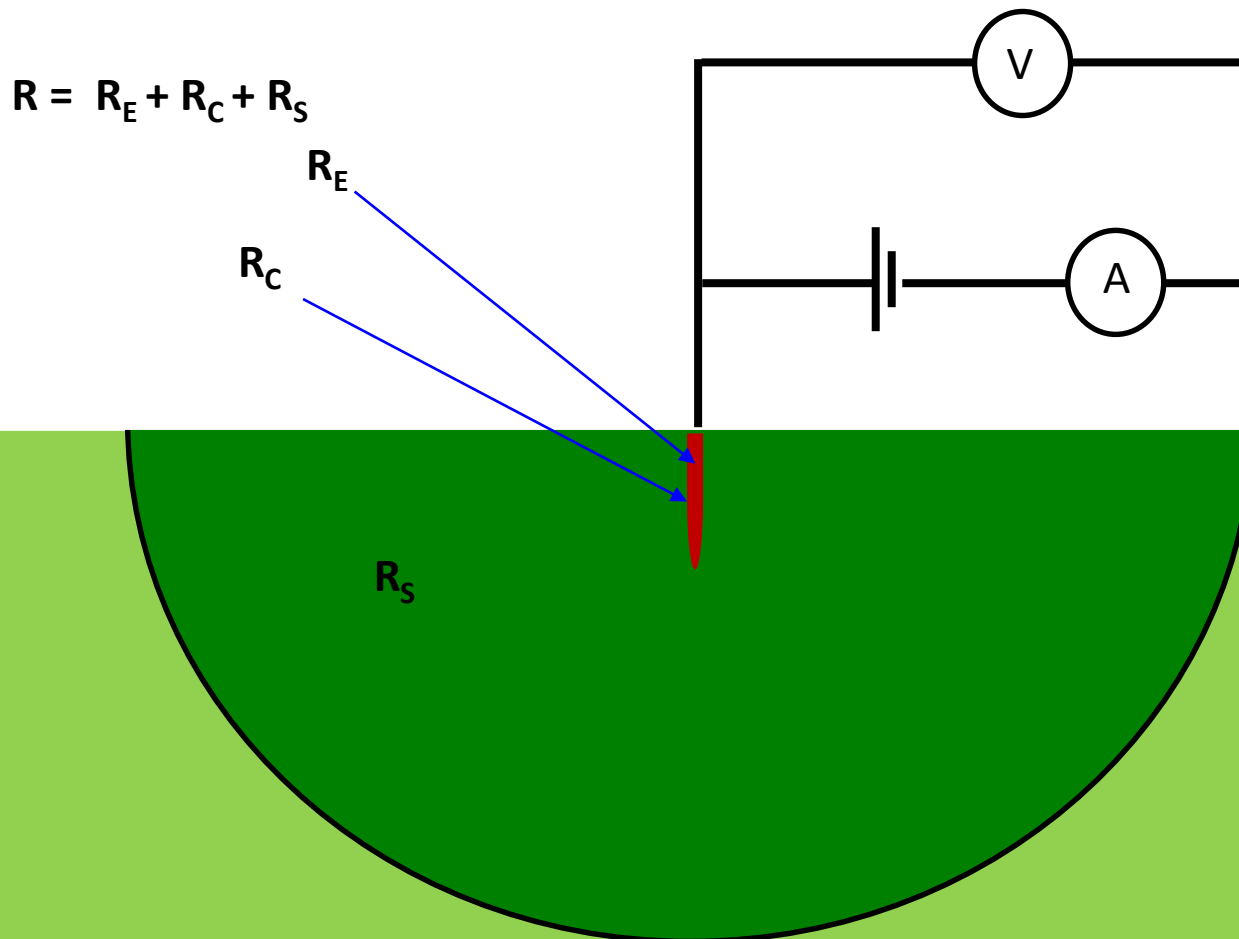


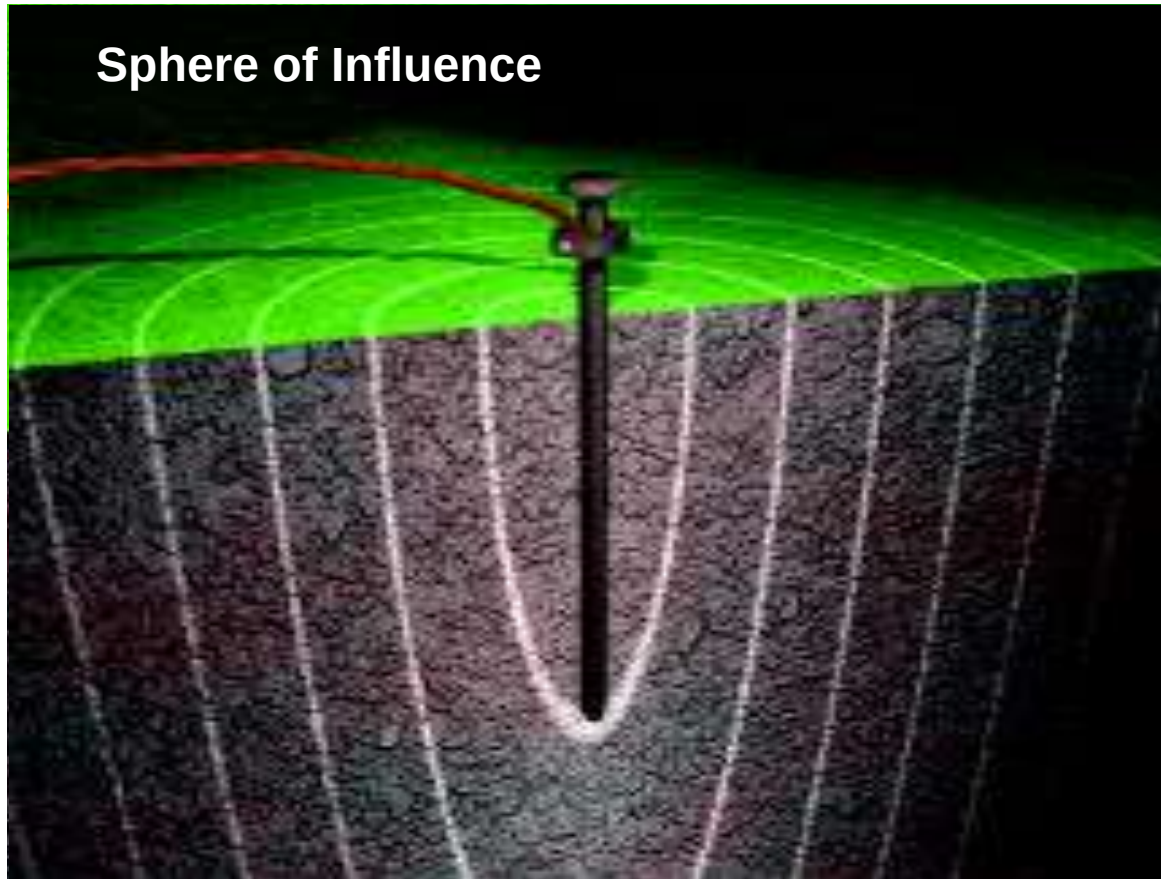
What is earth resistance of an electrode...?

What we get in this method is the Earth RESISTANCE under d.c. or low frequency conditions

Lightning is a HF phenomenon where RESISTANCE play a very insignificant role.

What you should measure is the **EARTH IMPEDANCE**





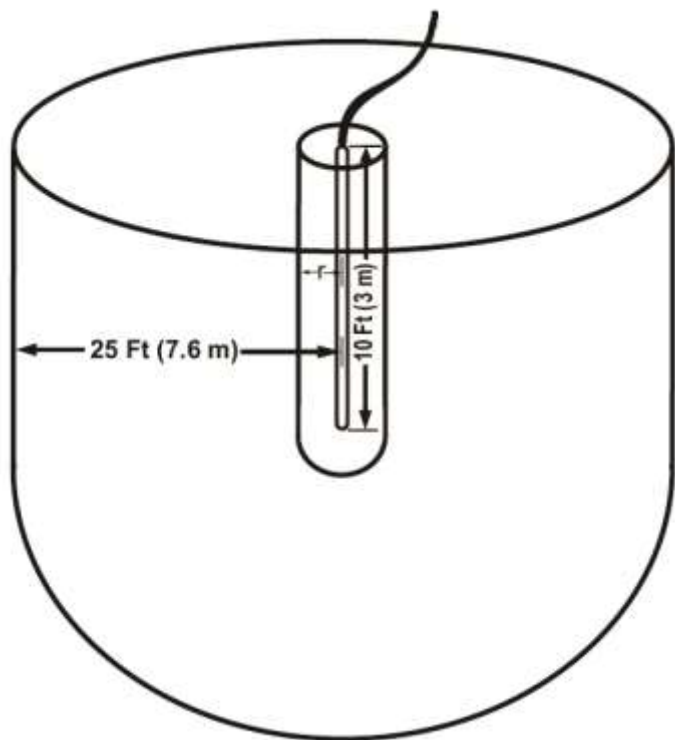
1. Earth Conductor
2. Connector
3. Electrode

This is only a part of Earthing system !!!!!!!!!!!!!

The resistance of a earthing system depends on

1. The resistance of the Electrode
2. The contact resistance of the electrode with soil
3. The resistance of soil surrounding Electrode

Electrode Resistance Development: IEEE 142



* These figures show that for the most practical reasons the majority of the resistance to remote earth occurs within 25 ft of the electrode, i.e., at 1000 ft the resistance is only 17% higher than that of 25 ft.

Distance from Electrode Surface (r)		Approximate Percentage of Total Resistance
mm	m	
30.48	0.03	25
60.96	0.06	38
91.44	0.09	46
152.4	0.15	52
304.8	0.3	68
1524	1.5	86
3048	3.0	94
4572	4.6	97
6096	6.1	99
7620	7.6	100
(30480)*	30.5	(104)
(304800)*	305.0	(117)

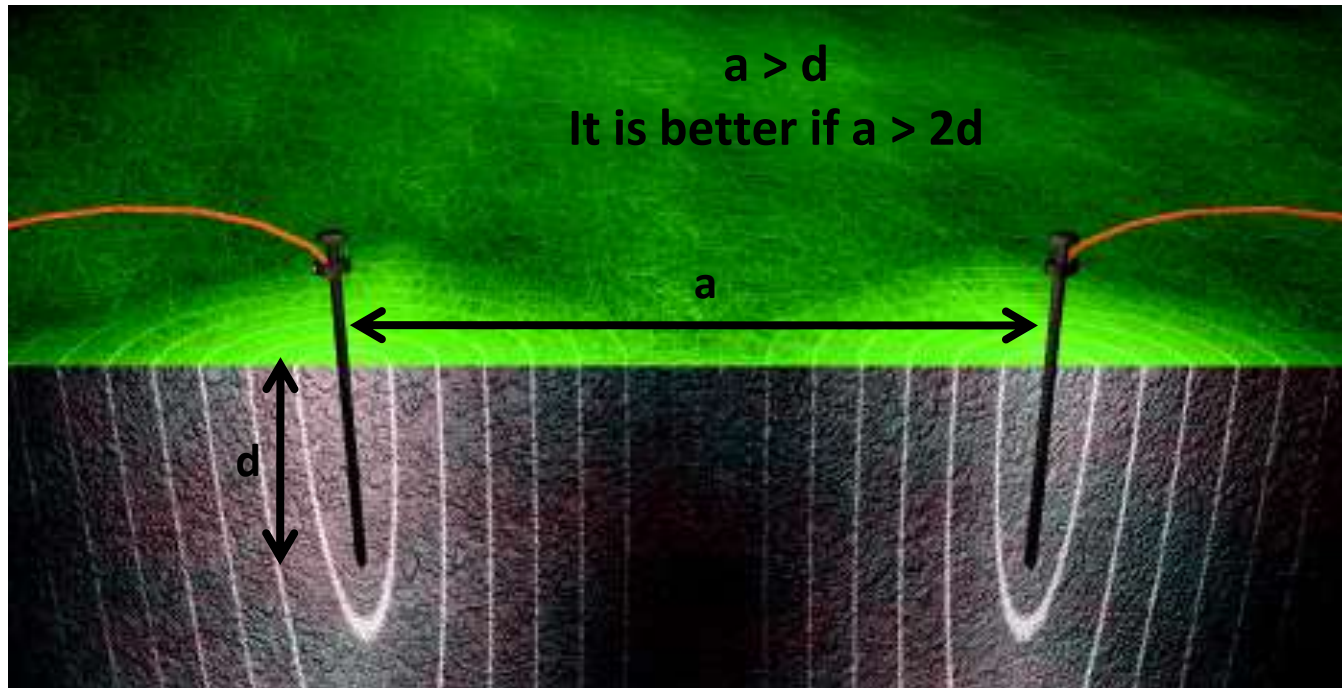
Table: Electrode Resistance at a Radius r in m from a 3 m Long, 16 mm Dia Rod
Total Resistance at $r = 7.6 \text{ m} = 100\%$

Multiple Earth electrodes.

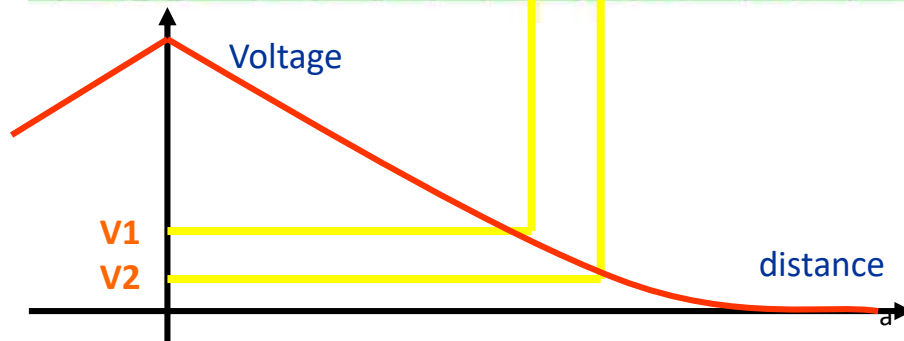
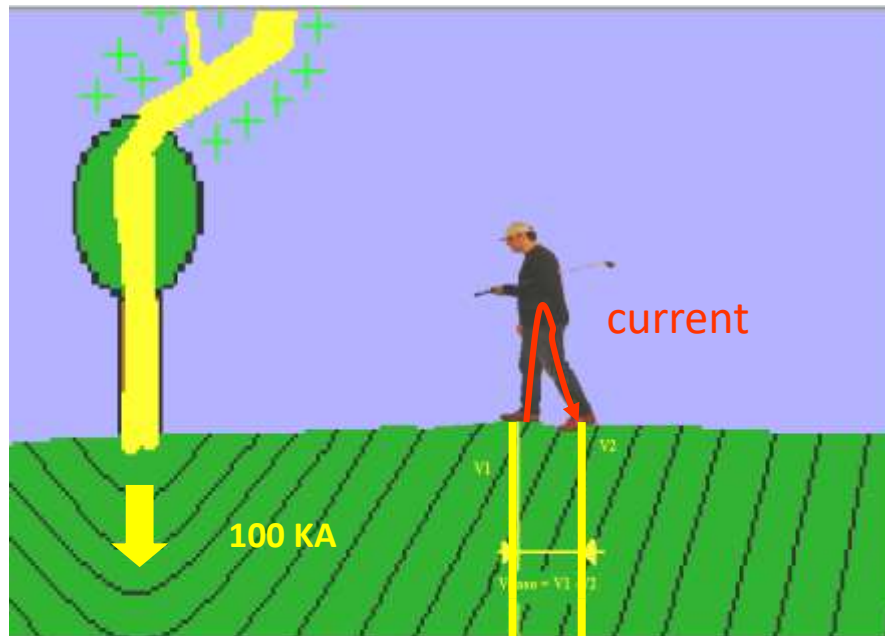
In this system more than one electrode is driven into the ground and connected in parallel to lower the resistance.

Each earth electrode has it's own sphere of influence and for additional electrodes to be effective the spacing of additional rods needs to be at least equal to the depth of the driven rod.

Without proper spacing of earth electrodes the spheres of influence will intersect and the lowering of the resistance will be minimal and of little value.



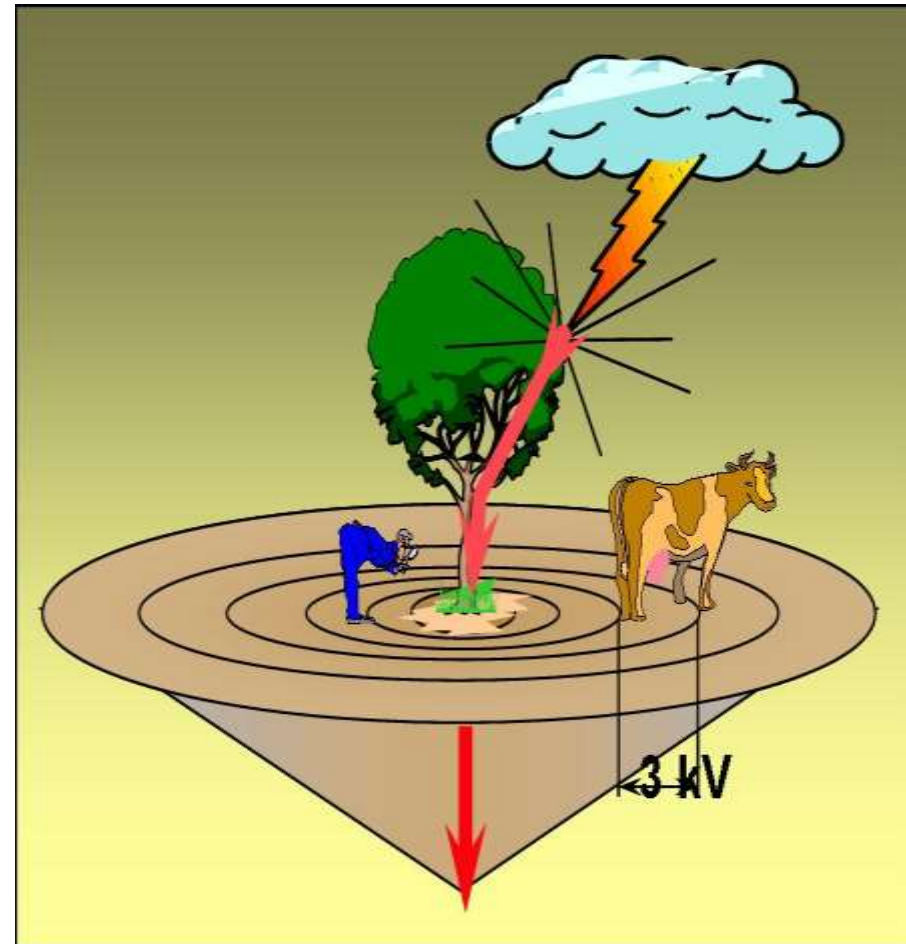
STEP VOLTAGE $V_s = V_1 - V_2$



Indirect Impact

People: High voltages on the ground

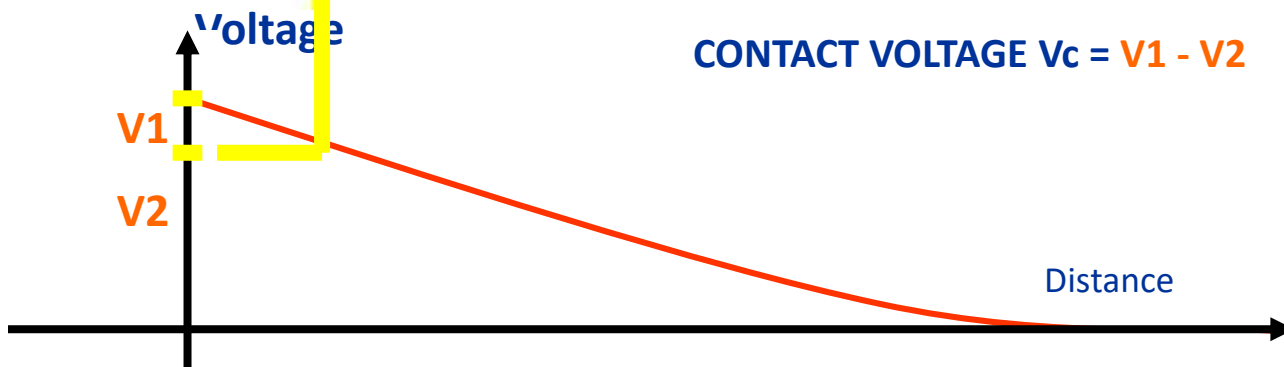
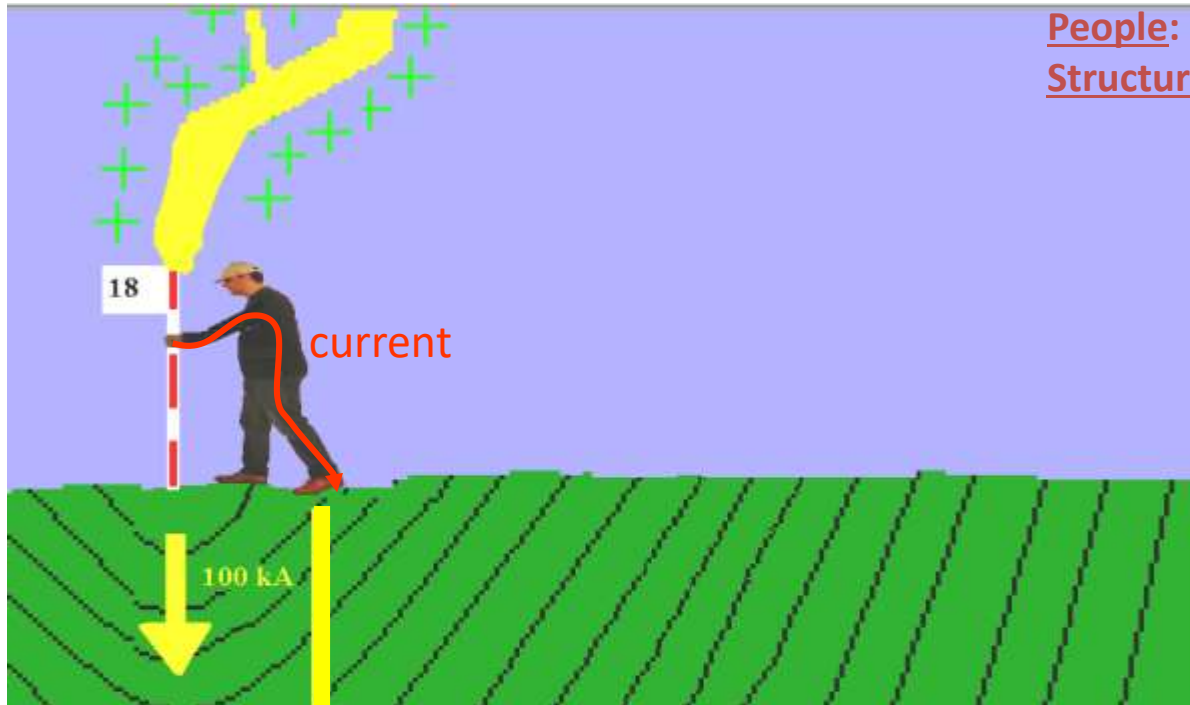
Structures: inductive and capacitive coupling



Indirect Impact

People: High voltages on the ground

Structures: inductive and capacitive joining



One system One earthing

- **Earthing for safety**
- **Double earthing for redundancy and reduction of fault loop resistance**
- **IS3043**

IS3043: Code practice of Earthing

(Reaffirmed 2006)

IS : 3043 - 1987
(Reaffirmed 2006)

Indian Standard

CODE OF PRACTICE FOR EARTHING

(First Revision)

Fourth Reprint JUNE 2007
(Including Amendment No. 1)

UDC 621.316.99 : 006.76

IS 3043 – took support from
CP1013 (British Standard)
IEEE 142
IEC 364

INTERNATIONAL
STANDARD

**IEC
60364-1**

9th edition
2005-11

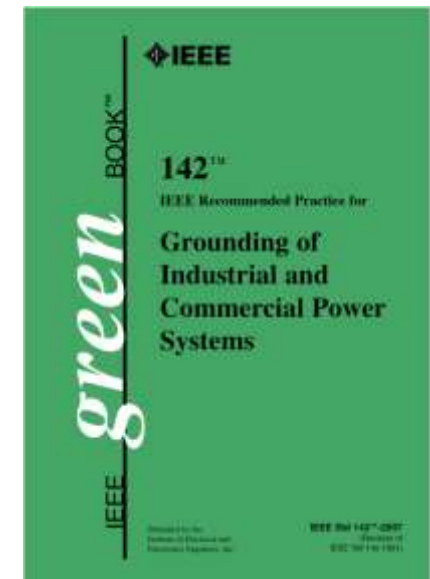
Low-voltage electrical installations –

Part 1:
Fundamental principles, assessment
of general characteristics, definitions

This English-language version is derived from the original
bilingual publication by deleting all French-language
pages. Missing page numbers correspond to the French-
language pages.

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BRITISH STANDARD CODE OF PRACTICE
CP 1013 : 1955
UDC 621.316.99

EARTHING

THE COUNCIL FOR CODES OF PRACTICE
BRITISH STANDARDS INSTITUTION

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9, BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

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September 1955

IS : 3043 - 1987

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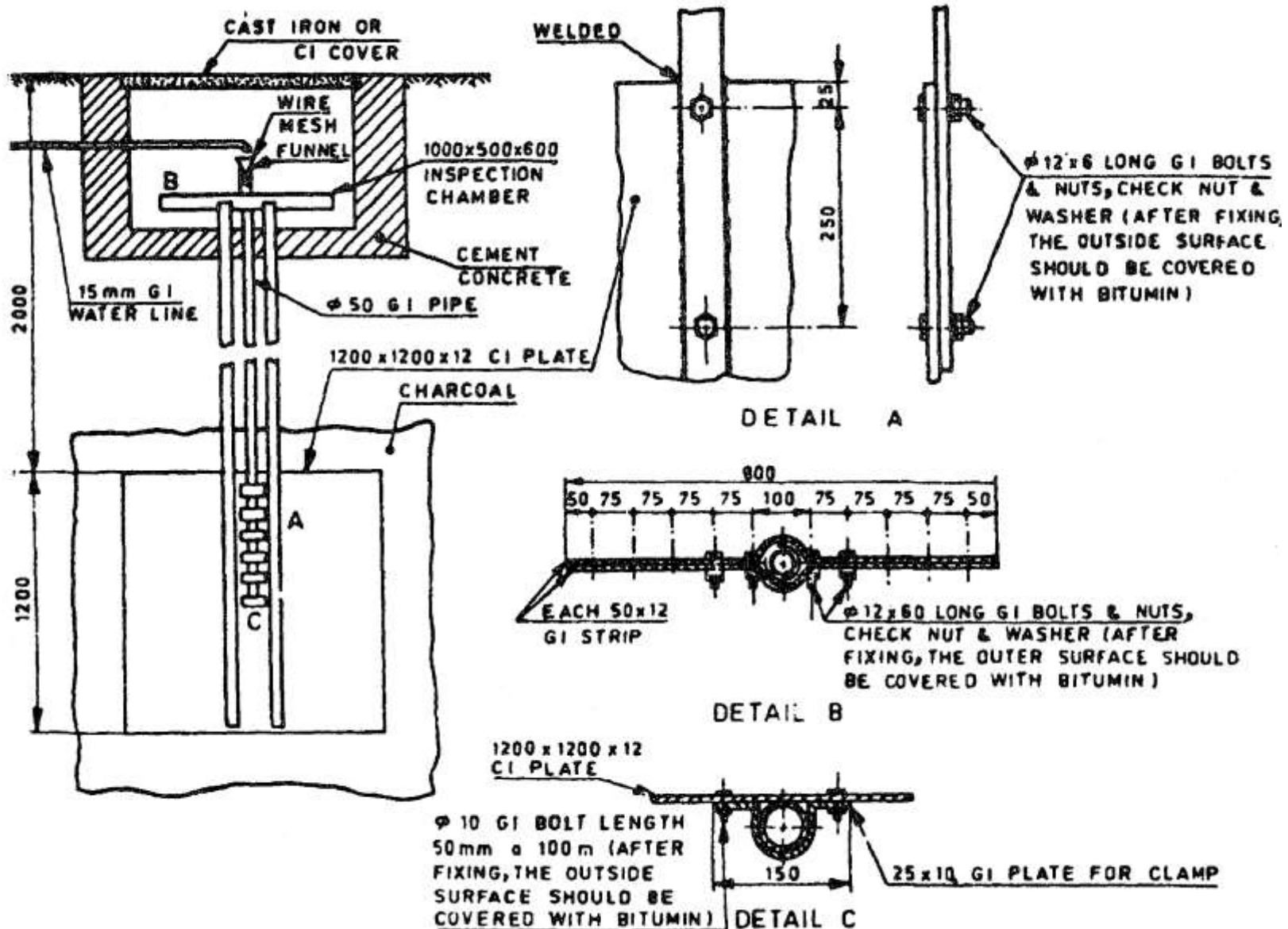
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Earth electrodes – IS3043 (Salt and Charcoal)



Protection against shock

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SECTION 3 EARTH FAULT PROTECTION IN CONSUMER'S PREMISES

18. EARTH FAULT PROTECTION IN INSTALLATIONS

18.0 Basic Philosophy of Earth Fault Protection

18.0.1 The rules given in this Section are applicable to installation below 1 000 V ac.

18.0.2 Amongst other things, protection against shock in case of a fault (protection against indirect contact) is provided by automatic disconnection of supply. This protective measure necessitates co-ordination of the types of system earthing and the characteristics of the protective devices. This Section discusses the basic criteria for achieving this protection.

18.0.3 Protection against electric shock both in normal service (protection against direct contact) and in case of fault (protection against indirect contact) can be achieved by several measures. Details of achieving protection through the choice of an appropriate protective measure is the subject of IS : 732*. One of such measures is protection by automatic disconnection of supply. Automatic disconnection is intended to prevent a touch voltage persisting for such time that a danger could arise. This method necessitates co-ordination of (a) the type of system earthing, and (b) characteristics of protective devices. Description of the types of system earthing permitted and the requirements for earthing arrangements and protective conductors *vis-a-vis* protection against shock is the subject of this code.

18.0.4 Protective measure by automatic disconnection of supply following an insulation fault relies on the association of two conditions given below:

- The existence of a conducting path (fault loop) to provide for circulation of fault current (this depends on type of system earthing); and
- The disconnection of this current by an appropriate device in a given time.

The determination of this time depends on various parameters, such as probability of fault, probability of a person touching the equipment during the fault and the touch voltage to which a person might thereby be subjected.

Limits of touch voltage are based on studies on the effects of current on human body (see IS : 8437-1977†).

18.0.5 The study of the electrical impedance of the human body as a function of touch voltage and magnitude of current flow in the body as a

function of its duration likely to produce a given effect are two components which help in establishing a relationship between prospective touch voltage and its duration which will not result in harmful physiological effects for any person.

Table 8 shows the values of disconnecting times *t* for given touch voltages for two most common conditions.

TABLE 8 DISCONNECTING TIMES FOR DIFFERENT TOUCH VOLTAGES

PROSPEC- TIVE TOUCH VOLTAGE <i>U_c</i>	CONDITION 1*			CONDITION 2†		
	<i>Z₁</i>	<i>I</i>	<i>t</i>	<i>Z₂</i>	<i>I</i>	<i>t</i>
(V)	(Ω)	(mA)	(s)	(Ω)	(mA)	(s)
25	—	—	—	075	23	5
50	1 725	29	5	925	54	0.47
75	1 625	46	0.60	825	91	0.30
90	1 600	56	0.45	780	115	0.25
110	1 535	72	0.36	730	151	0.18
150	1 475	102	0.27	660	227	0.10
220	1 375	160	0.17	575	383	0.035
280	1 370	204	0.12	570	491	0.020
350	1 365	256	0.08	565	620	—
500	1 360	368	0.04	560	893	—

*Dry or moist locations, dry skin and significant floor resistance.22

†Wet locations, wet skin and low floor resistance.

18.0.6 It is necessary, therefore, to apply these results emanating out of IS : 8437-1977* to the various earthing systems. The disconnecting times specified for different circuits in this code follows basically the summary in Table 8, in addition taking into account the likelihood of faults and likelihood of contact.

18.0.7 TN Systems— All exposed conductive parts shall be connected to the earthed point of the lower system by protective conductors. The protective conductors shall be earthed near each power transformer or generator of the installation. If other effective earth connections exist, it is recommended that the protective conductors also be connected to such points, wherever possible. Earthing at additional points as evenly as possible is desirable. It is also recommended that protective conductors should be earthed where they enter any buildings or premises.

The characteristics of the protective devices and the cross-sectional area of conductors shall be so chosen that if a fault of negligible impedance occurs any where between a phase conductor and

*Code of practice for wiring installations.

†Guide on effects of currents passing through the human body.

*Guide on effects of currents passing through the human body.

TABLE 8 DISCONNECTING TIMES FOR DIFFERENT TOUCH VOLTAGES

PROSPEC- TIVE TOUCH VOLTAGE <i>U_c</i>	CONDITION 1*			CONDITION 2†		
	<i>Z₁</i>	<i>I</i>	<i>t</i>	<i>Z₂</i>	<i>I</i>	<i>t</i>
(V)	(Ω)	(mA)	(s)	(Ω)	(mA)	(s)
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280	1 370	204	0.12	570	491	0.020
350	1 365	256	0.08	565	620	—
500	1 360	368	0.04	560	893	—

*Dry or moist locations, dry skin and significant floor resistance.22

†Wet locations, wet skin and low floor resistance.

220 volt fault
0.17 sec for dry condition
0.032 sec for wet condition

Method of Protection against shock

IS : 3043 - 1987

a residual current device or a voltage operated earth leakage circuit breaker with the former being preferred.

18.2 Earthing of Installations

18.2.1 Protection Against Indirect Contact (Against Electric Shock in Case of a Fault) — Protection against indirect contact is achieved by the adoption of one of the following protective measures:

- Safety extra low voltage;
- The use of Class II equipment or by equivalent insulation;
- A non-conducting location;
- Earth free local equipotential bonding;
- Electrical separation; and
- Earthed equipotential bonding and automatic disconnection of the supply.

NOTE 1 — The primary concern of this Code is (d) and (f) while other methods of protection against indirect contact are covered in other relevant Indian Standard Codes of Practice.

NOTE 2 — Item (a) requires that the nominal voltage of the circuit concerned does not exceed extra low voltage that the source has a high degree of isolation from higher voltage circuits (for example, a Class II safety isolation transformer) and that live parts also have a similar degree of isolation or separation from those circuits. The most important requirement, however, is that live parts and exposed conductive parts of a safety extra low voltage circuit should not be connected to earth, protective conductors or exposed conductive parts of another circuit. Where these general requirements are not met but the nominal voltage still does not exceed extra low voltage, the circuit is described as a functional extra low voltage circuit and one part of it may be connected to earth.

NOTE 3 — Item (b) is generally applicable and covers the selection and use of equipment complying with either insulation enclosed Class II equipment ('safety-insulated') or metal-cased Class II equipment. In some cases, such as factory built assemblies of switchgear and controlgear, the equivalent term used is 'total insulation'. Item (b) can also be achieved by the application of suitable supplementary or reinforced insulation to equipment on site.

Earthing of the equipment is not required; in fact, by definition there will be no facility for earthing provided in Class II equipment.

NOTE 4 — Items (c), (d) and (e) are of limited interest as they can be applied only in special situations and used under effective supervision. They all include a high degree of isolation from earth.

NOTE 5 — In this Section, detailed consideration is limited to earthed equipotential bonding and automatic disconnection of the supply.

18.2.2 Earthed Equipotential Bonding and Automatic Disconnection of the Supply — The two aims of this protective measure are to:

- ensure that when an earth fault occurs, the voltages appearing between exposed conductive parts and extraneous conductive parts in the location served by the installation concerned are minimized; and
- ensure rapid disconnection of the circuit in which that earth fault occurs.

In order to meet (a), a zone is created by first connecting all extraneous conductive parts by means of equipotential bonding conductors to the main earthing terminal or earth electrode(s) of the installation.

The zone is completed by the connection of all exposed conductive parts of the circuits in the installation and of current-using equipment fed from those circuits to the main earthing terminal (or installation earth electrode) using circuit protective conductors.

When such a zone is earthed an equipotential zone, this does not mean that voltages cannot exist between conductive parts in that zone when an earth fault occurs. The voltages referred to earlier (see 18.1) will still exist between the exposed conductive parts of perfectly sound equipment and between such parts and extraneous conductive parts, but the application of bonding minimizes these voltages in each case.

An installation may consist of a number of zones; for instance, when an installation supplies a number of buildings, equipotential bonding is necessary in each building so that each constitutes a zone having a reference point to which the exposed conductive parts of the circuits and current-using equipment in that building are connected.

The second aim of this protective measure is met by limiting the upper value of the earth fault loop impedance of each circuit to a value determined by the type and current rating of the protective device concerned such that, on the occurrence of an earth fault (assumed to be of negligible impedance), disconnection will occur before the prospective touch voltage reaches a harmful value.

18.2.3 Extraneous Conductive Parts — The extraneous conductive parts that are required to be bonded to the main earthing terminal of the installation (or to the earth electrode of the installation) include:

- gas pipes;
- other service pipes and ducting;
- risers and pipes of fire protection equipment;
- exposed metallic parts of the building structure; and
- lightning conductors (see Section 8).

NOTE — Connections to pipes, ducting and exposed metallic parts of building structure should be considered most carefully. In some types of earthing systems, especially TN-C or TN-C-S systems, effectively connect extraneous conducting metalwork in the supply system neutral and could cause continuously circulating currents and standing voltages that might result in electrochemical corrosion or random spark hazards in potentially flammable atmospheres.

The second aim of this protective measure is met by limiting the upper value of the earth fault loop impedance of each circuit to a value determined by the type and current rating of the protective device concerned such that, on the occurrence of an earth fault (assumed to be of negligible impedance), disconnection will occur before the prospective touch voltage reaches a harmful value.

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18.2.4 Exposed Conductive Parts — Exposed conductive parts that are required to be connected by means of protective conductors to the main earthing terminal (or earth electrode) of the installation are as follows:

a) All metalwork associated with wiring system (other than current-carrying parts) including cable sheaths and armour, conduit, ducting, trunking, boxes and catenary wires;

b) The exposed metalwork of all Class I fixed and portable current-using equipment. Even where at the time of the erection of the installation this equipment is of Class II construction or its equivalent, because there is a possibility that in the life of the installation the equipment may be replaced by Class I equipment, all fixed wiring accessories should incorporate an earthing terminal that is connected to the main earthing terminal by means of the protective conductors of the circuits concerned.

c) The exposed metalwork of transformers used in the installation other than those that are an integral part of equipment. The secondary windings of transformers should also be earthed at one point of the winding, unless the transformer is a safety isolating transformer supplying a part of the installation where the protective measure 'electrical separation' is being used).

Exposed conductive parts that (because of their small dimensions or disposition) cannot be gripped or contacted by a major surface of the human body (that is, a human body surface not exceeding 50 mm × 50 mm) need not be earthed if the connection of those parts to a protective conductor cannot readily be made and reliably maintained. Typical examples of such parts are screws and nameplate, cable clips and lamp caps. Fixing screws for non-metallic accessories need not be earthed provided there is no appreciable risk of the screws coming into contact with live parts.

Other exposed conductive parts not required to be earthed are:

- 1) Overhead line insulator brackets and metal parts connected to them if such parts are not within arm's reach; and
- 2) Short lengths of metal conduit or other metal enclosures used to give mechanical protection to equipment of Class II or equivalent construction.

18. Protection against Excessive Temperature Rise and Mechanical Damage

18.3.1 General — The protective circuit of an installation includes the following (see Fig. 20):

- a) Circuit protective conductors;

- b) Equipotential bonding conductors; and
- c) Earthing conductors.

Under certain circumstances, there may also be local equipotential bonding conductors.

The determination of cross-sectional areas of all these conductors is the subject of Section 2 (*also see 18.4*) and here consideration is limited to the types of conductor that can be used with some indication of the precautions that should be taken during erection, particularly those concerned with mechanical and chemical deterioration and electro-dynamic effects.

18.3.2 Earthing conductors — Copper earthing conductors, in general, need not be protected against corrosion when they are buried in the ground if their cross-sectional area is equal to or greater than 25 mm². In case of buried steel conductors, appropriate corrosion factors based upon the summed up corrosion indexes corresponding to different parameters connected with the material for grounding, environmental conditions, nature of soil, etc (see Section 4) should be applied in determining the size of the earthing conductor, however, the minimum size should not be less than 50 mm². If the earthing conductor is of tape or strip, the thickness should be adequate to withstand mechanical damage and corrosion.

It should be remembered that plain uncoated copper is positive to plain uncoated buried steel and when interconnected by a current carrying conductor, these metals will form an electrochemical cell that can cause accelerated corrosion of steel. As a rough guide, a dc current of 1 A leaving a buried steel structure can remove nearly 9 kg of metal in one year.

Where such conductors are protected against corrosion but are not mechanically protected, the minimum cross-sectional area is 16 mm² if the conductor is of copper or coated steel (Table 4). The determination of the cross-sectional area where the earthing conductor is both mechanically protected and protected against corrosion is considered in a later section.

Aluminium or copper clad aluminium conductors should not be used for final underground connections to earth electrodes. Where a copper conductor is to be joined to aluminium, the copper should be tinned, unless an approved conductor is used.

The connection of the earthing conductor to the earth electrode or other means of earthing should be readily accessible and soundly made by the use of soldered joints or substantial clamps of non-ferrous material. Where the earthing conductor is to be connected to the metal armour and sheath of a cable, the armour should be bonded to the metal sheath and the principal connection between the cable and the earthing conductor

carrying terminal by means of the protective conductors of the circuits concerned.

- c) The exposed metalwork of transformers used in the installation other than those that are an integral part of equipment. The secondary windings of transformers should also be earthed at one point of the winding, unless the transformer is a safety isolating transformer supplying a part of the installation where the protective measure 'electrical separation' is being used).

Exposed conductive parts that (because of

19. SELECTION OF DEVICES FOR AUTOMATIC DISCONNECTION OF SUPPLY

19.1 General - In general, every circuit is provided with a means of overcurrent protection. If the earth fault loop impedance is low enough to cause these devices to operate within the specified times (that is, sufficient current can flow to earth under fault conditions), such devices may be relied upon to give the requisite automatic disconnection of supply. If the earth fault loop impedance does not permit the overcurrent protective devices to give automatic disconnection of the supply under earth fault conditions, the first option is to reduce that impedance. It may be permissible for this to be achieved by the use of protective multiple earthing or by additional earth electrodes. There are practical limitations to both approaches.

22.1.2 Above 240 V should be designed as a PME system with separate protective conductor. The neutral of the transformer should be connected to be earth electrodes by duplicate connections and adequate number of earth electrodes should be provided with interlinking earth bus for getting an optimum value of the earth resistance depending upon the setting of the earth fault/earth leakage relays and also to limit the extent of rise of potential. The earth fault current can be of the order of symmetrical short-circuit current and hence the thermal design of the earth bus and the earthing system should depend upon the maximum symmetrical short circuit current available. The duration of the earth fault current according to the existing design practice is 3 seconds. However, in case of installations where adequate protective arrangements have been incorporated so as to instantaneously isolate the system in the event of an earth fault, a lesser duration can be considered for design purposes.

TN-S with PME

Transformer “Neutral” 2 connections to Earth “Double Earthing”

Interlinking Earth Bus

Short Circuit duration can be less than 3 seconds

22.1.3 As far as the value of the earth resistance is concerned, the objective from the point of safety consideration is not to attain minimum value of the earth resistance as is sometimes understood. But the consideration should be whether there is adequate co-ordination between the practically obtainable value of the earth resistance and setting of the protective devices. Placement of electrode, area and size of grid depend on electrical installation, earth grid continuity resistance with in the limit. However, in the case of a protective multiple earthing system where the neutral of the supply transformer and the non-current carrying metal parts in the system are interconnected by the common earth grid, which is designed for the prospective fault current, there is no reason to design the earth electrodes assuming that total earth fault current is dissipated through the earth electrodes. However, depending upon the value of the earth resistivity, a percentage of the current may flow through the mass of the earth as well.....

Earth resistance..... coordinated to tripping of protective device

Common Earth Grid interconnects Neutral of transformer and Body of load

No need to design earth electrode for total fault current

22.2.1 The main earthing conductor will be run in between standard earth electrodes conforming to specifications and distributed uniformly around the working area. **All the non-current carrying metal parts of the equipment, switchboards, etc., will be solidly connected to this earth grid and equipotential bonding conductor by duplicate earth connections of adequate size.** For interconnecting switchboards protected by HRC fuses to this earth grid, the size of interconnection need not be more than 75 mm² copper or its equivalent. In laying out the earth electrodes and the earth conductors, all efforts should be made to maintain a uniform potential gradient in and around the work area. **The transformer neutral should be solidly connected to this grid by duplicate earth connections, one going directly to earth electrodes and other going to the common earth bus.** The size of the neutral earthing conductor should in no case be less than that of the size of the main earthing conductor.

Double Earthing of equipment and switchboards

One to Grid, another to Earth Bonding Conductor (PE conductor)

Double Earthing of Transformer

One to Grid, another to Earth Bus bar

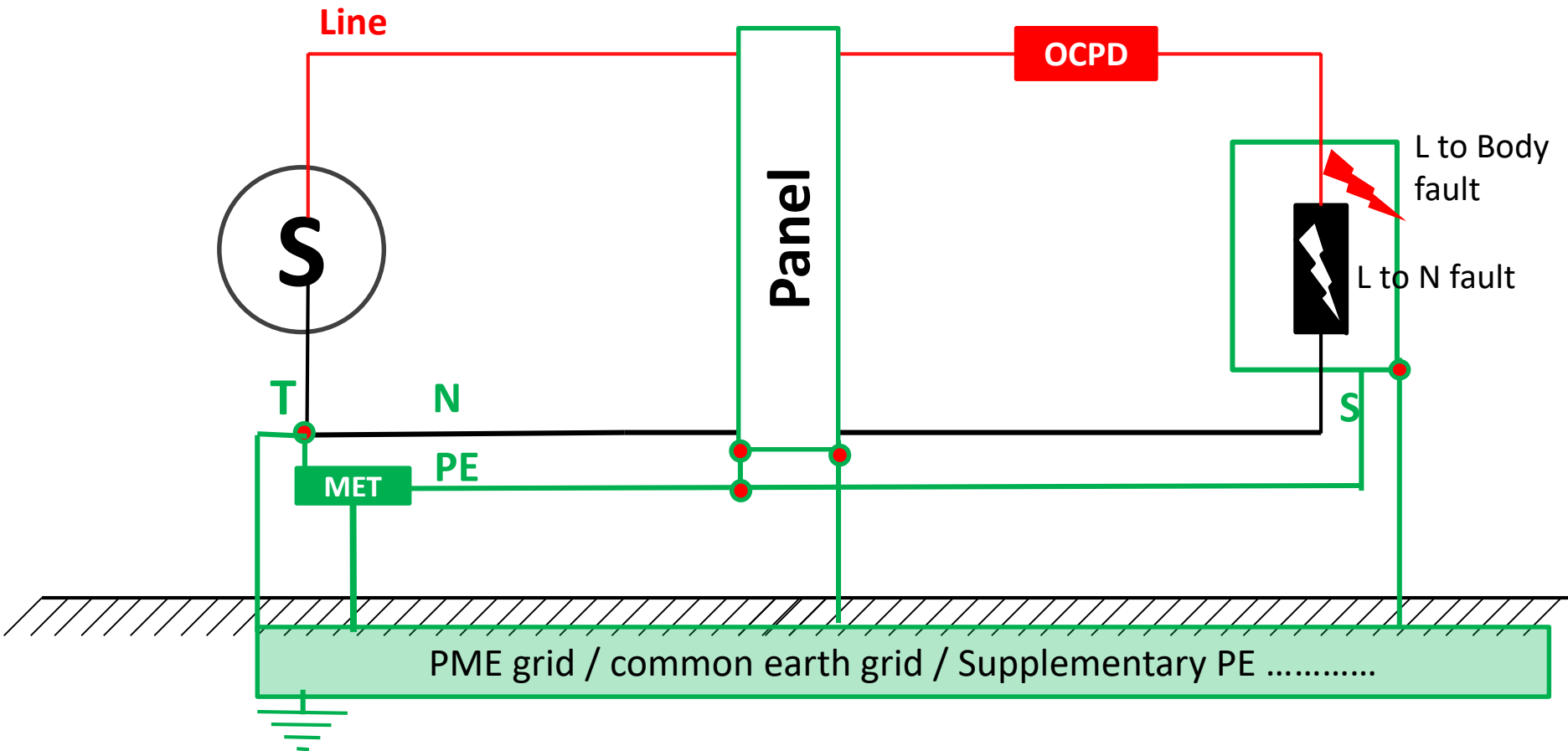
22.2.2 The earth grid should be run at a minimum depth of 50 cm below ground. When bare conductors are used as earth grid, this can also be assumed to dissipate the fault current to the mass of the earth

22.2.3 The continuity resistance of the earth return path through the earth grid should be maintained as low as possible and in no case greater than one ohm.

22.2.4 In the case of EHT substations, where there is possibility of the ground potential attaining very high values (of the order of 5 kV and above) in the event of an earth fault, the earth grid design should be based on the tolerable limits of the potential gradient in the substation area, and the step and touch potential due to fault conditions.

Earth Grid is the Earth Electrode

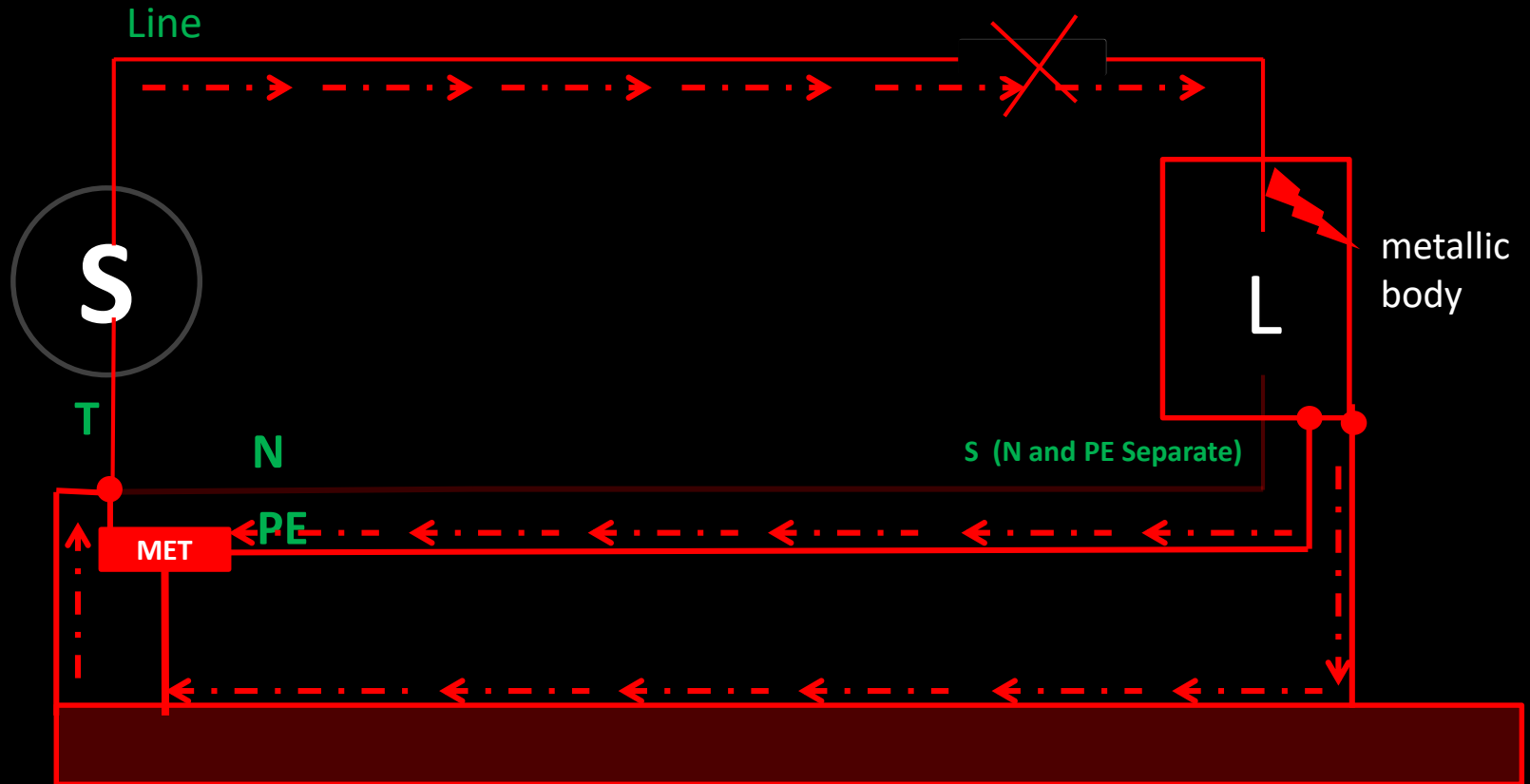
Periodic inspection of Continuity Resistance of Earth grid ensures the metal grid in soil is not corroded and disconnected



Two Parallel paths for the fault return current ensure Low loop impedance and fast tripping of protective device

PE conductors should be installed in close proximity (i.e. in the same conduits, on the same cable tray, etc.) along with the live cables of the related circuit

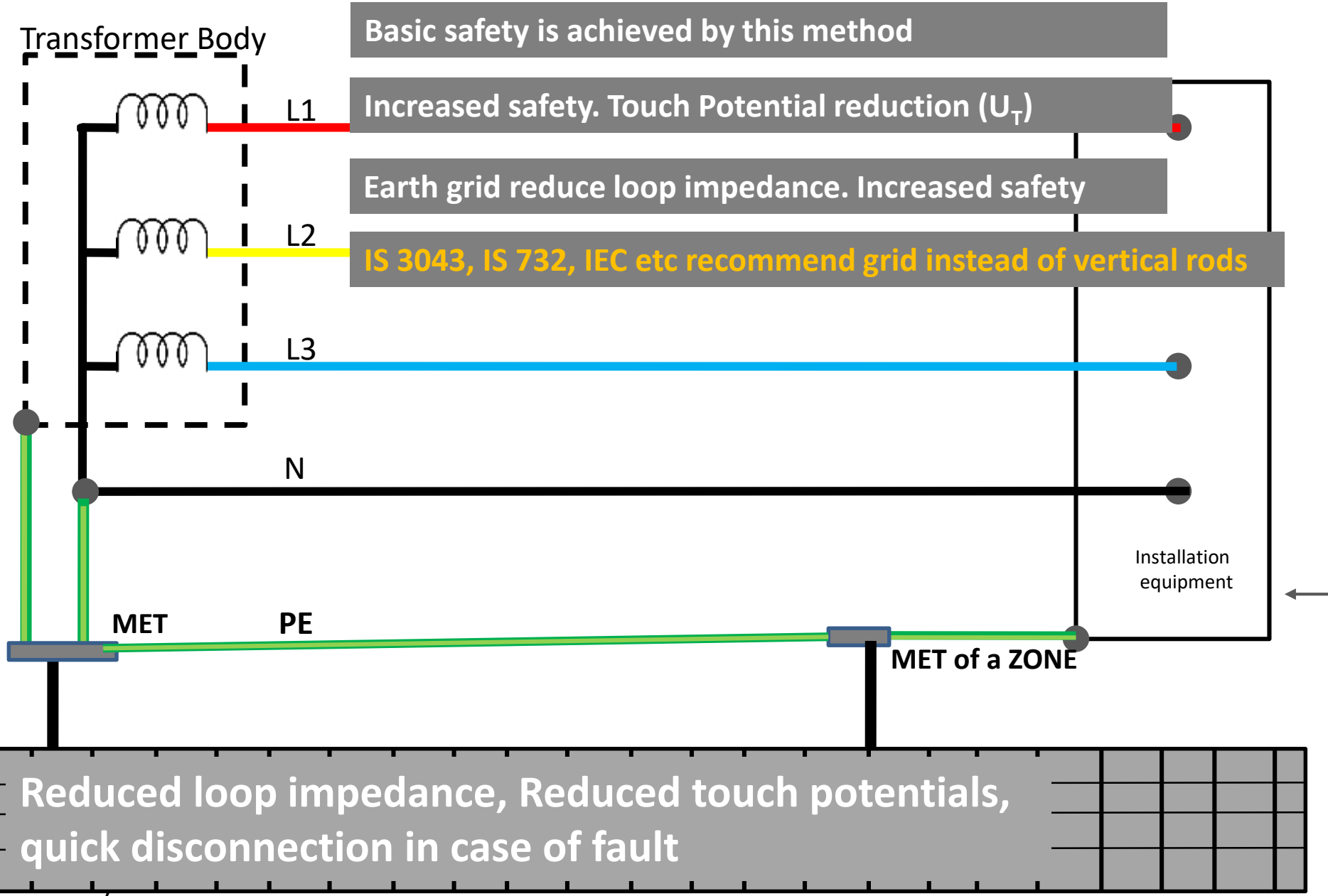
TN-S Network with separate protective conductor



Quick Disconnection of Power supply
More safe
Less chance of fire

Industrial and commercial 230/400 volts. TN - S with PME

Separate Neutral, PE conductor and PME grid



12. CONNECTIONS TO EARTH ELECTRODES – EARTHING AND PROTECTIVE CONDUCTORS

12.0 General

12.0.1 The materials used for making connections have to be compatible with the earth rod and the copper earthing conductor so that galvanic corrosion is minimized. In all cases, the connections have to be mechanically strong.

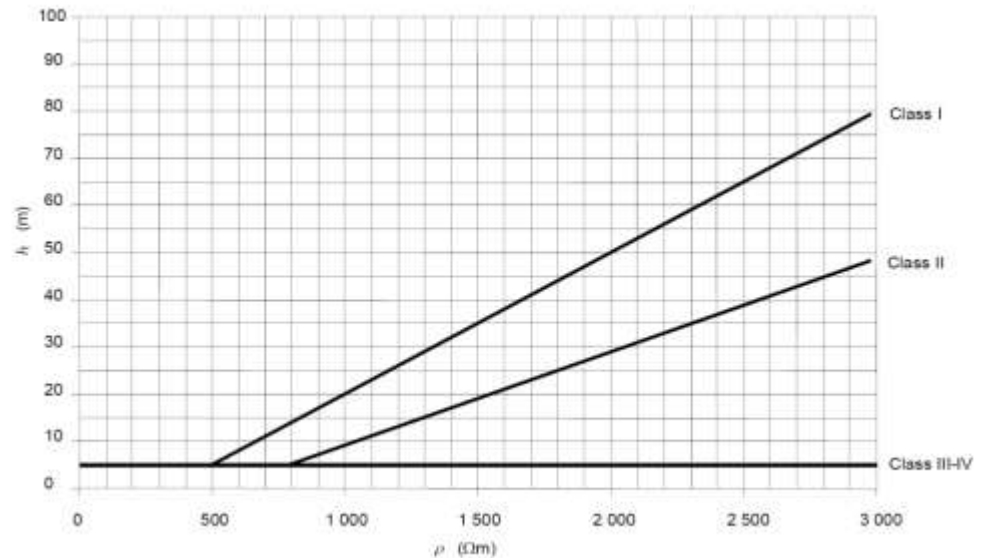
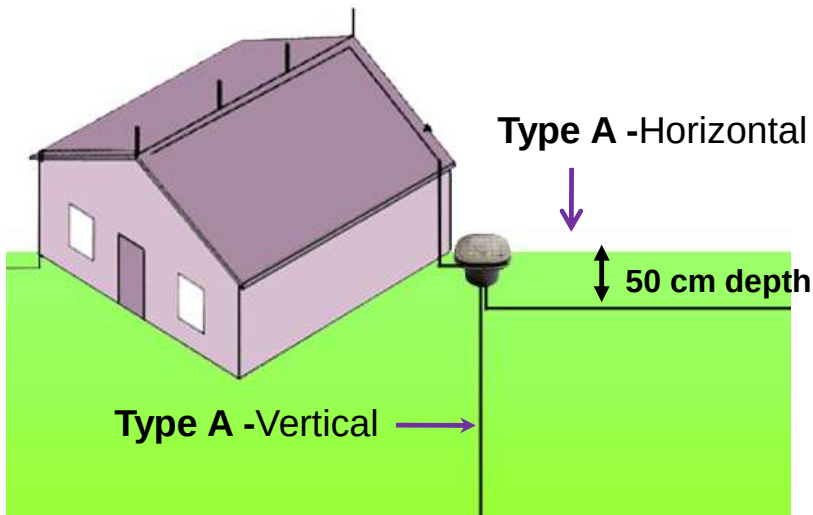
One system One earthing

Earthing for Lightning Protection

Equipotentialisation

This type of arrangement comprises horizontal or vertical earth electrodes installed outside the structure to be protected connected to each down conductor. In type A arrangements, the total number of earth electrodes shall be not less than two. (minimum 2 down conductors for a building)

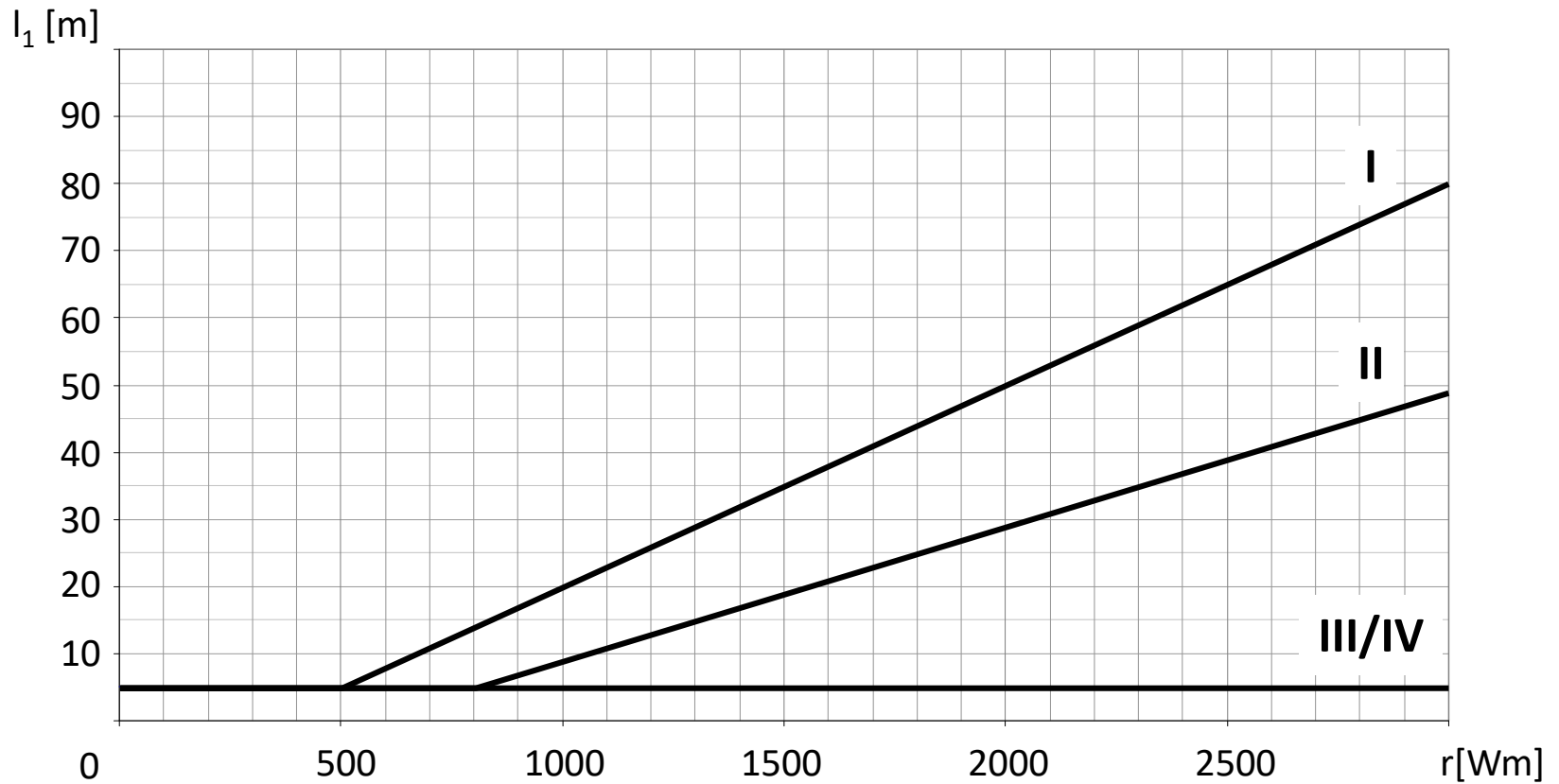
Suitable for small building with electricity connection

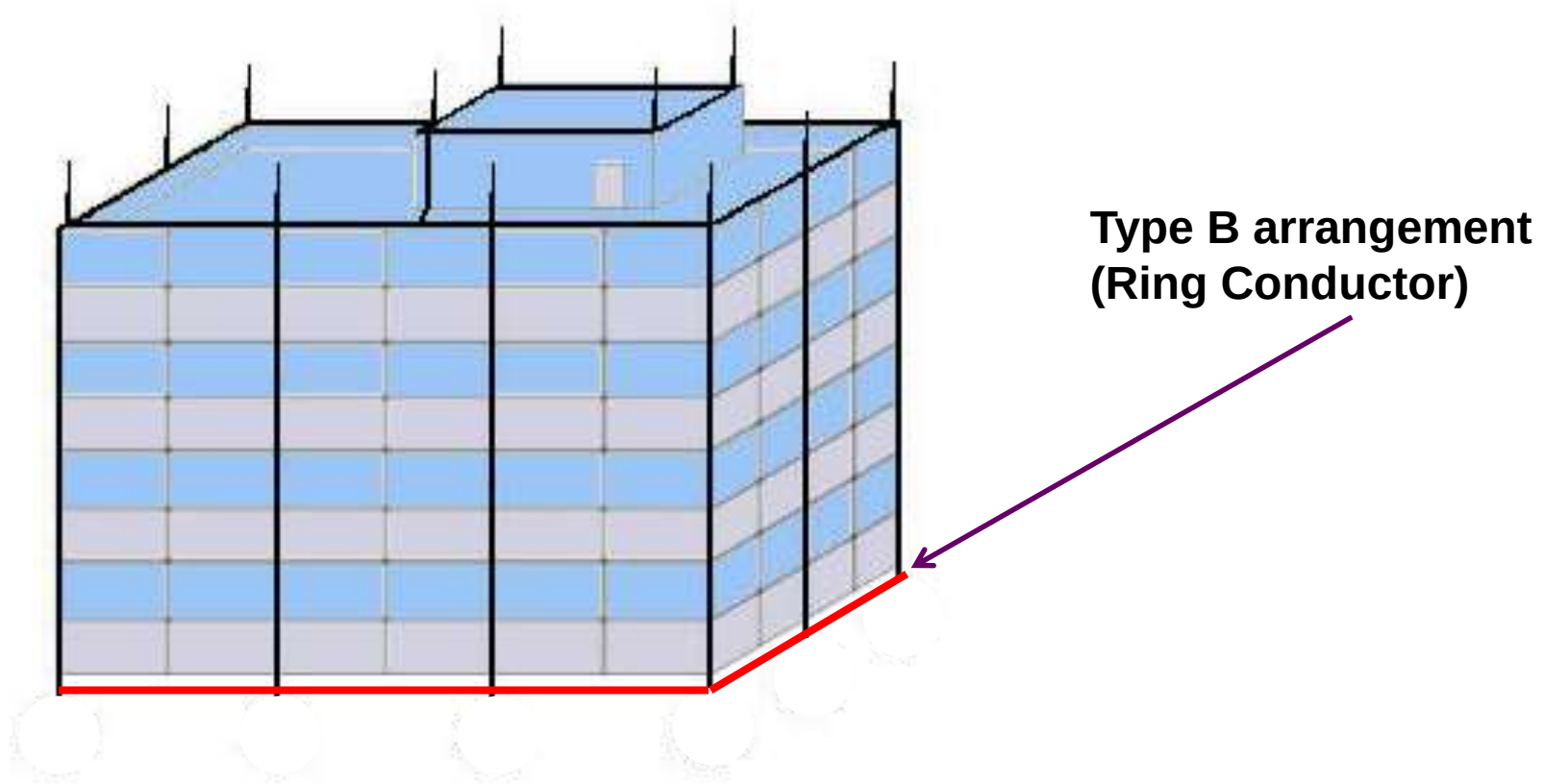


The minimum length of each earth electrode for each down-conductor as per graph – l_1 for horizontal electrodes, or – $0.5 l_1$ for vertical (or inclined) electrodes, For combined (vertical or horizontal) electrodes, the total length shall be considered

Resistance of 10 OHM is also recommended, but can be ignored with the above length

Minimum length l_1 of earth electrodes



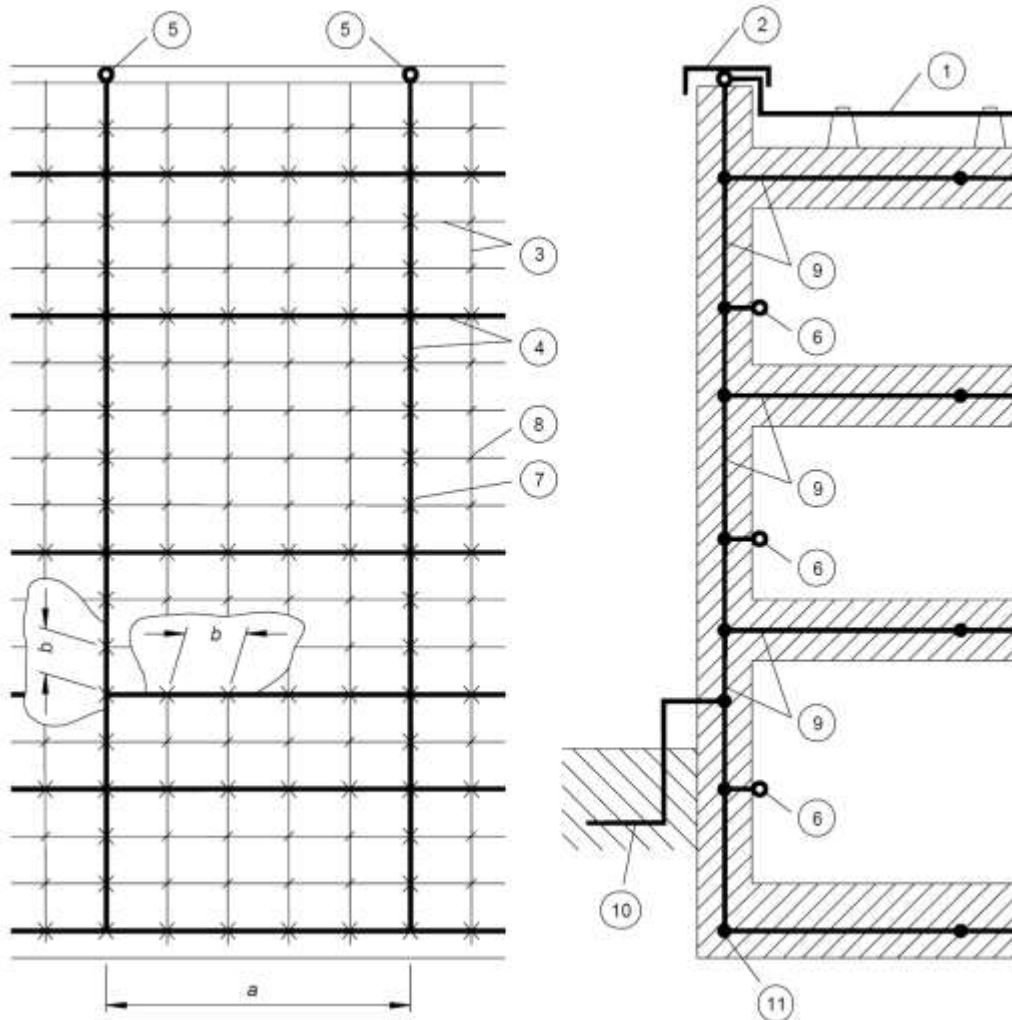


This type of arrangement comprises either a ring conductor external to the structure to be protected, in contact with the soil for at least 80 % of its total length, or a foundation earth electrode.

Such earth electrodes may also be meshed.

Recommended for buildings with Electronic Installation

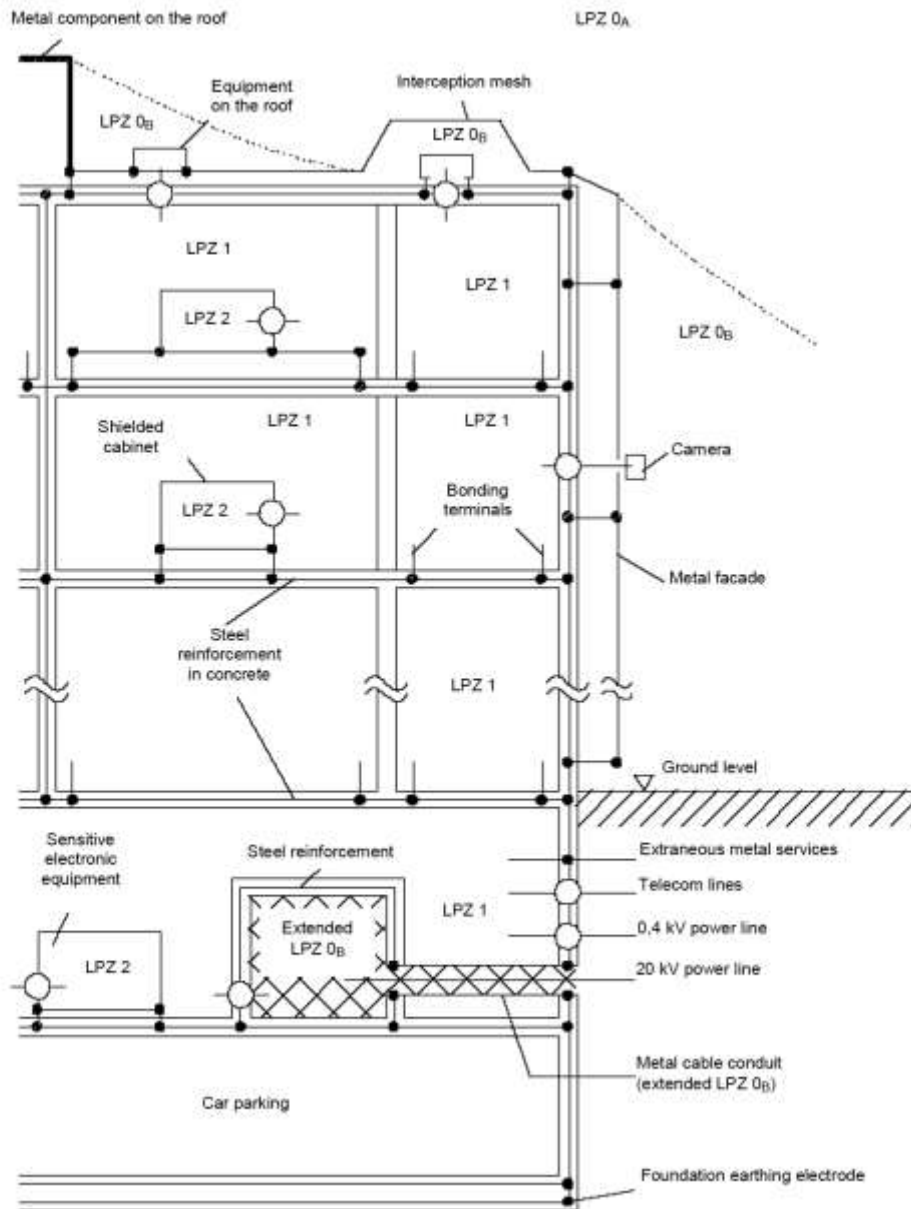
Conductor inside an RCC structure (or under an RCC structure) can be used for Earthing & equipotential bonding.



1. air-termination conductor
2. metal covering of the roof parapet
3. steel reinforcing rods
4. mesh conductors superimposed on the reinforcement
5. joint of the mesh conductor
6. joint for an internal bonding bar
7. connection made by welding or clamping
8. arbitrary connection
9. steel reinforcement in concrete (with superimposed mesh conductors)
10. ring earthing electrode (if any)
11. foundation earthing electrode

a typical distance of 5 m for superimposed mesh conductors
b typical distance of 1 m for connecting this mesh with the reinforcement

Nothing Exposed
Do not affect aesthetics
Most efficient for electronics



No extended vertical earth electrode in soil
Transformer and Dg inside building
All 5 purposes of earthing is achieved

Further questions are

How to Earth HT side of transformer

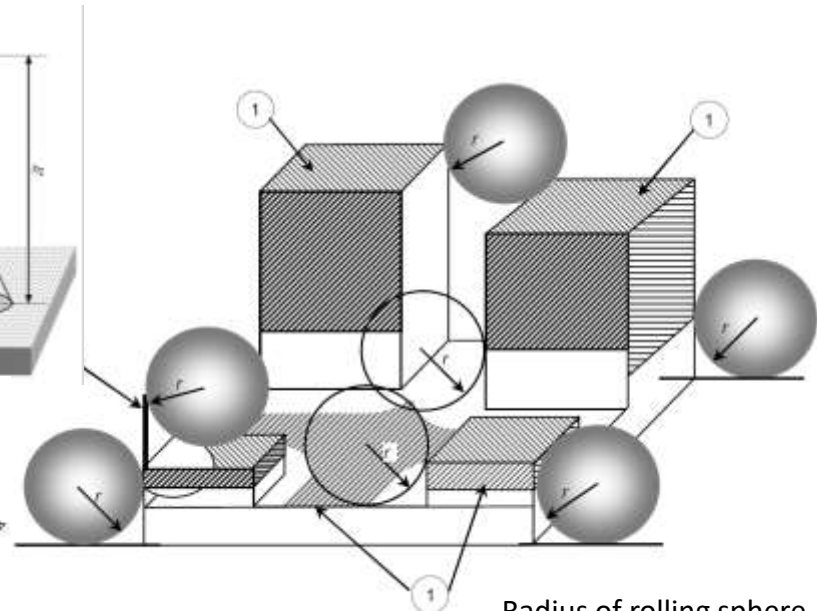
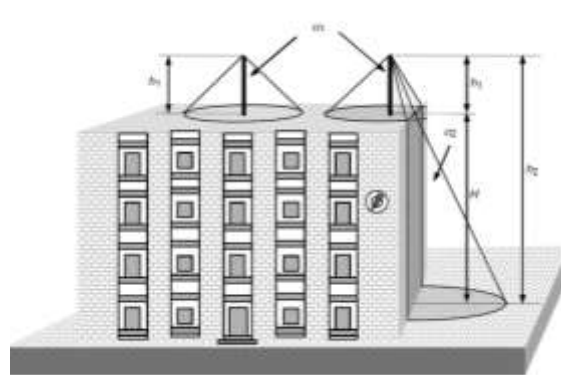
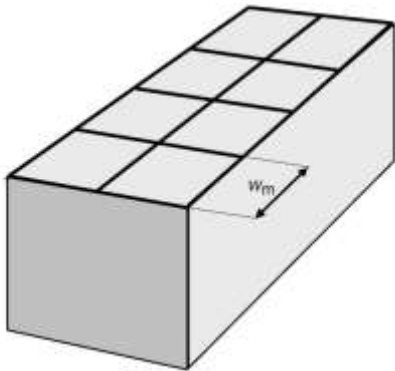
How to connect UPS

How to connect sensitive equipment

Design: Air Termination

Air Termination is design Based on the lowest possible Lightning strike of that LPL

Class of LPS	Mesh size in meters	Protection angle w.r.t height					Rolling Sphere radius
		10 meter	20 meter	30 meter	45 meter	60 meter	
1	5 x 5	45	23	Cannot be used			20
2	10 x 10	54	38	23	Cannot be used		30
3	15 x 15	62	48	36	23	Cannot be used	45
4	20 x 20	65	54	45	34	23	60



Radius of rolling sphere

$$r = 10 \times I^{0,65}$$

Positioning of down-conductor

Distance between down/ring conductors

Protection level	Typical distance [m]
I	10
II	10
III	15
IV	20

The consideration of the „**typically distances**“ doesn't relate to the the mathematical calculation of the separation distance!

$$\underline{d} \geq s$$

- d distance between lightning protection system and the electrical and metallic installations of on the other hand
- s separation distance

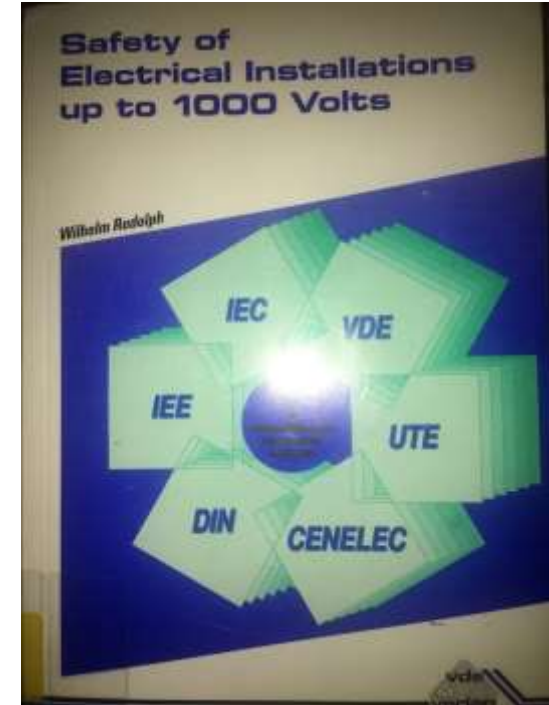
Electro chemical influence on and between earth electrodes

Metals laid in soil or water may be corroded by the following influence

- DC Leakage current in the soil
- Chemical components and their concentration in soil or water
- Galvanic cells formed by interconnected earth electrodes of different metals

Protection by means of non metallic covering of electrodes is not suitable because of the high electrical resistance of such materials.

Very important is the case where the earth electrodes form galvanic cells with other earth electrodes or with buried steel pipes and conduits. With underground metallic structures or building foundations provided they are metalically interconnected.



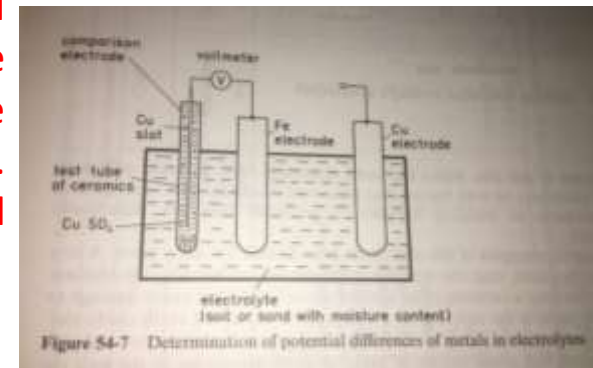
Corrosion Bi-Metallic & Galvanic effects

Potential Difference of metals in electrolyte

Table 54-C Potential differences of metals in electrolytes [5, 6, 7]

Metal in soil with moisture content	Cupric-sulphate, Cu/Cu SO ₄ , comparison electrode		
	Values acc. to K. Vögtli, Bern	Values acc. to E. Hönninger, Graz	DIN VDE 0151/06.86
Zinc and galvanized steel	-0,7 V ... -1,0 V	... -1,15 V	-0,9 V ... -1,1 V
Copper	0,0 V ... -0,2 V	... -0,15 V	0 V ... -0,1 V
Steel	-0,5 V ... -0,8 V	... -0,95 V	-0,5 V ... -0,8 V
Steel, rusty	-0,4 V ... -0,6 V		
Steel in humus		-0,6 V ... -0,8 V	
Steel in clean sand		-0,4 V ... -0,5 V	
Steel in concrete	-0,1 V ... -0,3 V	-0,05 V ... -0,2 V	-0,1 V ... -0,4 V
Zinc			-0,9 V ... -1,1 V
Lead			-0,5 V ... -0,6 V

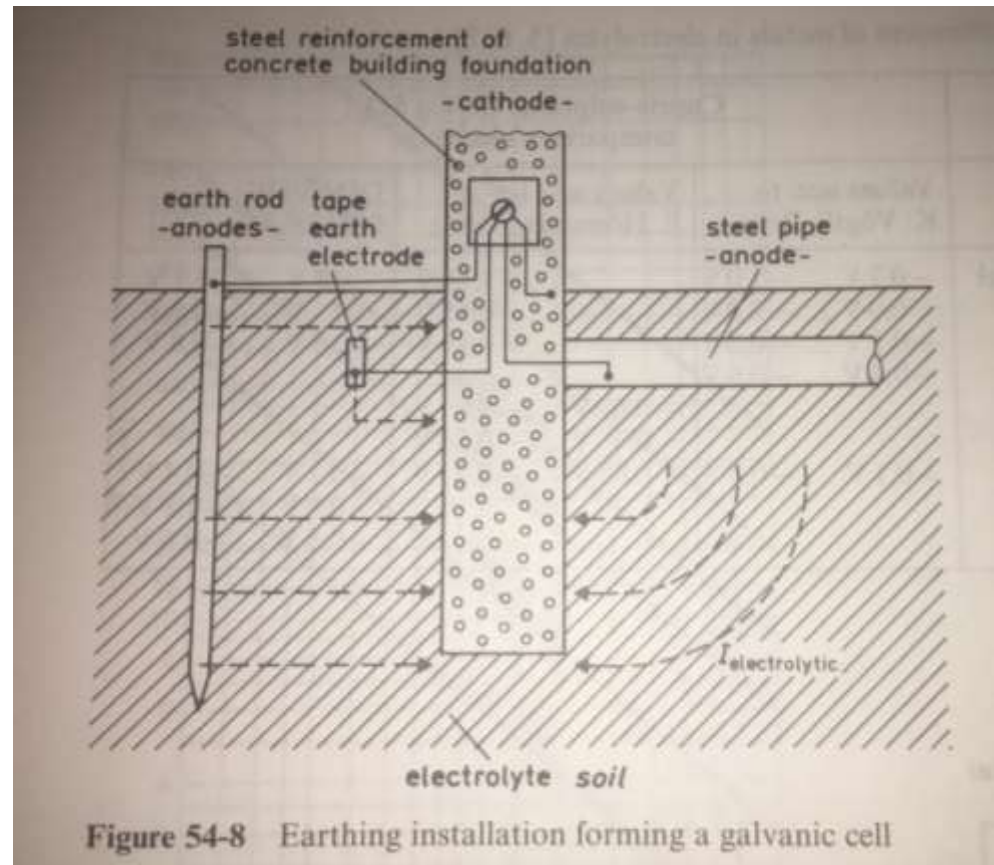
IS 3043: The possibility on damage to cables and other underground services and structural metalwork in the vicinity of earth-electrode due to electrolytic action between dissimilar materials should not be overlooked when the material for earth-electrodes is selected. Materials compatible with other metal structures in the vicinity should be selected or other remedial action taken



The fundamental principle of this effect have been known for many years.

A DC current between anode and cathode in the electrolyte results from the influence of the potential difference of the galvanic cell (fig 54-8), formed by the different materials of the earth electrodes by the electrolyte of the moisture content of the soil, and through the bonding of the anodic and cathodic earth electrodes. Within the anodic range, this current leaves the surface of the electrode towards the electrolyte, thus dissolving the metal of the anode. This current enters the metallic surface of the electrode in the cathodic area.

This electrolytic current passes losses of the mass of the anodic earth electrode. The speed of the the metallic losses (corrosion rate) depends mainly on the voltage of the galvanic cell.



Interconnection of all earth electrodes for different purpose with other earthed construction is recommended for increased electrical safety. Electrical isolation of various electrodes is no solution for protection of the electrodes against corrosive effect. The only reliable solution is the correct selection and installation of electrodes. The basic of this selection rule is the knowledge of the potential difference of different metals in an electrolyte.

A new experience is however that the metallic reinforcement of building foundations may likewise become a cathode of an electrolytic cell and thus cause damage to other metallic parts in the soil. Eg to water pipes, conduits or earth electrodes. With the ever increasing size of buildings and their foundations and the relatively reduced free metallic surface of pipes or earth electrodes in the soil, the surface relationship of anode to cathode becomes increasingly unfavorable. The danger of corrosion must necessarily increase when using metals with large potential differences according to table 54-C.

Metallurgy (anodic Index) (courtesy – www.corrosion-doctors.org)	Index (Volt)
Gold, solid and plated & Gold-platinum alloy	0.00
Silver, solid or plated, monel metal & High nickel-copper alloys	0.15
Nickel , solid or plated , titanium and alloys & Monel	0.30
Copper , solid or plated , low brasses or bronzes, copper-nickel-chromium alloys	0.35
Brass and bronzes	0.40
High brasses and bronzes	0.45
Lead, solid or plated, high lead alloys	0.70
Iron, wrought, gray or malleable, plain carbon and low alloy steels	0.85
Hot-dip-zinc plate, galvanized steel	1.20
Zinc, wrought, zinc-base die-casting alloys, zinc plated	1.25
Magnesium & magnesium-base alloys, cast or wrought	1.75

Corrosion is an electrochemical process. Copper plated materials are less corrosive than Galvanised steel. Corrosion resistance of copper plated material is equal to or better than copper (due to Nickel copper alloy)

Anodic Voltage of GI – 1.2 V

Anodic Voltage of copper – 0.30

Due to 0.9 to 1 volt difference copper (or steel in concrete) will absorb GI in soil

E.4.3.4 Materials

The **Galvanised Material inside concrete is a problem** protection purposes: steel, mild steel, galvanized steel, stainless steel, copper and copper coated steel .

The behavior of a galvanized layer on steel in concrete is very complicated, particularly in concrete with chlorides, the zinc will corrode quickly on contact with the reinforcement, and can under certain conditions cause damage to the concrete. Galvanized steel should therefore not be used in coastal areas and where there may be salt in the ground water. As the use of galvanized steel in concrete requires evaluation of many external factors this material should be used only after careful analysis. With this in mind the use of the other mentioned materials is preferred over the use of galvanized steel.

In order to avoid confusion between the different types of steel rods in concrete, it is recommended that round steel rods of at least 8 mm diameter with a smooth surface be used as additional conductors in contrast to the ordinary ribbed surface of the reinforcing rods.

Galvanic corrosion between steel in RCC and steel in soil

From Clause E.5.4.3.2

A further problem arises from electrochemical corrosion due to galvanic currents. Steel in concrete has approximately the same galvanic potential in the electrochemical series as copper in soil. Therefore, when steel in concrete is connected to steel in soil, a driving galvanic voltage of approximately 1 V causes a corrosion current to flow through the soil and the wet concrete and dissolve steel in soil.

Copper / Stainless Steel / Copper coated Steel are the best

TN-S PME for multi floor buildings

Design and execution need deep knowledge in the subject

Don't use GI in soil

Don't use GI inside concrete

All connections in soil must be exothermically welded

Use only tested components

Beware of

Lightning protection and earthing manufacturers from Europe.

They know earthing for the purpose of Lightning Protection

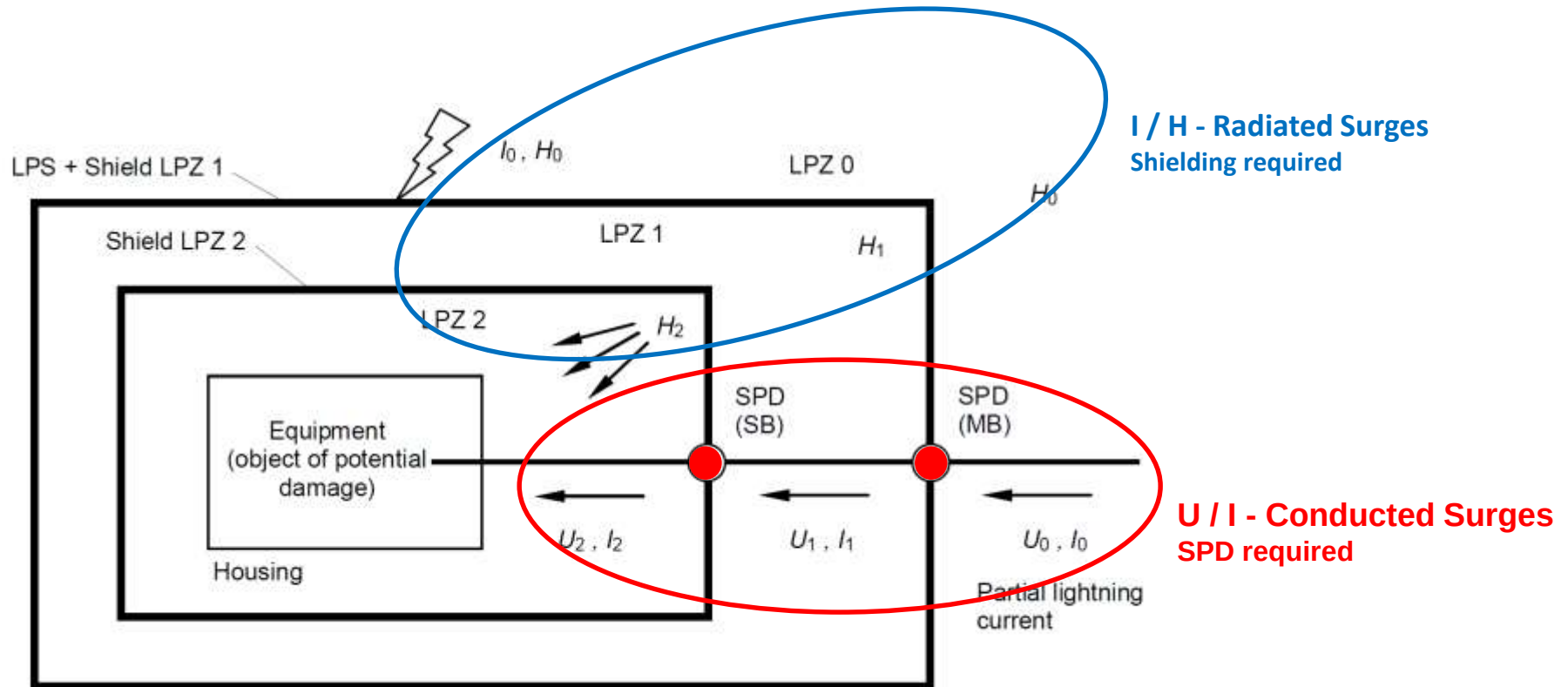
Manufacturer don't do design and installation in Europe / USA

(eg – German manufacturers don't do exothermic welding)

(Eg – British manufacturers recommend micro pile earthing) (Damage caused to plastic waterproof membrane will create serious issues in future)

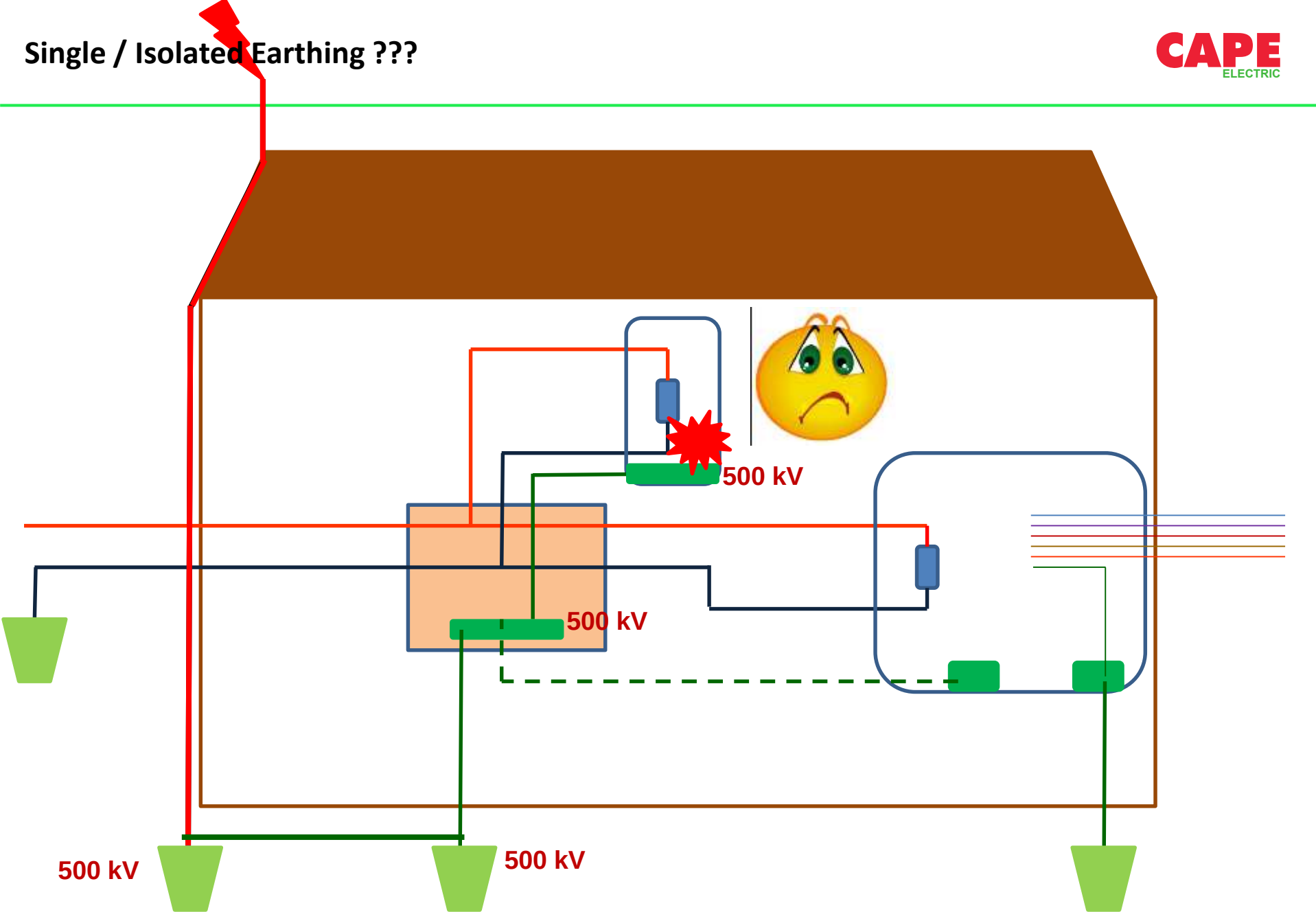
One system One earthing

Equipotentialisation

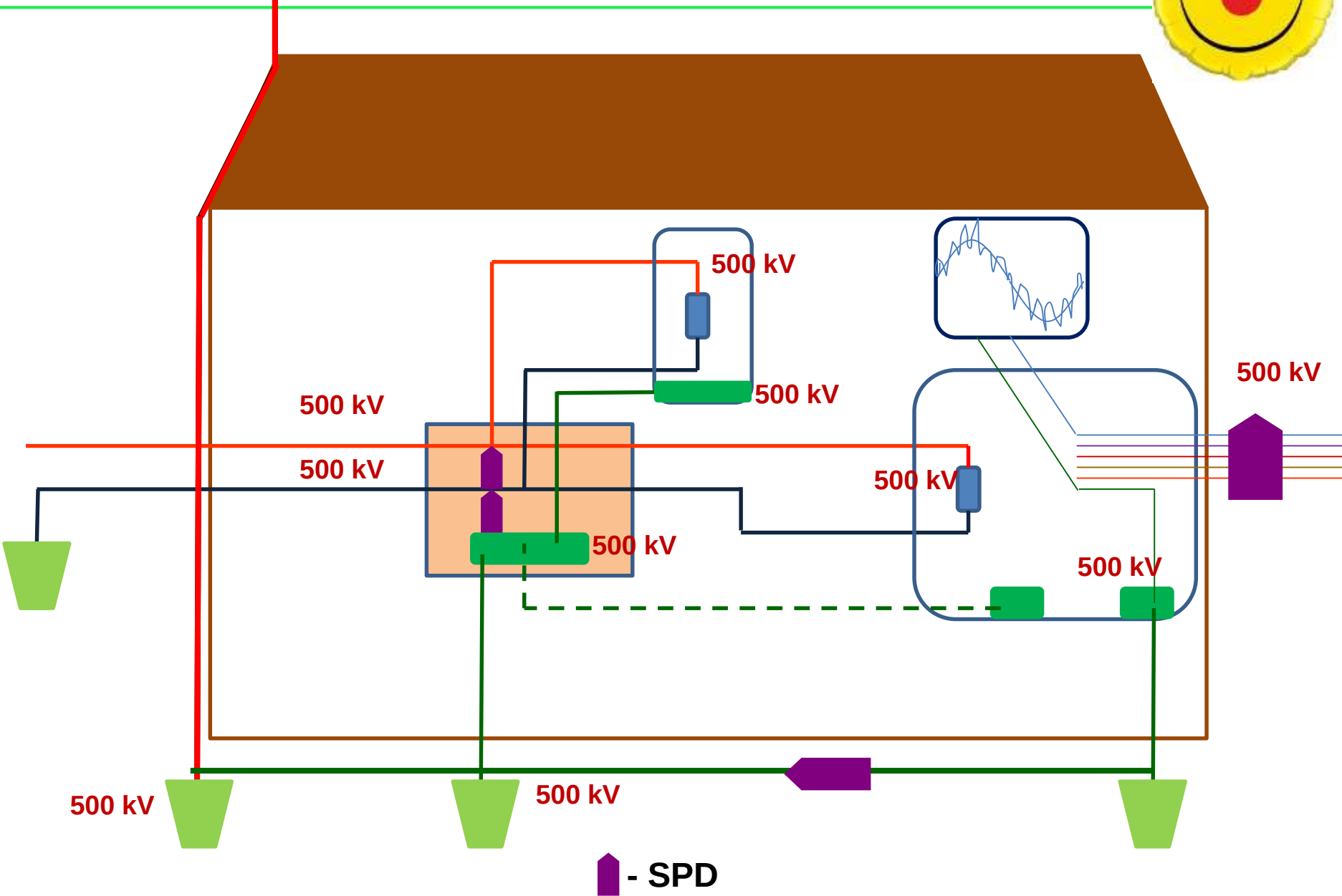


**Equipotentialisation – NO Potential difference between metallic installations / parts
Equal at every ZONE Boundaries**

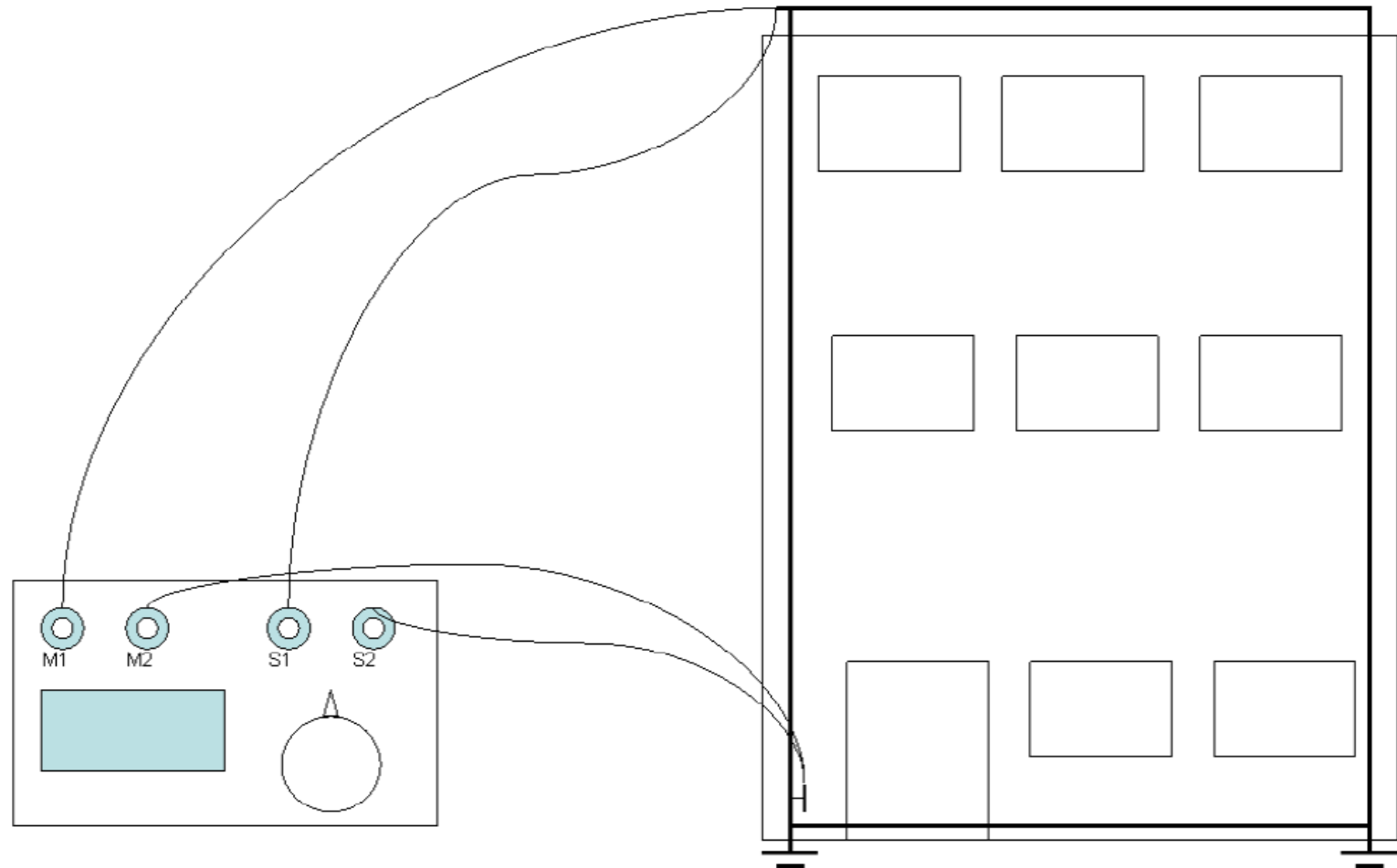
Single / Isolated Earthing ???



Single Integrated Earthing with SPD is the best

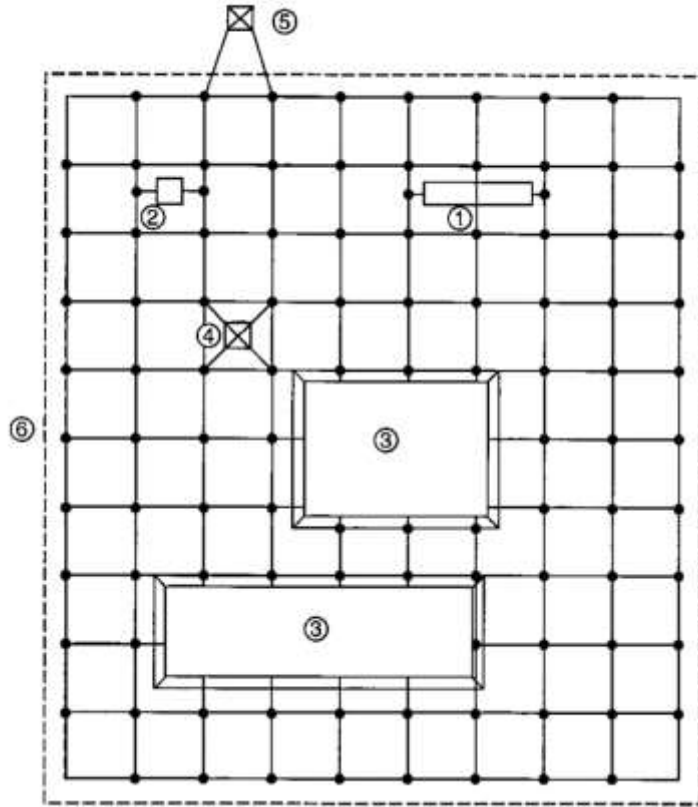


Structural Earthing – Testing Provisions need to be provided at top and bottom

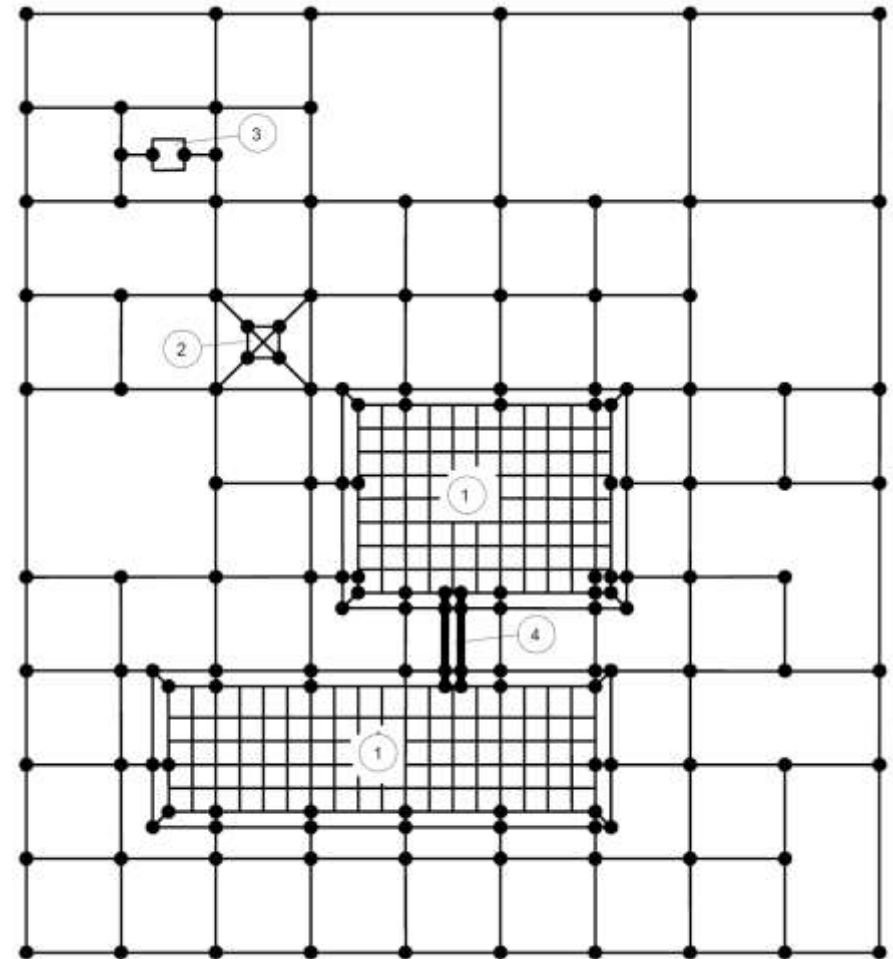


Resistance Less than 0.2 Ohm

Schematic plan view of a typical earth electrode for an industry / building complex



- ① Metallic frame
- ② Stand-alone apparatus or system
- ③ Building
- ④ High- or medium-voltage tower inside the plant
- ⑤ High- or medium-voltage tower near the plant
- ⑥ Fence



- 1) building with meshed network of the reinforcement
- 2) tower inside the plant
- 3) stand-alone equipment
- 4) cable trenches

Signal Earthing

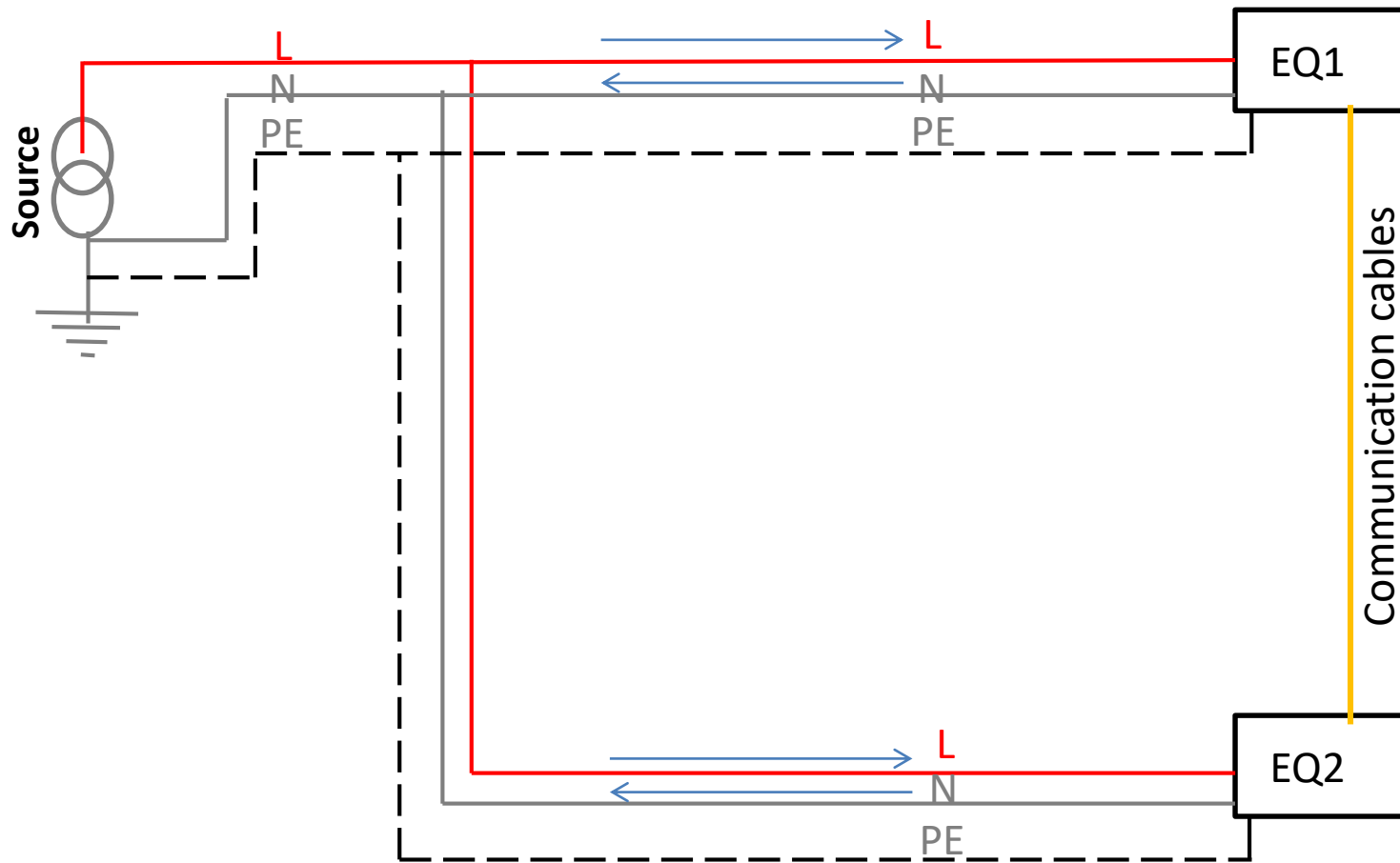
Communication / Sensitive electronic Equipment Earthing

microprocessor-based systems such as computers, PLC's,
industrial process plant DCS's,
telecommunications systems,
medical diagnostic imaging
other related sensitive electronic equipment

Isolated equipment ground:

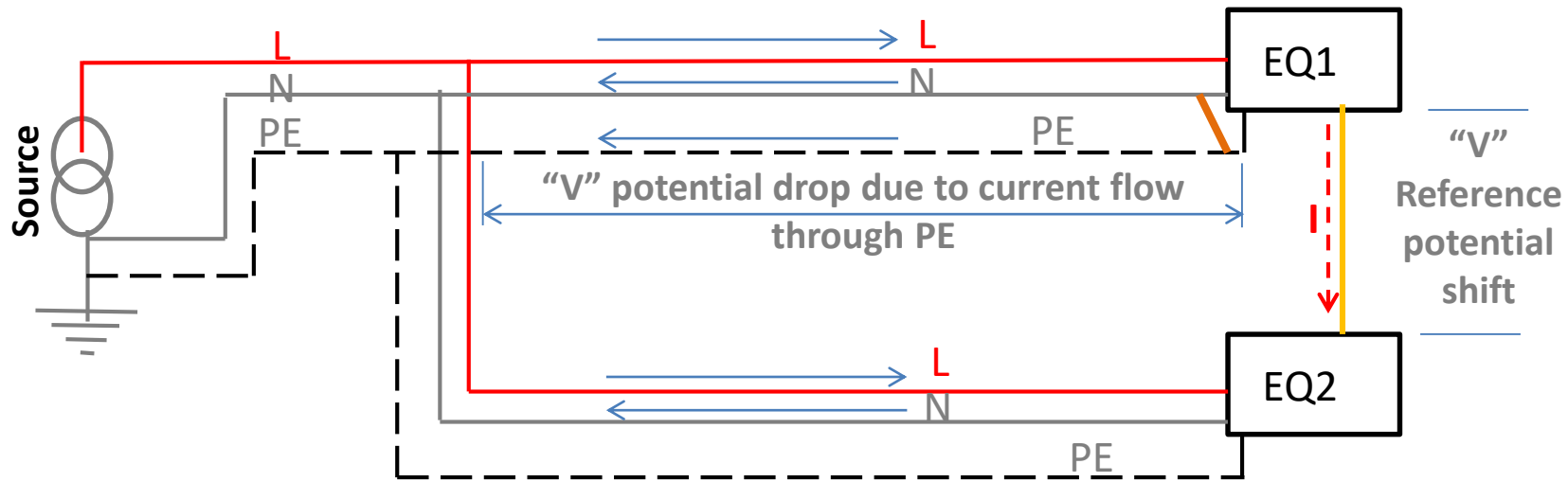
An insulated equipment grounding conductor (EGC) run in the same conduit or raceway as the supply conductors. This conductor is insulated from the metallic raceway and all ground points throughout its length. It originates at an isolated ground type receptacle or equipment input terminal block. The conductor may pass through one or more panel boards without connection to the panel-board grounding terminal, and terminates at an EGC terminal at the applicable derived system or service.

TN-S system. Two electronic devices connected with a communication cable powered from the same source



Microprocessors in sensitive electronic equipment operate at very low voltage. They need reference voltage. Reference of voltage is mostly the Earth Terminal of the equipment. PE conductor used for safety purpose will substitute signal reference also during normal operation

Signal Earthing requirement.



Under **several conditions**, signal reference may change to measurable voltage difference in the earth conductor.

Eg Connection between N and PE near EQ1. Partial Neutral current flows back to source through PE conductor

"V" voltage drop in PE conductor is created

This potential difference "V" is experienced between EQ1 and EQ2. Reference potential of the equipment changes

Reference potential shift creates an undesired current flow in the signal cable.

Quote from IEEE 142:

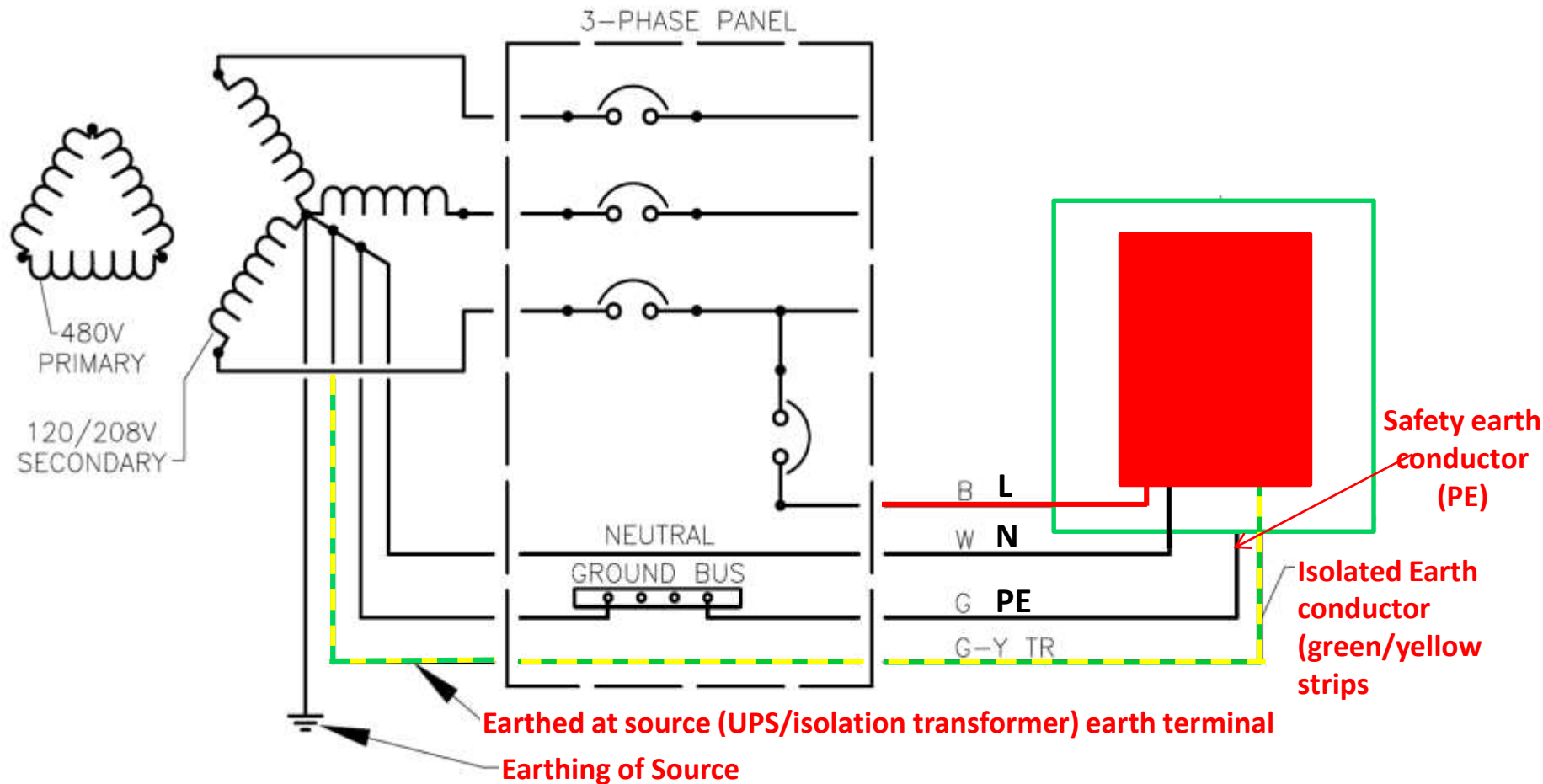
It is not unusual to find the branch circuit distribution panel neutral bus bar connected to the metallic panel frame (ground), which is a violation of the NEC. One study showed 20% of the neutral conductors accidentally faulted to ground in circuits supplying lighting fixtures. With multiple connections of the neutral conductor to ground on the same power system, a portion of the load current flowed on the equipment grounding system to which the electronic equipment was referenced. This current flow transferred voltages into the grounding system of the electronic equipment, causing errors or worse.

Electronic equipment: Like the electrical power supply system, electronic equipment has diversified systems to be grounded, as follows:

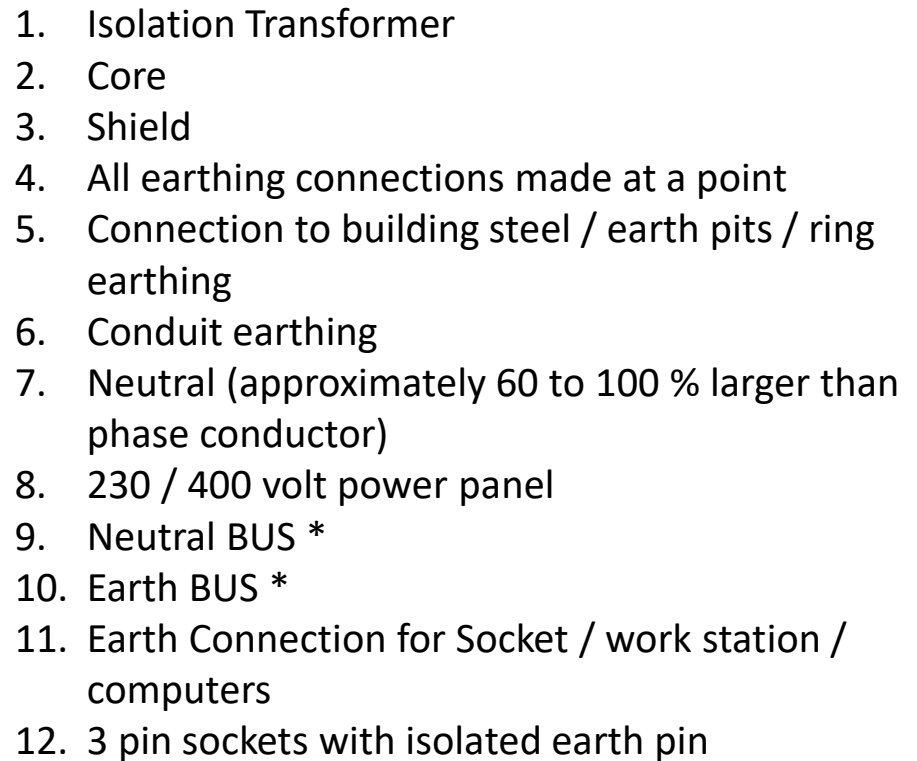
- a) *Signal common grounding.*** The signal common is also referred to as the *dc signal common*. The zero reference system for data lines, and the signal portion in general, represents the sensitive neutral of the electronic equipment. This is one of the systems that is sensitive to transient voltages and requires a stable reference point, with respect to a voltage potential.
- b) *DC power supply reference ground bus.*** The electronic equipment may have several different dc voltage systems, such as + 12 / 0 / – 12 V, + 5 / 0 / – 5 V.
- c) *Equipment ground bus.*** This is the metallic enclosure, or frame, of the electronic equipment. This may include the chassis of the electronic equipment elements, as well as the outer enclosure or cabinet. Some electronic equipment manufacturers refer to the equipment ground bus as the *safety ground bus*.

In addition to these terms for the various ground bus systems, you may encounter such terms as: *ac safety (mains) grounds*, *computer reference ground*, *dc signal common*, *earth common*, *dc ground bus*, *dc master ground point*, and *power supply common ground point*. It appears that each electronic equipment company has generated its own term for various grounded parts of their systems. There is no uniformity in the terminology, although as you will see later, they all must end up connected together.

Solution: Single Point Earthing. (courtesy IEEE142)



To prevent stray continuous currents or circulating currents from affecting the electronic equipment signals and operation, it is necessary to keep the electronic equipment ground system separate from the equipment ground components and connected together *at only one point*



Note:

- a. Each Branch circuit shall have a separate neutral and earth wire. No daisy Chaining Permitted
- b. Only computer or control system should be served from this panel

Special Installations

Hospitals & IT Installations

Wrong - Separate Connection to a low resistance earth electrode in soil

Right - Low resistance and separate connection to an earthing system



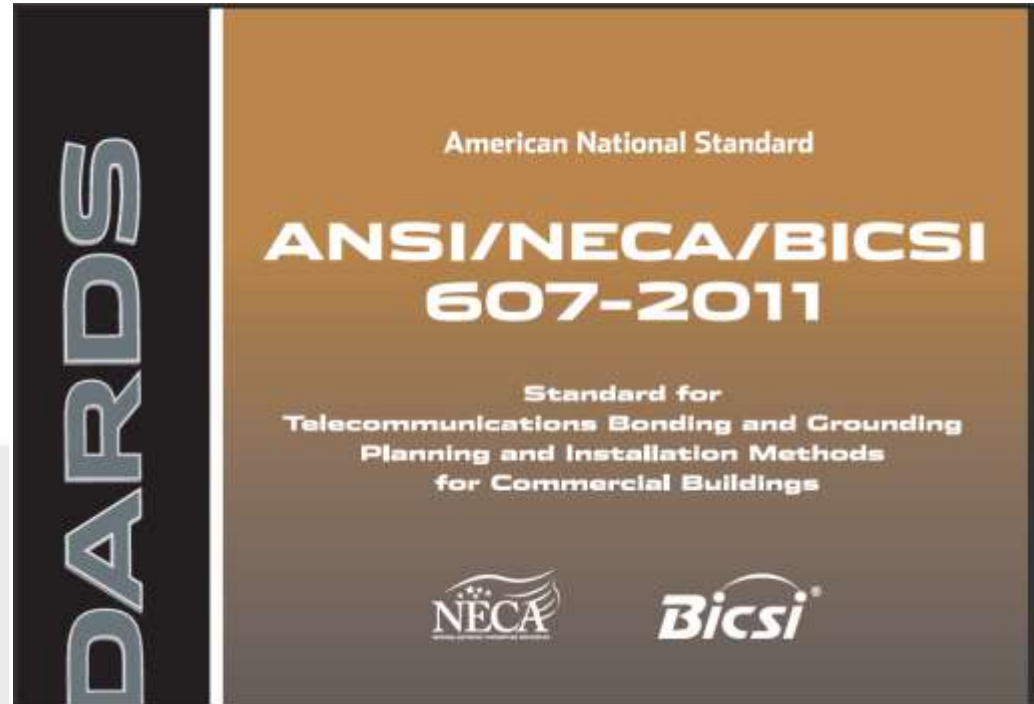
ISO/IEC 30129

Edition 1.0 2015-10

INTERNATIONAL STANDARD



Information technology – Telecommunications bonding networks for buildings
and other structures



Telecommunications Bonding and Grounding Planning and Installation Methods for Commercial Buildings

NECA/BICSI 607

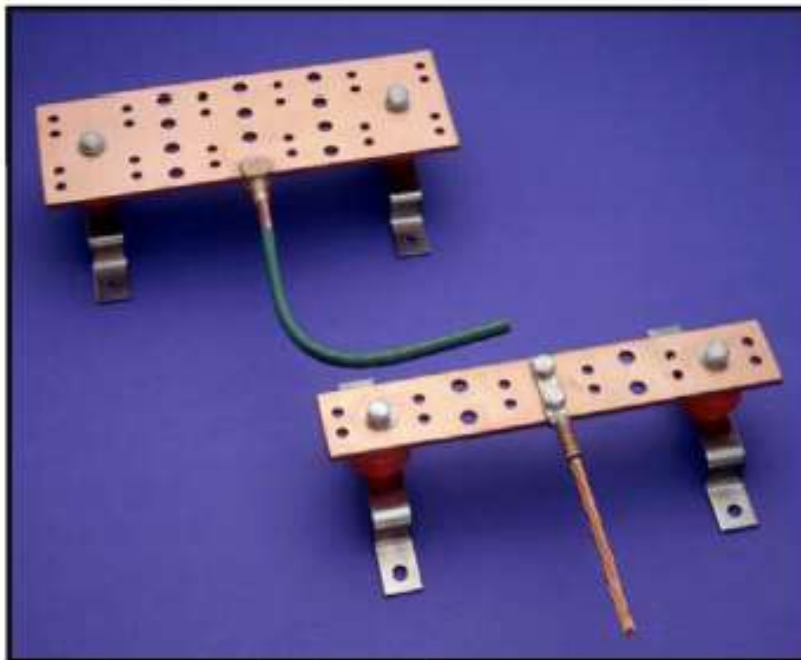


Figure 9. Example of an exothermic connection and a two-hole lug connection to a busbar

- a) Trim the insulation back so that the bared conductor is slightly longer (recommended not to exceed 1.6 mm [0.0625 in] of the barrel) than the barrel (see figure 11).

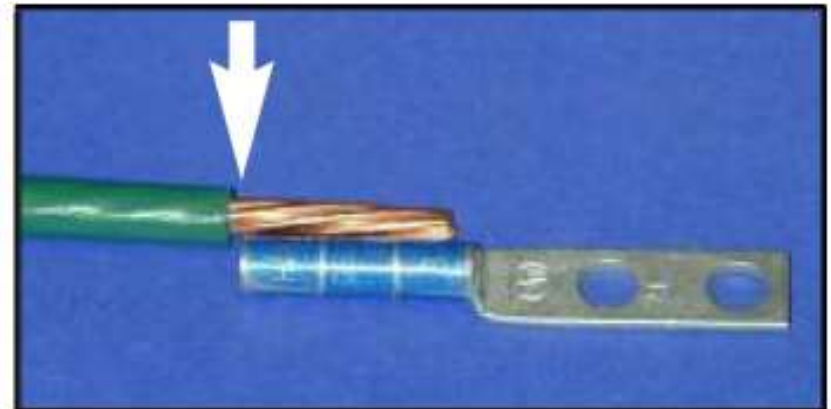


Figure 11. Example of trimmed insulation from a conductor

conductor. The two-hole lug shall also be the long barrel type with an inspection port (see figure 10).

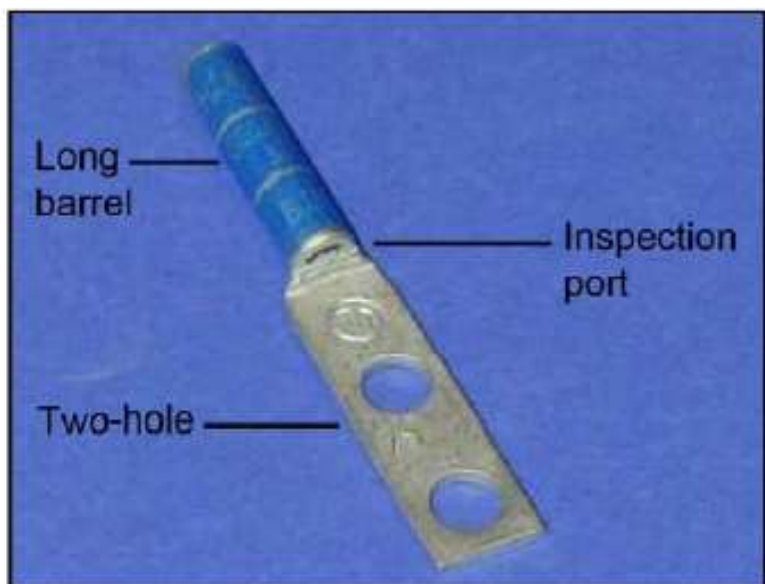


Figure 10. Example of a two-hole lug

7.5.2 Installation

7.5.2.1 Crimp connection of a two-hole lug to a bonding conductor

Select the lug size, configuration, and material applicable to the conductor size and fastening conditions. The following steps ensure a reliable connection between the two-hole lug and the bonding conductor using a crimp connection:

b) After applying an antioxidant compound on the exposed conductors, insert the conductor so that it “butts up” against the end of the barrel as viewed through the inspection port. The inspection port allows the installer to visually ensure appropriate conductor insertion (see figure 12).

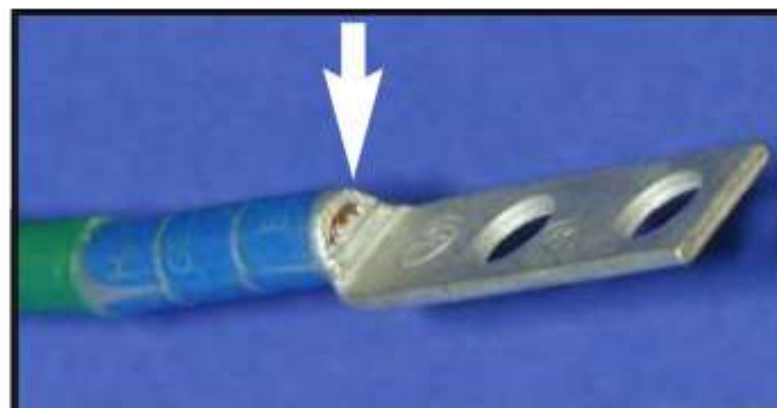


Figure 12. Example of a conductor seen through the inspection port

c) Ensure the end of the conductor remains at the end of the barrel; make the first crimp nearest the tongue end (flat end), working toward the conductor with the remaining crimps (see figure 13). The lug manufacturer's instructions shall be followed for the number of crimps and their location on the barrel (see figure 14).

Separate and Low resistance connection to Earthing system

NECA/BICSI 607 Telecommunications Bonding and Grounding Planning and Installation
Methods for Commercial Buildings



Figure 13. Example of crimping a conductor in the barrel of the lug

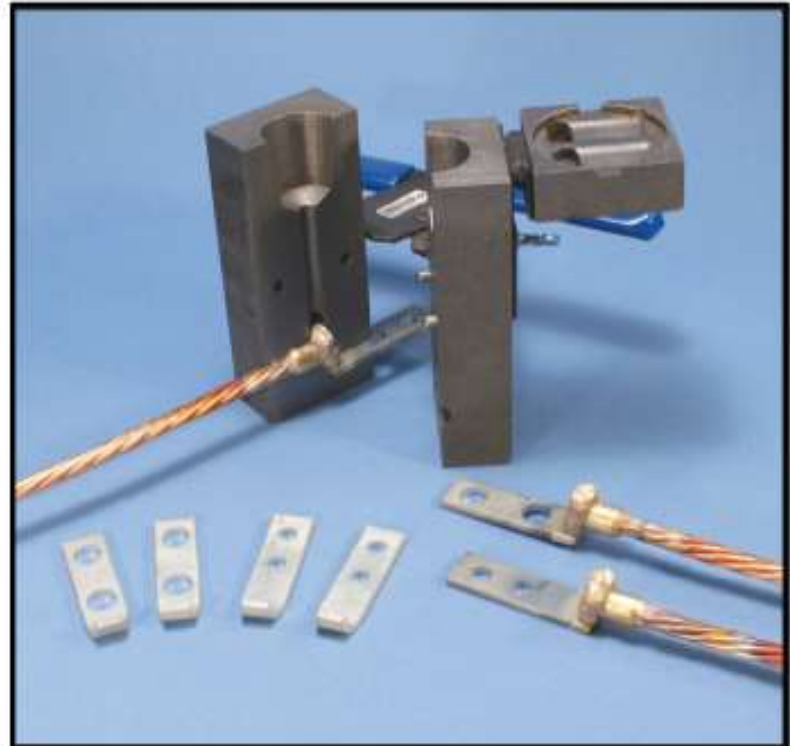


Figure 15. Example of a mold for an exothermic weld



Figure 14. Example of a barrel with three crimps

7.5.2.2 Exothermic weld of a two-hole lug to a bonding conductor

The following steps ensure a reliable connection between the two-hole lug and the bonding conductor using an exothermic weld:

- a) Select the mold and weld metal applicable to the conductor size and lug configuration.
- b) Clean and torch-dry the conductor and the mold. Then insert the conductor and lug into the mold (see figure 15).

- c) Close the handle clamp, lock the mold and then insert the disk into the mold (see figure 16).



Figure 16. Mold being locked and disk inserted

- d) Pour the weld metal into the mold. Sprinkle starting material over the weld metal and on the lip of the mold (see figure 17).

Rules - IS3043 and NEC: 2011

Electrical & Electronic systems design in health care facility is an extraordinary challenge. Required technical knowledge exceeds typical residential and industrial construction.

Patients may be undergoing surgery and in life support systems. Any break in electrical supply for more than few seconds could be fatal for them.

Further some patients may have conductive instruments in contact with the bloodstream or heart muscle where the possibility for serious injury and/or death if that metal becomes energized (even to a very low level). Other dangers include wet areas, hazards due to flammable liquids and the presence of oxygen.

Technical requirement of equipment are not considered / provided while designing electrical system. Most equipment are connected to redundant UPS / Generators, to take care in case of a supply failure, but we forget to think what will happen if a circuit breaker trips due to a fault. This means electrical work must be designed and installed to an unusual level of safety and redundancy.

Rules - IS3043 and NEC: 2011

Similar to normal buildings, grounding and bonding issues create much confusion. Earthing and bonding systems distinctly differ from residential and industrial wiring systems. NEC-2011 (National Electric Code - Bureau of Indian standards) explains a hospital environment as

- The patient may not be in a condition to react normally to the effects of hazardous events
- The electrical resistance of the skin, which is normally an important protection against harmful electric currents is bypassed in certain examinations or treatments
- Medical electrical equipment may often be used to support or substitute vital body functions, the breakdown of which may cause a dangerous situation
- Specific locations in medical establishments where flammable atmosphere exists, call for special treatment
- Electric and magnetic interference may disturb certain medical examinations or treatments.

CAPE
ELECTRIC



- 1 = Feeder from the main service entrance (main distribution board)
- 2 = Distribution of the floor
- 3 = Operating theatre distribution panel
- 4 = Safety supply system
- 5 = Medical isolating transformer
- 6 = Insulation monitoring device
- 7 = Special safety supply system, E_2
- 8 = Special safety supply system E_1
- 9 = Central heating
- 10 = Metal window-frame
- 11 = Metal cabinet for instruments
- 12 = Meal washing-basin and water supply
- 13 = Ceiling stand with outlets for gas supply
- 14 = Ceiling stand with mains socket outlets (with terminals for equipotential bonding, enclosure connected to the protective conductor bar)
- 15 = Alarm device for the insulation monitoring device (example)
- 16 = Operating table (electrically driven)
- 17 = Operating lamp
- 18 = Ampere meter for special safety supply system
- 19 = X-ray equipment
- 20 = Sterilizer
- 21 = Residual-current protective device
- 22 = Protective conductor bar
- 23 = Equipotential conductor bar
- 24 = Terminals for equipotential bonding

N = neutral conductor

- 25 = Operation
- 26 = Warning
- 27 = Green
- 28 = Red
- 29 = Buzzer
- 30 = Stop button for buzzer
- 31 = Test button

IS3043 & NEC 2011 - Recommendations

LV Earthing TN-S with PME double earthing: IT earthing for specific areas

- Ground potential equipotentialisation in critical care areas using special bonding Techniques.
- IT system earthing increases reliability of power supply in areas where an interruption of power supply may cause hazard to patient.
- Separate circuits shall be provided for X-ray, electrotherapy, diathermy, electrocardiograph, etc
- Main Earthing Terminal for individual rooms as well as Local Equipotential bonding for patient environments.
- Screening requirements against interference in rooms where measurements of bioelectric potentials are performed.
- Electrical wiring inside walls, floor and ceiling of rooms should be screened by means of metal shielding of cables to reduce Interference Caused by Mains-Induced Electric Fields
- Room screening inside wall structure for ECG and EEG monitoring.

Hospital – Special Earthing Requirements

Special Bonding and Earthing requirements are mandatory for hospitals as per NEC - 2011 & IS 3043. But are generally not followed due to unawareness of system requirements. Implementing these special bonding requirements is easy if structural PME earthing measures are implemented while constructing the building.

Implementing these bonding and earthing requirements, patient's exposure to unwanted voltage on normally non-current-carrying conductive surfaces can be controlled so that their lives and health are not endangered by the electrical environment

Hospital – Special Earthing Requirements

Safety Provisions

#	Provisions	Principal Requirements	Installation Measures
i)	P_0	Duration of touch voltages restricted to a safe limit	TN-S, TT or IT system
ii)	P_1	As P_0 but additionally: Touch voltages in patient environment restricted to a safe limit	Additional to P_0 : Supply system with additional requirements for protective earthing, etc.
iii)	P_2	As P_1 but additionally: Resistance between extraneous conductive parts and the protective conductor bus bar of the room not exceeding 0.1Ω	Additional to P_1 : Supplementary equipotential bonding
iv)	P_3	As P_1 or P_2 but additionally: Potential difference between exposed conductive parts, extraneous conductive parts and the protective conductor bus bar not exceeding 10 mV in normal condition	As P_1 or P_2 : Measurement necessary, corrective action possibly necessary
v)	P_4	As P_1 or P_2 . Additional protection against electric shock by limitation of disconnecting time	Additional to P_1 or P_2 : Residual current operated protective device
vi)	P_5	Continuity of the mains supply maintained in case of a first insulation fault to earth and currents to earth restricted	Additional to P_1 , P_2 or P_3 : IT supply system with insulation monitoring

Hospital – Special Earthing Requirements

Safety Provisions

#	Provisions	Principal Requirements	Installation Measures
vii)	P_6	Reduction of fault currents and touch voltages in case of a fault in the basic insulation	Additional $P1$ or $P2$: Medical isolating transformer supplying one individually piece of equipment
viii)	P_7	Prevention of dangerous touch voltages in normal condition and in single fault condition	Additional to $P1$ or $P2$: Supply with medical safety, extra low voltage
ix)	$G E$	No interruption of the power supply of the essential circuits of the hospital for more than 15 s	Safety supply system
x)	E_1	No interruption of the power supply of life-supporting equipment for more than 15 s	Special safety supply system
xi)	E_2	No interruption of the power supply of the operating lamp for more than 0.5 s	Special safety supply system for operating lamp
xii)	A	Prevention of explosions, fire and electrostatic charges	Measures concerning explosion and fire hazards
xiii)	I	No exercise interference from electric and magnetic fields	Layout of building and installation, screening

Hospital – Special Earthing Requirements

Examples of Application of Safety Provisions

#	Medically Used Room	Protective Measures							Safety Supply System			Explosions & Fire	Measures Against EM Fields
		P_0/P_1	P_2	P_3	P_4	P_5	P_6	P_7	GE	$E1$	$E2$		
i)	Massage room	M	O					O	X				
ii)	Operating wash room	M	X					O	X				
iii)	Ward general	M	O					O	X				
iv)	Delivery room	M	X		X	O		O	X	O	X	O	O
v)	ECG, EEG, EMG room	M	X		X			O	X				X
vi)	Endoscopic room	M	X		X			O	X		O		
vii)	Examination or treatment room	M	O		X	O		O	X		O		
viii)	Labour room	M	X		X	O		O	X				O
ix)	Operating sterilization room	M	O		X			O	X				
x)	Urology room	M	X		X			O	X		O		
xi)	Radiological diagnostic and therapy room, other than mentioned under SI No. (xx) and (xxiv)	M	X		X			O	X				
xii)	Hydrotherapy room	M	X		X		O	O	X				
xiii)	Physiotherapy room	M	X		X	O		O	X				

M = Mandatory measure, X = Recommended measure, X1 = As X, additionally insulation resistance measurement, O = Additional measure, may be considered desirable.

Hospital – Special Earthing Requirements

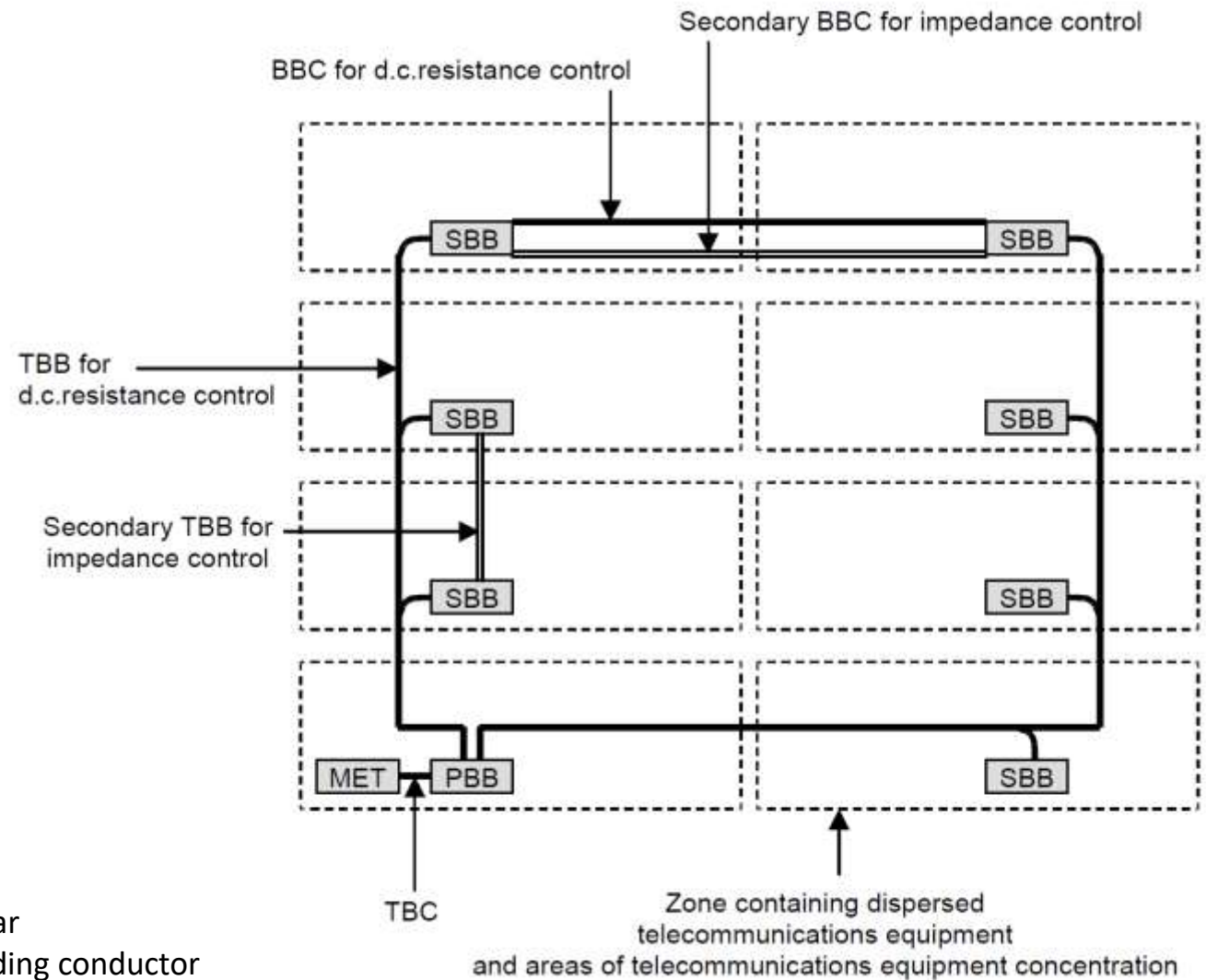
Examples of Application of Safety Provisions

#	Medically Used Room	Protective Measures							Safety Supply System			Explosions & Fire	Measures Against EM Fields
		P_0/P_1	P_2	P_3	P_4	P_5	P_6	P_7	GE	$E1$	$E2$	A	I
xiv)	Anaesthetic room	M	X	X	X_1	X		O	X	X	X	O	O
xv)	Operating theatre	M	X	X	X_1	X		O	X	X	X	X	X
xvi)	Operating preparation room	M	X	X	X_1	X		O	X	X	X	X	X
xvii)	Operating plaster room	M	X		X_1	X		O	X	X	X	X	X
xviii)	Operating recovery room	M	X	X	X_1	X		O	X	X	X	X	X
xix)	Outpatient operating theatre	M	X		X_1	X		O	X	X	X	X	X
xx)	Heart catheterization room	M	X	X	X_1	X		O	X	X	X		X
xxi)	Intensive care room	M	X	O	X_1	X		O	X	X	X		X
xxii)	Intensive examination room	M	X	O	X_1	X		O	X	O	O		X
xxiii)	Intensive monitoring room	M	X	O	X_1	X		O	X	X	X		X
xiv)	Angiographic examination room	M	X	O	X_1	X		O	X	O	O		O
xxv)	Hemodialysis room	M	X	X	X_1	X			X				
xxvi)	Central monitoring room	M	X	O	X_1	X		O	X				O

M = Mandatory measure, X = Recommended measure, X_1 = As X, additionally insulation resistance measurement, O = Additional measure, may be considered desirable.

ISO / IEC 30129: IT infrastructure – Telecommunication Bonding Network

example of a large building



PBB - primary bonding bus bar

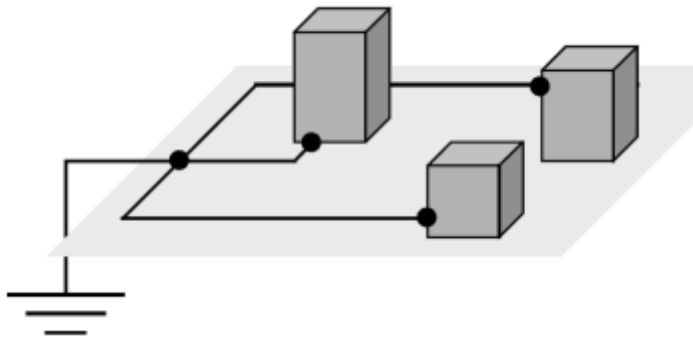
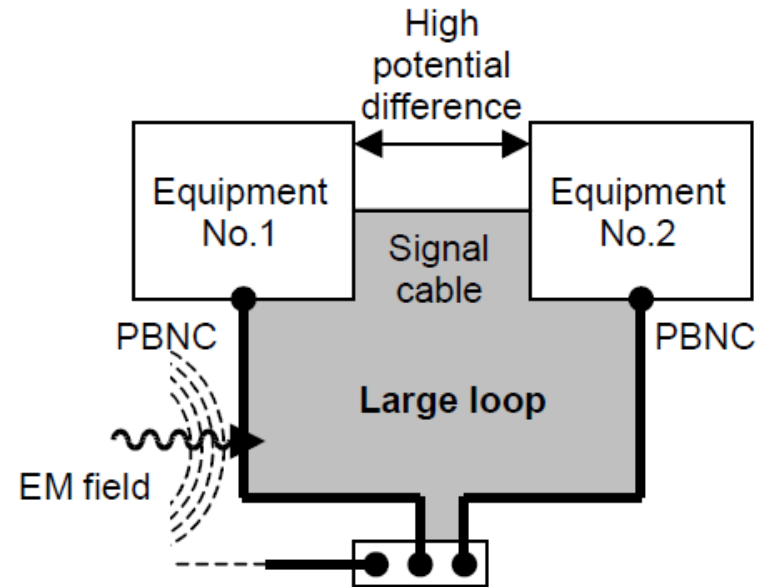
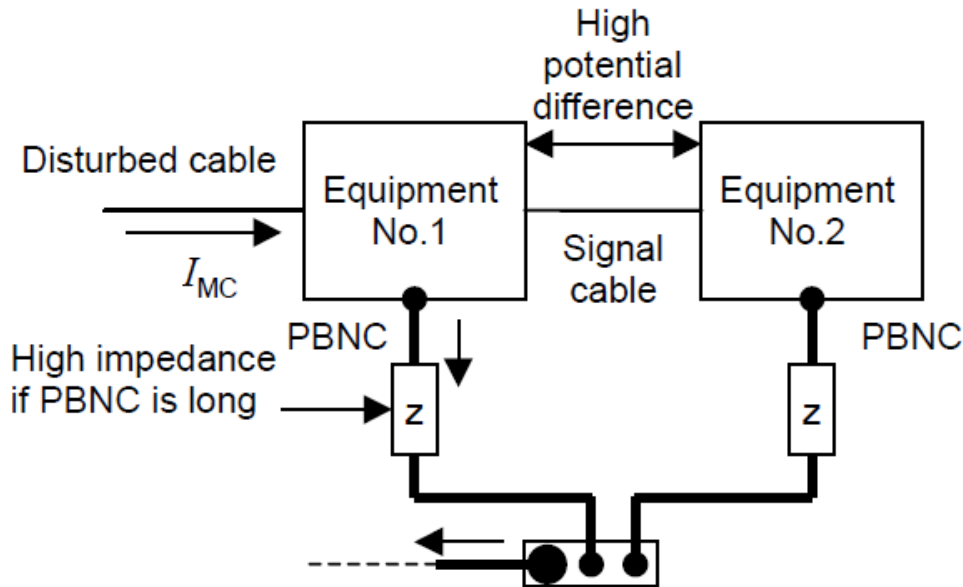
SBB - secondary bonding bus bar

TBC - telecommunications bonding conductor

TBB - telecommunications bonding backbone

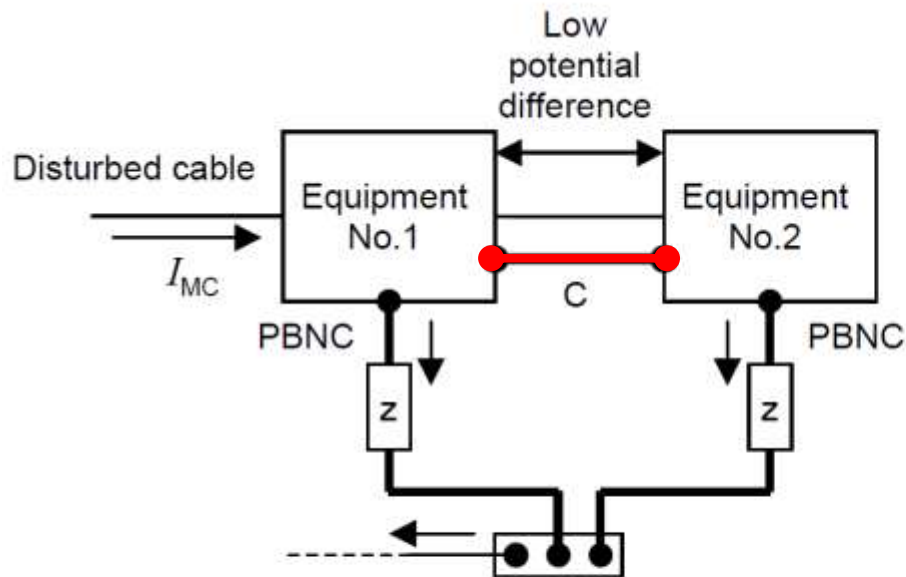
BBC - backbone bonding conductor

ISO / IEC 30129 Equipotential Bonding Networks **STAR – BN (STAR Topology)**

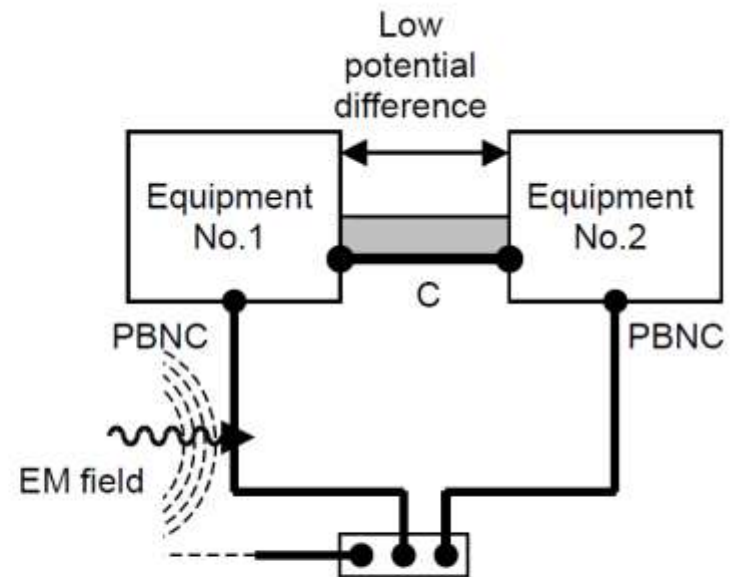


More vulnerable to electromagnetic disturbances
Can be used for Residential / Less Electronics

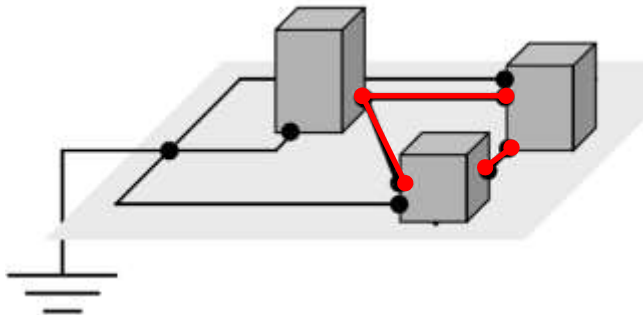
ISO / IEC 30129 Equipotential Bonding Networks **STAR – BN (STAR Topology)** with Supplementary Bonding Conductors

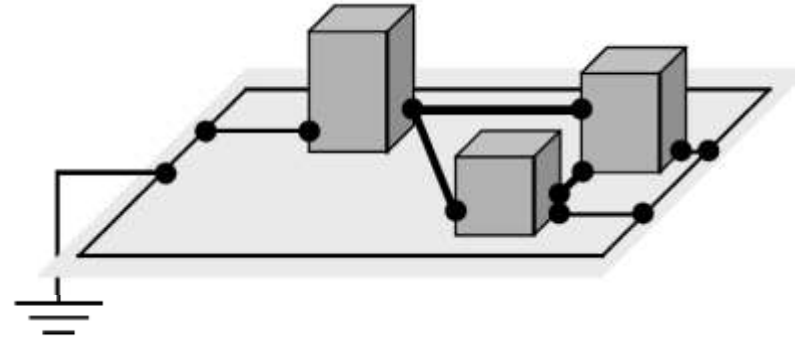
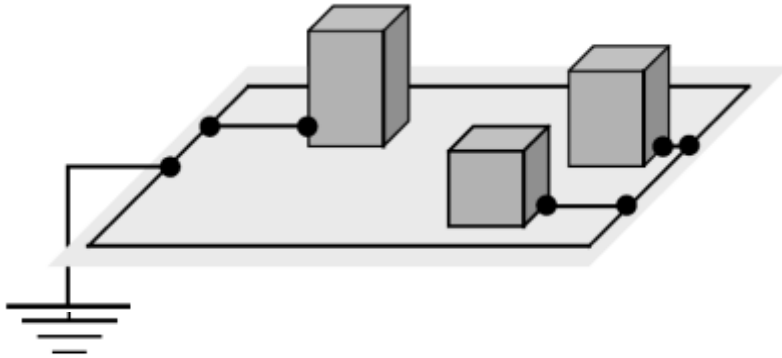


An additional bonding conductor (C) reduces the common impedance between equipment

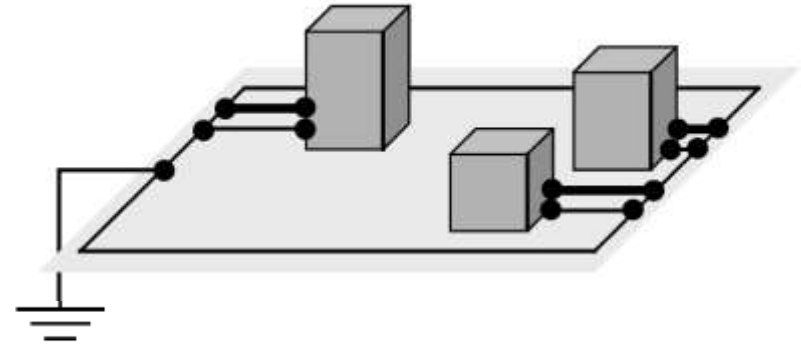


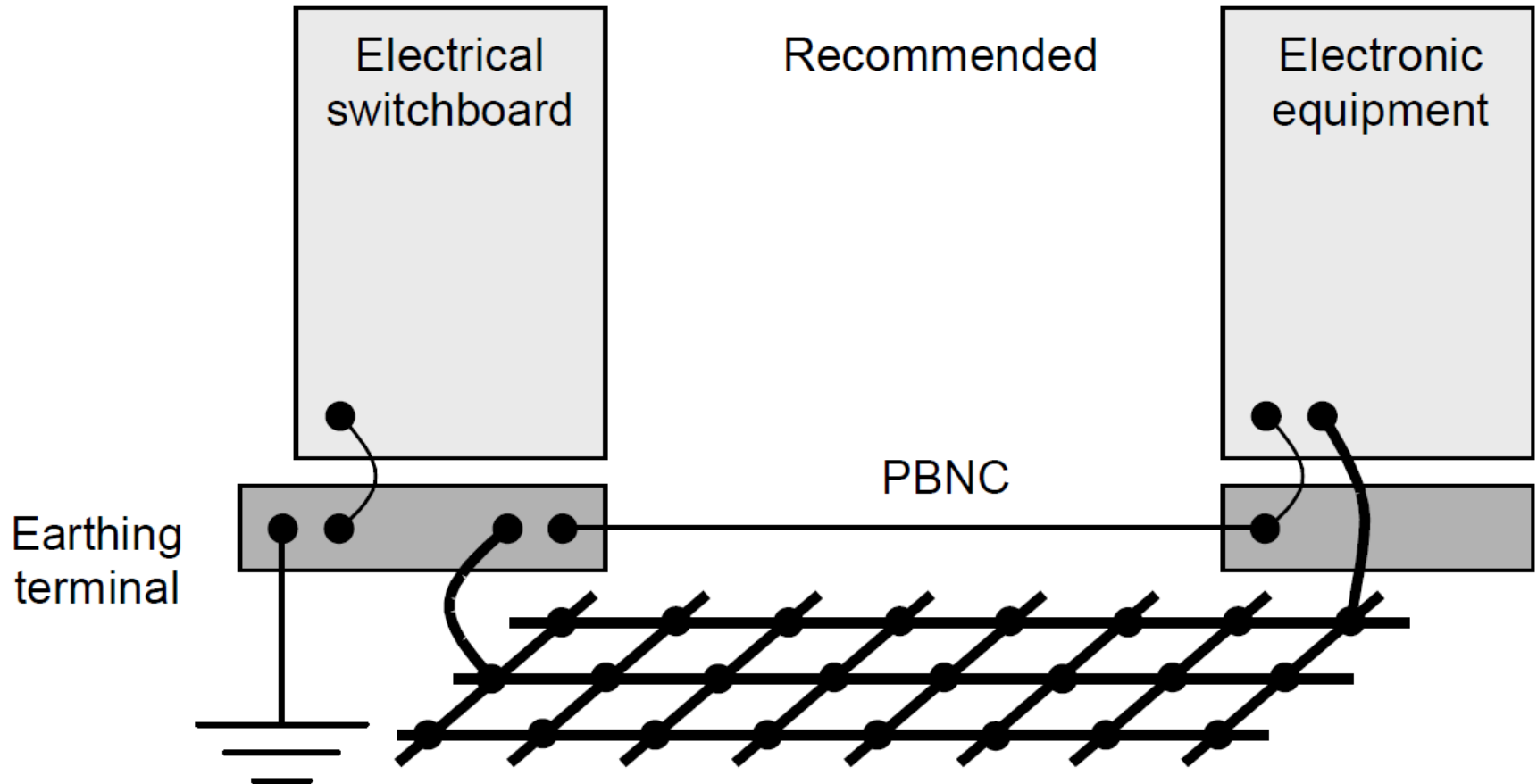
An additional bonding conductor (C) reduces the area of the sensitive loop





In addition, it is recommended that bonding conductors should be connected between conductive cable management systems and the bonding ring conductor





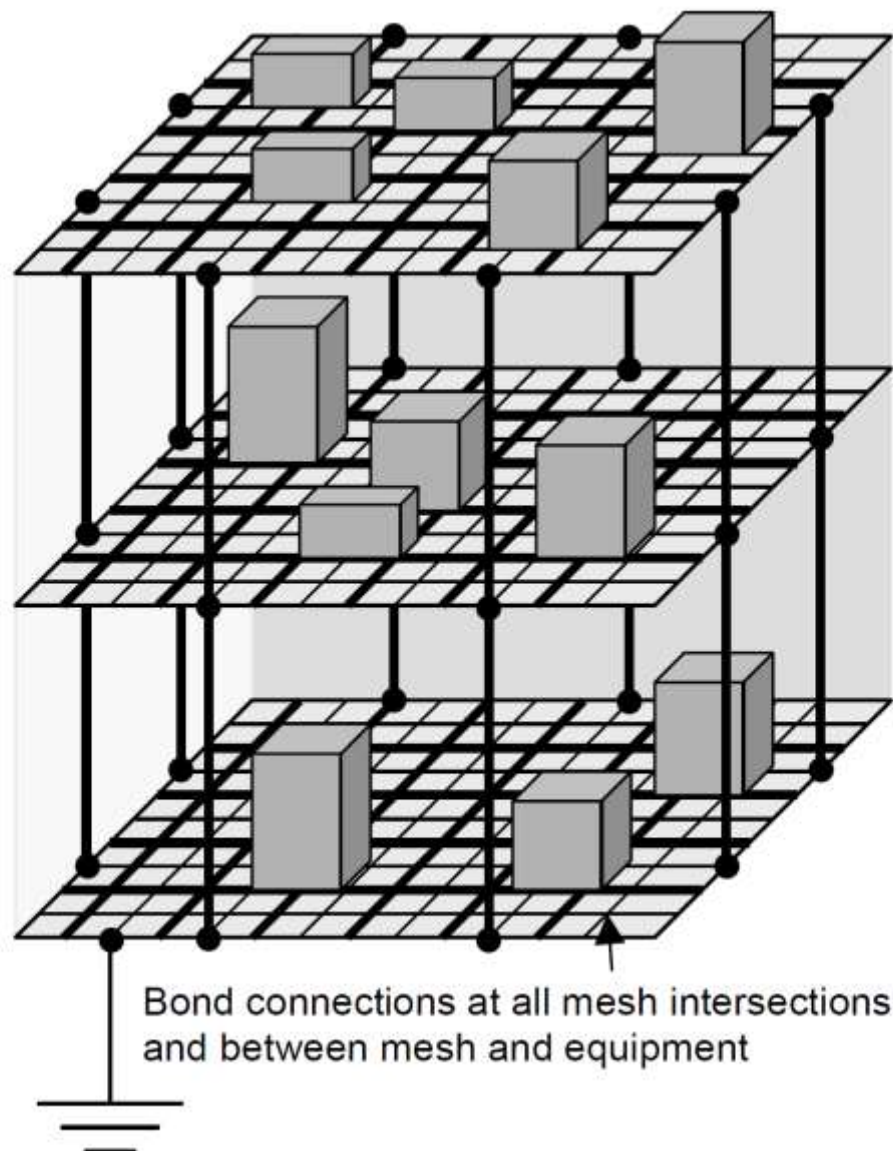
Multiple bonds, including those using building structures, should be used rather than a single bond since this reduces the impedance (inductance) of the resulting bond

Mesh Bonding Network - (MESH-BN) for a multi floor building

A MESH-BN with equipment cabinets, frames, racks and CBN bonded together

A merged MESH-BN and protective bonding network in buildings shall include the interconnections to the following installations, if present

- a. “integrated lightning protection system” according to IEC 62305-4,
- b. bonding measures of antenna installations (including satellite receiving equipment under private property) and cable networks according to series IEC 60728,
- c. bonding measures of information technology cabling according to ISO/IEC 14763-2,
- d. bonding in hazardous areas, e.g. according to IEC 60079-14.



Wrong Practices in India

- **Lightning Protection: ESE Lightning Protection is used widely in India**
- **This is not confirming IS standards**
- **NBC-2016 Banned usage of ESE rods for lightning protection**
- **CEA safety regulation – 2016 draft recommend to install lightning protection as per IS/IEC 62305 for building higher than 15 meters**



ESE rods shall not be used. NBC-2016

भारत की राष्ट्रीय
भवन निर्माण संहिता 2016
खण्ड 2
NATIONAL BUILDING CODE
OF INDIA 2016
VOLUME 2

- Vertical rods (offers certain angle of protection);
- Catenary wires; and
- Meshed/Gridd conductors.

All types of air-termination systems shall be positioned in accordance with 11.5.1.2. The individual air-termination rods should be connected together at roof level to ensure current division. Radiomagnetic air-terminals shall not be allowed. Any other kind of air-terminal like dissipation system/ESE air-terminal/CSE air-terminal shall not be acceptable.

11.5.1.2 Positioning

Air-termination components installed on a structure shall be located at corners, exposed points and edges (especially on the upper level of any facades) in accordance with one or more of the following methods (see also Figs. 16 to 20 in conjunction with good practice [8-2(45)]):

- Protection angle method;
- Rolling sphere method; and
- Mesh method.

The protection angle method is suitable for simple shaped buildings but it is subject to limits of air-termination height indicated in Table 6. The mesh method is a suitable form of protection where plane/pitch roof surfaces are to be protected. The rolling sphere method is suitable in all cases. See good practice [8-2(45)] for details.

11.5.1.3 Roof mounted electrical/electronic equipment (for example, chillers, antennas, cameras and bill boards) need vertical air-termination to avoid direct flashover. All parts of lightning protection should maintain separation distance from these electrical/electronic equipment. Power and data connection to these equipment should have proper Class I and Class II SPD's (see 11.6.5) to avoid failures. Overhead cables such as cable TV lines from one building to the other should be avoided.

11.5.1.4 Unearthed metallic roofs should be avoided. Metallic roofs shall be connected either to steel

reinforcement or to other earthed steel parts of the building satisfying the requirements of number of down-conductors (see 11.5.2). Small buildings with metallic roofs less than 100 m² shall be earthed at least not less than 2 places.

11.5.1.5 Structures of height less than 60 m and more than 60 m

On structures lower than 60 m in height, generally flashes to the side may not occur, hence air-termination protection on sides will not be required (see Fig. 21A).

On structures taller than 60 m, flashes to the side may occur, especially to points, corners and edges of surfaces. In general, the risk due to these flashes is low, but electrical and electronic equipment on walls or outside structures may be destroyed even by lightning flashes with low current peak values.

An air-termination system shall be installed to protect the upper part of tall structures (that is, typically the topmost 20 percent of the height of the structure as far as this part exceeds 60 m in height) and the equipment installed on it. The rules for positioning the air-termination systems on these upper parts of a structure shall meet at least the requirements for LPL IV with emphasis on the location of air-termination devices on corners, edges, and significant protrusions (such as balconies, viewing platforms, etc (see Fig. 21B)).

11.5.1.6 Buildings with roof top solar PV and water heaters

Vertical air-terminals are required for protecting roof mounted installations such as solar PV, water heaters, chillers as well as water tanks. Protection angle should be considered as per Table 6. Vertical air-terminals need to be connected to the air-termination mesh/down-conductors. Metal support structure of these installations shall be bonded to the air-termination mesh/down-conductors. Class I/Class II surge protection devices (SPDs) (see 11.6.5) should be installed in the electrical lines to protect the installations inside the building typically d.c. SPD for solar PV output at inverter or junction box level and a.c. SPD

Table 6 Maximum Values of Mesh Size and Protection Angle Corresponding to the Class of LPS
(Clause 11.5.1.2)

Sl No.	Class of LPS	Mesh Size	Rolling Sphere Radius	Protection Angle with respect to Height				
				10 m	20 m	30 m	45 m	60 m
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	I	5 × 5	25	45	23	Cannot be used		
ii)	II	10 × 10	30	54	28	23	Cannot be used	
iii)	III	15 × 15	45	62	48	36	3	Cannot be used
iv)	IV	20 × 20	60	69	54	45	34	23

Earthing is for Safety: Return of fault current and quick operation of switchgear.
Earthing for Dissipation of Lightning current to soil
Earthing for Voltage reference and Equipotential base
Earthing for EMI and protection of electronics

Improvement in Bonding Techniques is necessary for Protection of electronics

THANK YOU

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